Discovery-Mode Flavor Constraint Catalog

May 16, 2026

Abstract

Placeholder abstract for the discovery-mode flavor-constraint catalog companion to the rc1.1 quark-scan paper.

1 Kaon

2 K001: ε_K

Process

Standard notation: ε_K. This entry covers indirect CP violation in neutral-kaon mixing, namely the CP-violating part of the K^0-$\bar{K}^0$ $\Delta F = 2$ amplitude. It is the headline kaon CP constraint for anarchic warped flavor because generic new phases in s-d mixing are tightly limited by the observed value.

PDG / equivalent value(s)

The canonical experimental value is the PDG kaon CP-violation magnitude (`PDG2026:EpsilonK`),

$$|\varepsilon| = (2.228 \pm 0.011) \times 10^{-3},$$

with the related phase $\phi_\varepsilon = (43.5 \pm 0.5)^\circ$. The local text snapshot is `flavor_catalog/references/K001/pdg2026_epsilon_k.txt`. For the SM comparison used by the legacy code, Brod–Gorbahn–Stamou (`BrodGorbahnStamou:EpsilonK2020`) quote

$$|\varepsilon_K|_{\rm SM} = (2.161 \pm 0.153_{\rm param.} \pm 0.076_{\text{non-pert.}} \pm 0.065_{\rm pert.}) \times 10^{-3} = 2.16(18) \times 10^{-3}.$$

The relevant FLAG kaon-bag input (`FLAG2024:BK`) is the $N_f = 2+1$ average $\hat{B}_K = 0.7533(91)$, with $B_K^{\overline{\rm MS}}(2\,{\rm GeV}) = 0.5503(66)$.

Relevance to the RS / anarchic-flavor pipeline

In anarchic RS models, KK-gluon and electroweak gauge exchange generate misaligned four-quark operators for K^0-$\bar{K}^0$ mixing. The left-right matrix elements are chirally enhanced, and generic complex phases make ε_K one of the strongest constraints on the KK scale. The CFW baseline

(`CsakiFalkowskiWeiler:RSFlavor2008`) quotes KK-gluon masses of about 21 TeV in standard RS and about 33 TeV in the composite pseudo-Goldstone variant.

Post-2008 developments

Since the 2008 CFW baseline, the experimental central value has remained precise. The main updates are theory and lattice inputs. BGS (`BrodGorbahnStamou:EpsilonK2020`) reformulated the SM prediction with manifest CKM unitarity and reduced the charm short-distance uncertainty, giving the 2.161×10^{-3} central value used in the repository's legacy $\Delta F = 2$ path. FLAG 2024 (`FLAG2024:BK`) updates the kaon bag averages. The catalog should therefore keep the experimental number, the SM theory budget, and the lattice inputs separate.

Constraint validity and model dependence

As an RS constraint this is robust for generic tree-level $\Delta F = 2$ new physics with arbitrary CP phases. Model dependence enters through the matching convention, QCD running, hadronic matrix elements, and the choice of how much of $|\varepsilon_K^{\rm exp} - \varepsilon_K^{\rm SM}|$ is assigned as the allowed NP budget. It is not a standalone measurement of one Wilson coefficient; it constrains the imaginary part of the full K-mixing amplitude.

Code coverage in this repo

Coverage status: YES. The legacy implementation registers `epsilon_k` in `quarkConstraints/deltaf2.py:209–212`, stores the PDG and BGS constants at `quarkConstraints/deltaf2.py:622–623`, and evaluates the observable in `quarkConstraints/deltaf2.py:729`. The QCD-running wrapper is at `quarkConstraints/deltaf2.py:775`. The modern policy surface also enumerates `epsilon_K` at `quarkConstraints/modern/phenomenology.py:23` and routes it through the bridge evaluator at `quarkConstraints/modern/phenomenology.py:458` and `quarkConstraints/modern/phenomenology.py:1320`. Tests exercise the evaluator at `tests/test_epsilon_k_physics.py:166` and `tests/test_epsilon_k_physics.py:`

Implementation difficulty

Implementation difficulty: LOW. The needed $\Delta F = 2$ operator basis, hadronic contraction, QCD-running wrapper, and pass/fail budget already exist. A future production cleanup should reconcile the legacy and modern frozen input constants, but no new operator, lattice machinery, or exclusive-mode calculation is needed for catalog-level coverage.

Key references

Process-local source keys before bibliography consolidation: PDG2026:EpsilonK, BrodGorbahn-Stamou:EpsilonK2020, FLAG2024:BK, and CsakiFalkowskiWeiler:RSFlavor2008.

3 K002: Δm_K

Process

Standard notation: $\Delta m_K \equiv m_{K_L} - m_{K_S}$. This entry covers the CP-even magnitude of neutral-kaon mixing, separate from the CP-violating ε_K entry.

PDG / equivalent value(s)

The canonical PDG-sidecar value is the 2026 pdgLive fit assuming CPT (`pdg_live_S013D_20260516`),

$$\Delta m_K = (0.5293 \pm 0.0009) \times 10^{10}\,\hbar\,\mathrm{s}^{-1},$$

with the quoted PDG uncertainty including a scale factor of 1.3. The same snapshot gives the companion fit not assuming CPT, $(0.5289{\pm}0.0010){\times}10^{10}\,\hbar\,\mathrm{s}^{-1}$. Both values trace to `flavor_catalog/references/K00`

Relevance to the RS / anarchic-flavor pipeline

In anarchic warped flavor models, tree-level KK-gluon exchange generates $\Delta S = 2$ four-quark operators after rotating to the quark-mass basis. Δm_K constrains the real part of the new $K^0 - \bar{K}^0$ mixing amplitude, while ε_K constrains its imaginary part. The same VLL, VRR, SLL/SLR, and LR operator structures used for the repo's neutral-meson mixing lane apply here, and the LR matrix elements remain chirally enhanced. The important physics caveat is that the measured mass splitting contains a large long-distance Standard Model component, so the catalog should treat Δm_K as a conservative magnitude bound unless a controlled SM subtraction is supplied.

Post-2008 developments

Relative to the CFW 2008 baseline (`cfw_2008`), the experimental side was updated by KTeV's final neutral-kaon analysis (`ktev_2011`), which the PDG fit still uses. The more important post-2008 shift is theoretical: lattice calculations began attacking the long-distance contribution directly. Christ et al. (`christ_2013_ld_delta_mk`) developed the long-distance method, and Bai et al. (`bai_2014_delta_mk`) reported a first complete lattice calculation, $\Delta M_K = 3.19(41)(96) \times 10^{-12}$ MeV, at heavier-than-physical masses. Later physical-quark-mass studies (`wang_2023_delta_mk`) reached statistical uncertainty near 9% while retaining important systematics. For BSM interpretation, FLAG 2024 (`flag2024`) and the physical-mass BSM kaon-mixing calculation (`boyle_2024_bsm_kaon_mixing`) update the bag inputs used to translate Wilson coefficients into matrix elements.

Constraint validity and model dependence

The experimental number is precise, but Δm_K is long-distance limited as a new-physics constraint. A conservative RS scan can require the absolute new-physics contribution not to exceed the measured splitting, or use only a fraction of it by policy. A sharper bound would require specifying the short-distance charm-loop treatment, the long-distance lattice input, and the CPT-fit convention. This entry is therefore robust as a conservative $\Delta F = 2$ magnitude check, but not as a clean SM-subtracted exclusion.

Code coverage in this repo

PARTIAL. The methodology note lists Δm_K among the five $\Delta F = 2$ scan observables at `docs/quark_scan_methodology_note.tex:118`. The legacy kaon helper sets the PDG mass splitting at `quarkConstraints/deltaf2.py:615` and implements `evaluate_delta_mk` at `quarkConstraints/deltaf2.` The modern lane also has a `K` input displayed as Δm_K at `quarkConstraints/modern/inputs.py:958` and a bridge evaluator at `quarkConstraints/modern/phenomenology.py:493`. I keep the status partial because the legacy default kaon input is still the ε_K entry at `quarkConstraints/deltaf2.py:209`, and the catalog has not audited a standalone long-distance policy.

Implementation difficulty

LOW. No new operator basis is needed for a conservative catalog or scan-side magnitude bound: the existing $\Delta F = 2$ VLL/VRR/LR machinery covers the required Wilson coefficients. A future precision SM-subtracted Δm_K likelihood would be a separate high-theory-effort project.

Key references

Process-local source keys: `pdg_live_S013D_20260516`, `ktev_2011`, `christ_2013_ld_delta_mk`, `bai_2014_delta_mk`, `wang_2020_physical_delta_mk`, `wang_2023_delta_mk`, `boyle_2024_bsm_kaon_mixing`, `flag2024`, and `cfw_2008`.

4 ε'/ε in $K \to \pi\pi$

Process

Standard notation: $\mathrm{Re}(\varepsilon'/\varepsilon)$. This entry covers direct CP violation in the two-pion decay amplitudes of neutral kaons, commonly written through $\mathrm{Re}(\varepsilon'/\varepsilon) = (1 - |\eta_{00}/\eta_{+-}|)/3$. The process id is $K003$.

PDG / equivalent value(s)

The canonical PDG-sidecar value is

$$\mathrm{Re}(\varepsilon'/\varepsilon) = 0.00166 \pm 0.00023 = (16.6 \pm 2.3) \times 10^{-4}.$$

This number is recorded in the YAML `pdg_or_equivalent` block and traces to `flavor_catalog/references/K003/` The companion PDG DataBlock snapshot records `OUR FIT` $= 1.67 \pm 0.23$ and `OUR AVERAGE` $= 1.68 \pm 0.20$, both in units of 10^{-3}. The same sidecar lists the two dominant experimental inputs: KTeV $(19.2 \pm 2.1) \times 10^{-4}$ and NA48 $(14.7 \pm 2.2) \times 10^{-4}$.

Relevance to the RS / anarchic-flavor pipeline

Unlike ε_K, this is a $\Delta S = 1$ nonleptonic observable. It tests CP-odd short-distance contributions to the $K \to \pi\pi$ amplitudes after they are filtered through long-distance QCD. In warped or composite flavor models with anarchic Yukawas, the dangerous effects are not the repo's current neutral-meson-mixing operators but new penguin, scalar/tensor, and dipole structures. BSM master-formula work identifies strong sensitivity to electroweak left-right penguin Q_8, QCD penguin Q_6, and chromo-magnetic O_{8g}-type operators, making the channel a natural stress test for UV completions with non-minimal chirality structure.

Post-2008 developments

The experimental average has not been replaced by a newer kaon experiment, but KTeV's final $K \to \pi\pi$ result was published after the CFW-era baseline. The main post-2008 changes are theoretical. RBC/UKQCD produced a physical-kinematics lattice calculation of A_0 and ε', with $\mathrm{Re}(\varepsilon'/\varepsilon) = 21.7(2.6)(6.2)(5.0) \times 10^{-4}$. Aebischer, Bobeth, and Buras then reanalysed the ratio with NNLO electroweak-penguin corrections and ChPT isospin-breaking inputs, obtaining $(17.4 \pm 6.1) \times 10^{-4}$ in the octet treatment and $(13.9 \pm 5.2) \times 10^{-4}$ in the nonet treatment. In parallel, 2019 BSM master-formula papers organized the $\Delta S = 1$ operator dependence in WET/SMEFT language rather than treating ε'/ε as a single phenomenological number.

Constraint validity and model dependence

As a catalog constraint, $K003$ is highly informative but not plug-and-play. The experimental number is cleanly established, while the SM subtraction and BSM interpretation depend on hadronic matrix elements, isospin breaking, operator matching, and RG evolution. It should be classified as a theory-limited, model-dependent $\Delta S = 1$ amplitude constraint rather than a robust tree-level exclusion like the current $\Delta F = 2$ kaon lane.

Code coverage in this repo

NO. Greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no direct implementation of ε'/ε, $K \to \pi\pi$, $\Delta S = 1$, Q_6, Q_8, or O_{8g}. The nearby kaon code is explicitly a $\Delta F = 2$ benchmark slice in `quarkConstraints/deltaf2.py:1`, with default inputs for ε_K at `quarkConstraints/deltaf2.py:209` and an ε_K evaluator at `quarkConstraints/delt`
The modern policy surface likewise enumerates only ε_K, K, B_d, B_s, and D^0 systems at `quarkConstraints/modern/`

Implementation difficulty

HIGH. A production constraint would require a new $\Delta S = 1$ operator basis, matching from the warped/composite UV model to WET/SMEFT coefficients, RG running to hadronic scales, and lattice/ChPT inputs for $K \to \pi\pi$ matrix elements. The current $\Delta F = 2$ VLL/VRR/LR machinery cannot be reused as a direct evaluator.

Key references

Local source keys: `pdg_live_S013_20260516`, `pdg_live_S013EPS_datablock_20260516`, `ktev_2011`, `na48_2002`, `rbc_ukqcd_2020`, `aebischer_buras_2020`, `bsm_master_2019`, `bsm_anatomy_2019`, and `cfw_2008`.

5 K004: $K^+ \to \pi^+ \nu\bar{\nu}$

Process

Standard notation: $K^+ \to \pi^+ \nu\bar{\nu}$. This entry covers the charged "golden" kaon decay governed by the short-distance $s \to d\nu\bar{\nu}$ flavor-changing neutral current, with emphasis on the real-part sensitivity that complements $K_L \to \pi^0 \nu\bar{\nu}$.

PDG / equivalent value(s)

The YAML sidecar uses the latest NA62 experimental-equivalent combination, rather than an older PDG average:

$$\mathrm{BR}\big(K^+ \to \pi^+ \nu\bar{\nu}\big) = \big(9.6^{+1.9}_{-1.8}\big) \times 10^{-11},$$

from NA62 data collected in 2016–2024, traced to `flavor_catalog/references/K004/na62_2026_arxiv2604_1264` The published 2016–2022 observation anchor is $\big(13.0^{+3.3}_{-3.0}\big) \times 10^{-11}$, with 51 signal candidates and an expected background of 18^{+3}_{-2} events; see `flavor_catalog/references/K004/na62_2025_arxiv2412_12015.txt`. For SM comparison, the sidecar records the Buras–Venturini benchmark $\mathrm{BR}_{\mathrm{SM}} = (8.60 \pm 0.42) \times 10^{-11}$ and the Brod–Gorbahn–Stamou update $\mathrm{BR}_{\mathrm{SM}} = 7.73(61) \times 10^{-11}$.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models can generate misaligned s-d neutral currents through KK gauge exchange, Z-coupling shifts, or related electroweak matching effects. Unlike the current quark lane's $\Delta F = 2$ kaon observables, this is a $\Delta F = 1$ semileptonic process. It therefore tests whether the model has controlled short-distance neutral currents beyond the four-quark K-mixing sector. The charged mode is especially useful because it is theoretically clean, uses a hadronic matrix element tied to semileptonic kaon data, and can interfere with the SM amplitude rather than only setting an incoherent upper bound.

Post-2008 developments

The CFW-era RS discussion used $\Delta F = 2$ flavor bounds as the central pressure point, quoting a generic RS KK-gluon mass scale of about 21 TeV and a composite pseudo-Goldstone scenario scale of about 33 TeV. Since that era, NA62 reported the published observation in 2025 and a newer 2026 preliminary combination with better-than-20% precision. On the theory side, post-2008 SM updates have sharpened the short-distance prediction and exposed the role of CKM-input choices: Brod–Gorbahn–Stamou quote $7.73(61) \times 10^{-11}$, while Buras–Venturini quote $(8.60 \pm 0.42) \times 10^{-11}$ using a strategy designed to avoid the direct $|V_{cb}|$ and $|V_{ub}|$ tensions.

Constraint validity and model dependence

As an experimental branching-ratio constraint this mode is clean, but its use inside an RS scan is model-dependent. A production constraint needs Wilson coefficients for $(\bar{s}\gamma_\mu P_{L,R} d)(\bar{\nu}\gamma^\mu(1-\gamma_5)\nu)$, a choice of neutrino-flavor treatment, and a convention for SM–NP interference. The Buras–Venturini SM number also assumes no new-physics infection in the $\Delta F = 2$ inputs used to remove CKM ambiguities, so a checker should keep that assumption visible if the catalog later turns this process into a hard bound.

Code coverage in this repo

NO. The required grep sweep finds kaon mixing, generic neutrino-mass utilities, and the catalog planning row, but no implementation of $K^+ \to \pi^+\nu\bar{\nu}$, no $\Delta F = 1$ $s \to d\nu\bar{\nu}$ operator, and no branching-ratio acceptance gate. The closest quark-code surfaces are the $\Delta F = 2$ entries in `quarkConstraints/deltaf2.py` and the modern policy list `MODERN_PHENOMENOLOGY_SYSTEM_IDS = (epsilon_K, K, B_d, B_s, D0)`.

Implementation difficulty

MEDIUM. The observable needs a new $\Delta F = 1$ semileptonic operator basis and a branching-ratio formula with SM–NP interference, but the hadronic input is standard and this channel does not require the angular likelihood machinery needed for many $b \to s\ell\ell$ observables. The difficulty would become HIGH only if the catalog requires a full RS tower and neutrino-sector matching rather than an EFT Wilson input.

Key references

Process-local reference keys before bibliography consolidation: NA62:KpPipNunu2026, NA62:KpPipNunu2025, BurasVenturini:KpPipNunu2022, BrodGorbahnStamou:KpPipNunu2021, and CsakiFalkowskiWeiler:RSFlavor2008

6 $K_L \to \pi^0\nu\bar{\nu}$

Process

Standard notation: $K_L \to \pi^0\nu\bar{\nu}$. This entry covers the CP-violating neutral-kaon golden mode, a $\Delta S = 1$ flavor-changing neutral-current decay with an invisible neutrino pair. The process id is $K005$.

PDG / equivalent value(s)

The sidecar value is

$$\mathcal{B}(K_L^0 \to \pi^0\nu\bar{\nu}) < 2.2 \times 10^{-9} \quad (90\%\ \mathrm{CL}),$$

from the 2026 PDG pdgLive S013R40 table, traced to `flavor_catalog/references/K005/pdg2026_pdgLive_S013R`
The same sidecar records the direct KOTO 2025 input behind the PDG value: the 2021 KOTO data set had expected background $0.252 \pm 0.055_{\text{stat}}{}^{+0.052}_{-0.067}{}_{\text{syst}}$, single-event sensitivity $(9.33 \pm 0.06_{\text{stat}} \pm 0.84_{\text{syst}}) \times 10^{-10}$, and zero signal-region events, yielding the same 2.2×10^{-9} limit.

For SM comparison the sidecar gives two current theory benchmarks: Brod–Gorbahn–Stamou quote $\mathcal{B}_{\text{SM}} = 2.59(29) \times 10^{-11}$, while the Buras–Venturini $|V_{cb}|$- and $|V_{ub}|$-independent determination quotes $(2.94 \pm 0.15) \times 10^{-11}$ under its no-new-physics assumptions in $|\varepsilon_K|$ and $S_{\psi K_S}$. The related charged golden mode has the 2026 NA62 value $\mathcal{B}(K^+ \to \pi^+ \nu\bar{\nu}) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$, included only as program context and not as a direct K005 measurement.

Relevance to the RS / anarchic-flavor pipeline

This mode is one of the cleanest probes of short-distance flavor violation: the hadronic matrix element is tied to semileptonic kaon data, and the neutral mode isolates the imaginary part of the $s \to d\nu\bar{\nu}$ amplitude. In RS/anarchic-flavor language it tests flavor-changing Z, Z', or other neutral-current couplings induced by fermion localization and KK mixing, plus loop-level contributions if the ultraviolet completion generates them. It is therefore complementary to ε_K: both are CP-sensitive kaon observables, but K005 is a $\Delta S = 1$ rare decay rather than a $\Delta F = 2$ mixing constraint.

Post-2008 developments

Since the Csaki–Falkowski–Weiler era, the experimental frontier moved from older E391a/KOTO pilot limits to the KOTO direct search now mirrored by PDG. KOTO's 2025 PRL result improves the previous KOTO direct bound and reports a background expectation below one event for the 2021 analysis. On the charged side, NA62 has moved $K^+ \to \pi^+ \nu\bar{\nu}$ from evidence/observation to a combined 2016–2024 value with better-than-20% precision, sharpening the cross-check between the charged and neutral golden modes. Theory updates also matter: Brod–Gorbahn–Stamou updated the SM $K \to \pi\nu\bar{\nu}$ prediction, and Buras–Venturini recast the prediction in ratios designed to reduce $|V_{cb}|$ and $|V_{ub}|$ dependence.

Constraint validity and model dependence

The SM channel is short-distance dominated and unusually clean, but the present catalog value is still an upper limit. The interpretation is robust for high-scale new physics that modifies semileptonic $s \to d\nu\bar{\nu}$ operators. It becomes model-dependent if the missing energy is a light non-neutrino particle, if neutrino flavor structure is non-universal, or if a global kaon fit is used to correlate K005 with the charged mode.

Code coverage in this repo

NO. The PKA searched the requested code areas, excluding notebooks, for $K_L \to \pi^0 \nu\bar{\nu}$, KOTO, NA62, $s \to d$, penguin, and generic rare-kaon strings. No rare-decay implementation was found. The only relevant K_L hit was the mixing input `quarkConstraints/deltaf2.py:615` for Δm_K, and the broader kaon code is $\Delta F = 2$ mixing machinery rather than a $\Delta S = 1$ semileptonic decay calculation.

Implementation difficulty

MEDIUM. A production implementation would need a new $s \to d\nu\bar{\nu}$ Wilson/operator path and the CP-odd K_L projection, but the observable uses standard clean hadronic inputs. No new lattice, RG, or long-distance calculation has been identified; the existing audited code can supply kaon-mixing context, but not the rare-decay observable itself.

Key references

Process-local keys deposited in `flavor_catalog/references/K005/`: `PDG2026:K005`, `KOTO2025:KLPi0NuNu`, `BrodGorbahnStamou2021:KPiNuNu`, `BurasVenturini2022:RareK`, `NA622026:KPlusPiNuNu`, and `BuchallaBuras19`

7 $K_L \to \mu^+\mu^-$

Process

Standard notation: $K_L \to \mu^+\mu^-$. This entry covers the long-lived neutral-kaon dimuon decay, catalog id $K006$. It is a strangeness-changing weak-neutral-current mode. The observed rate is dominated by the two-photon long-distance amplitude, but the dispersive component leaves a short-distance window for $s \to d$ Z-penguin and electroweak-box effects.

PDG / equivalent value(s)

The YAML sidecar source of truth lists the current PDG value as

$$\mathcal{B}(K_L \to \mu^+\mu^-) = (6.84 \pm 0.11) \times 10^{-9},$$

from the PDG K_L^0 listing, Γ_{24}, in the 2024 Review of Particle Physics. The local snapshot is `flavor_catalog/references/K006/pdg2024_kl_decay_modes.txt`, accessed 2026-05-16. The sidecar also records the primary E871 measurement, $\mathcal{B} = (7.18 \pm 0.17) \times 10^{-9}$, as experimental provenance rather than the catalog average.

Relevance to the RS / anarchic-flavor pipeline

In warped or anarchic-flavor models this channel is relevant through the same flavor-changing $s \to d$ neutral-current structures that appear in rare kaon phenomenology: modified Z couplings, heavy gauge-boson penguins, and box topologies. It is less clean than $K \to \pi\nu\bar{\nu}$, because the measured branching fraction is almost saturated by long-distance two-photon physics, but it is still a useful diagnostic for large semileptonic short-distance effects. The Isidori-Unterdorfer dispersive analysis quoted in the sidecar gives a conservative bound $\mathcal{B}(K_L \to \mu^+\mu^-)_{\text{short}} < 2.5 \times 10^{-9}$, which is the right order-of-magnitude target for any future catalog-to-code translation.

Post-2008 developments

No newer high-precision $K_L \to \mu^+\mu^-$ branching-fraction measurement was found beyond the PDG average used above. The main post-2008 movement is theoretical: D'Ambrosio-Kitahara identified direct-CP and K_L-K_S interference handles for $K \to \mu^+\mu^-$; Dery-Ghosh-Grossman-Schacht showed that time-dependent information can isolate a clean short-distance K_S observable with hadronic uncertainty below 1%; and Chao-Christ developed a lattice-QCD framework for the long-distance two-photon amplitude, with a quoted 10% target accuracy. These papers do not replace the PDG branching fraction, but they sharpen how this family of modes could become a cleaner short-distance probe after the CFW-era treatment.

Constraint validity and model dependence

As a direct rate constraint this is long-distance-limited. New physics with large scalar or pseudoscalar lepton operators would require separate treatment; the short-distance interpretation above is aimed at vector/axial semileptonic operators generated by Z-penguins and electroweak boxes. A production constraint should therefore use either a conservative dispersive allowance or a future lattice-assisted SM subtraction, not the total PDG branching fraction as a clean short-distance observable.

Code coverage in this repo

NO. A targeted grep across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`, excluding notebooks and binary-like assets, found no $K_L \to \mu^+\mu^-$, dimuon, or $s \to d\ell^+\ell^-$ implementation. The nearest kaon hit is `quarkConstraints/deltaf2` which defines the $K_L - K_S$ mass difference for $\Delta F = 2$ mixing, not this decay.

Implementation difficulty

HIGH. Implementing K006 would need a new rare-kaon semileptonic observable, short-distance $s \to d\mu^+\mu^-$ matching, and an explicit treatment of the long-distance two-photon dispersive amplitude or a conservative external bound. Those ingredients are not present in the audited $\Delta F = 2$ code path.

Key references

- `PDG2024:K006`: PDG K_L^0 listing, Γ_{24}.
- `Ambrose2000:E871KLMuMu`: primary branching-ratio measurement.
- `IsidoriUnterdorfer2004:KLMuMuSD`: short-distance bound.
- `DAmbrosioKitahara2017:KMuMuCP`: post-2008 interference/CP handle.
- `DeryGhoshGrossmanSchacht2021:KMuMuClean`: time-dependent clean probe proposal.
- `ChaoChrist2024:KLMuMuLattice`: lattice two-photon framework.

8 K008: $K_L \to \pi^0 e^+ e^-$

Process

Standard notation: $K_L \to \pi^0 e^+ e^-$. This rare semileptonic neutral-kaon mode is a $\Delta S = 1$ weak-neutral-current channel. Its Standard-Model rate is conventionally decomposed into indirect CP violation, direct CP violation, their interference, and a small CP-conserving two-photon piece (`isidori_smith_unterdorfer2004_kl_pi0ll_arxiv.txt`).

PDG / equivalent value(s)

The canonical PDG/API limit is

$$\mathcal{B}(K_L \to \pi^0 e^+ e^-) < 2.8 \times 10^{-10} \quad (90\% \text{ CL}),$$

from the 2026 PDG pdgLive entry S013.20, accessed 2026-05-16 (`pdg2026_kl_pi0ee_pdgLive.txt`). PDG traces the value to the KTeV combined result in PRL 93, 021805. The KTeV 1999–2000 sample observed one candidate with 0.99 ± 0.35 expected background events and set a standalone 3.5×10^{-10} limit at 90% CL; the combination with the 1997 data gives the PDG value (`ktev2004_kl_pi0ee_arxiv.txt`, `pdg2026_kl_pi0ee_pdgLive.txt`).

Relevance to the RS / anarchic-flavor pipeline

This decay probes $s \to d e^+ e^-$ electroweak-penguin and semileptonic contact structures rather than the $\Delta F = 2$ KK-gluon lane already present in the repository. In anarchic or warped models it is a diagnostic for flavor-changing Z couplings, heavy neutral gauge exchange, and dipole or photon-penguin effects. The KTeV theory summary quotes a Standard-Model rate of order 3×10^{-11}, a CP-violating range of 8–45 $\times 10^{-12}$, a direct-CP estimate of 3–6 $\times 10^{-12}$, and a 40% direct-CP interference contribution (`ktev2004_kl_pi0ee_arxiv.txt`). The present experimental limit is therefore above, but near, the scale where short-distance effects become interesting.

Post-2008 developments

No post-2008 experiment in the PKA snapshot supersedes the KTeV/PDG $K_L \to \pi^0 e^+ e^-$ limit. The important developments are interpretive. The PDG 2026 listing records the CP-conserving contribution inferred from $K_L \to \pi^0 \gamma\gamma$ as $(0.0047^{+0.0022}_{-0.0018}) \times 10^{-10}$ (`pdg2026_kl_pi0ee_pdgLive.txt`), supporting the standard view that the electron mode is CPV dominated. The supporting $K_S \to \pi^0 e^+ e^-$ input for the indirect-CP amplitude remains the NA48 value $(3.0^{+1.5}_{-1.2} \pm 0.2) \times 10^{-9}$ for $m_{ee} > 0.165\,\text{GeV}$, with an extrapolated full-region value $5.8^{+2.9}_{-2.4} \times 10^{-9}$ (`na48_ks_pi0ee_pdgLive.txt`). Since 2008, lattice work has begun addressing long-distance $K \to \pi \ell^+ \ell^-$ amplitudes (`ChristEtAl2016:KPiLLLatti` and the Snowmass rare-kaon white paper keeps $K_{L,S} \to \pi^0 \ell^+ \ell^-$ branching ratios and q^2 spectra in the short-distance rare-kaon program (`AebischerBurasKumar2022:RareKaons`).

Constraint validity and model dependence

Class: long-distance-limited, CP-decomposition-dependent, and not yet a clean single-Wilson-coefficient bound. A useful implementation must keep the indirect CPV piece tied to $K_S \to \pi^0 e^+ e^-$, the direct CPV short-distance piece, their sign-dependent interference, and the small two-photon CPC component separate. The Isidori–Smith–Unterdorfer decomposition gives $C^e_{\rm mix} = (15.7 \pm 0.3)|a_S|^2$, $C^e_{\rm int} = (6.2 \pm 0.3)|a_S|$, $C^e_{\rm dir} = 2.4 \pm 0.2$, $C^e_{\rm CPC} \simeq 0$, $|a_S| = 1.2 \pm 0.2$, and $({\rm Im}\lambda_t/10^{-4})_{\rm SM} = 1.36 \pm 0.12$, entering the rate in units of 10^{-12} (`isidori_smith_unterdorfer2004_kl_pi0ll_arxiv.txt`).

Code coverage in this repo

NO. Targeted PKA greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $K_L \to \pi^0 e^+ e^-$, $K_S \to \pi^0 e^+ e^-$, a_S, or $s \to d e^+ e^-$ implementation. Existing kaon coverage is $\Delta F = 2$ only, e.g. `quarkConstraints/delta` and `quarkConstraints/modern/phenomenology.py:23`.

Implementation difficulty

HIGH. K008 requires a new $\Delta S = 1$ semileptonic observable, short-distance matching to $s \to d e^+ e^-$, treatment of indirect/direct CP separation and interference, and long-distance inputs from K_S and $K_L \to \pi^0 \gamma\gamma$. The existing $\Delta F = 2$ operator basis is not sufficient.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:K008`, `KTeV2004:KLPi0EE`, `NA482003:KSPi0EE`, `IsidoriSmithUnterdorfer2004:KLPi0LL`, `CFW2008:RSFlavor`, `ChristEtAl2016:KPiLLLatt` and `AebischerBurasKumar2022:RareKaons`. Numeric claims above are tied to the sidecar-listed local snapshots.

9 K009: $K_L \to \pi^0 \mu^+ \mu^-$

Process

Standard notation: $K_L \to \pi^0 \mu^+ \mu^-$. This entry covers the long-lived neutral-kaon rare semileptonic muon channel, a $\Delta S = 1$ weak-neutral-current process receiving indirect-CP, direct-CP, interference, and CP-conserving two-photon contributions.

PDG or equivalent value

The canonical measured limit is

$$\mathcal{B}(K_L \to \pi^0 \mu^+ \mu^-) < 3.8 \times 10^{-10} \quad (90\%\ \text{CL}),$$

from the 2026 PDG pdgLive entry $S013.16/S013R30$, accessed 2026-05-16 (`pdg2026_kl_pi0mumu_pdgLive.txt`). PDG traces the active limit to the KTeV search in PRL 84, 5279. The KTeV snapshot records two observed events, 0.87 ± 0.15 expected background events, and the same 3.8×10^{-10} upper limit (`ktev2000_kl_pi0mumu_arxiv.txt`).

Relevance to RS / anarchic flavor

This mode probes $s \to d\mu^+\mu^-$ semileptonic and electroweak-penguin structures, not the $\Delta F = 2$ KK-gluon mixing lane currently implemented. In RS or anarchic-flavor variants it is sensitive to flavor-changing Z couplings, heavy neutral gauge exchange, and short-distance vector/axial operators. The muon mode's CP-conserving two-photon contribution is numerically important in the PKA theory decomposition, so a future signal would require separating CPV short-distance information from long-distance hadronic input.

Post-2008 developments

No post-2008 experimental result was found that supersedes the KTeV/PDG $K_L \to \pi^0\mu^+\mu^-$ limit. The supporting short-lived mode remains the NA48/PDG $K_S \to \pi^0\mu^+\mu^-$ value $(2.9^{+1.5}_{-1.2} \pm 0.2) \times 10^{-9}$, based on six observed events and $0.22^{+0.18}_{-0.11}$ expected background events (`na48_ks_pi0mumu_pdgLive.txt`, `na48_ks_pi0mumu_arxiv.txt`). Since arXiv:0804.1954, the main developments are interpretive: exploratory lattice work has started on long-distance $K \to \pi\ell^+\ell^-$ amplitudes (`christ_feng_juettner_lawson_porte`
and the Snowmass rare-kaon white paper keeps $K_{L,S} \to \pi^0\ell^+\ell^-$ branching ratios and spectra in the short-distance rare-kaon program (`aebischer_buras_kumar2022_rare_kaon_arxiv.txt`).

Constraint validity / model dependence

Class: long-distance-limited and CP-decomposition-dependent. The Isidori–Smith–Unterdorfer decomposition writes the rate as mixing, interference, direct, and CP-conserving pieces multiplying 10^{-12}. For the muon mode the coefficients are

$$C^{\mu}_{\text{mix}} = (3.7 \pm 0.1)|a_S|^2, \quad C^{\mu}_{\text{int}} = (1.6 \pm 0.1)|a_S|, \quad C^{\mu}_{\text{dir}} = 1.0 \pm 0.1, \quad C^{\mu}_{\text{CPC}} = 5.2 \pm 1.6,$$

with $|a_S| = 1.2 \pm 0.2$. The same paper quotes the constructive-interference SM expectation as $(1.5 \pm 0.3) \times 10^{-11}$, well below the current experimental limit. These numerical theory inputs are recorded in the YAML `theory_decomposition` block and trace to `isidori_smith_unterdorfer2004_kl_pi0mumu_arxiv.txt`

Code coverage in this repo

NO. Targeted greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $K_L \to \pi^0\mu^+\mu^-$, $K_S \to \pi^0\mu^+\mu^-$, a_S, or $s \to d\mu^+\mu^-$ implementation. Existing nearby kaon coverage is $\Delta F = 2$, e.g. `quarkConstraints/deltaf2.py:20`
and the modern policy list at `quarkConstraints/modern/phenomenology.py:23`.

Implementation difficulty

HIGH. A production constraint needs a new $\Delta S = 1$ semileptonic observable, short-distance $s \to d\mu^+\mu^-$ matching, CPV/interference bookkeeping tied to $K_S \to \pi^0\mu^+\mu^-$, and a long-distance two-photon/CPC treatment. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR basis is not sufficient.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:K009`, `KTeV2000:KLPi0MuMu`, `NA482004:KSPi0MuMu`, `IsidoriSmithUnterdorfer2004:KLPi0MuMu`, `CFW2008:RSFlavor`, `ChristEtAl2016:KPiLL`
and `AebischerBurasKumar2022:RareKaons`.

10 K010: $K_S \to \pi^0 e^+ e^-$

Process

Standard notation: $\mathcal{B}(K_S \to \pi^0 e^+ e^-)$. This entry covers the short-lived neutral-kaon rare semileptonic mode $K_S \to \pi^0 e^+ e^-$, a $\Delta S = 1$ weak-neutral-current decay. It is the electron-mode K_S companion to $K008$ because it fixes the indirect-CP input used in $K_L \to \pi^0 e^+ e^-$ phenomenology.

PDG or equivalent value

The canonical PDG 2026 pdgLive listing $S012.10$, accessed 2026-05-16, gives the NA48/1 measurement

$$\mathcal{B}(K_S \to \pi^0 e^+ e^-;\, m_{ee} > 0.165 \text{ GeV}) = \left(3.0^{+1.5}_{-1.2}\,\text{stat} \pm 0.2\,\text{syst}\right) \times 10^{-9}.$$

The same listing records the NA48/1 full-kinematic extrapolation, using a constant form factor and vector matrix element,

$$\mathcal{B}(K_S \to \pi^0 e^+ e^-) = \left(5.8^{+2.9}_{-2.4}\right) \times 10^{-9}.$$

Both values trace to `pdg2026_ks_pi0ee_pdgLive.txt` in the YAML `pdg_or_equivalent` block. The NA48/1 arXiv snapshot reports seven observed events with an expected background of 0.15 events (`na48_2003_ks_pi0ee_arxiv.txt`).

Relevance to RS / anarchic flavor

This mode probes $s \to de^+e^-$ semileptonic neutral-current structure, not the $\Delta F = 2$ kaon-mixing lane already implemented in the repository. In warped or anarchic-flavor models it is sensitive, through a hadronic long-distance amplitude, to flavor-changing Z, photon-penguin, and heavy neutral-gauge effects. Its most immediate catalog role is as an experimental normalization for the indirect-CP part of $K_L \to \pi^0 e^+ e^-$, so it should be kept distinct from short-distance theory prefactors or NDA normalizations.

Post-2008 developments

No post-2008 measurement was found that supersedes the NA48/1 value used by PDG. The important developments are interpretive: the rare-kaon Snowmass white paper emphasizes $K_{L,S} \to \pi^0\ell^+\ell^-$ branching ratios and q^2 distributions as new-physics probes, with correlations to ε'/ε, ε_K, and ΔM_K. The modern literature also treats this K_S input as part of the indirect-CP normalization for the K_L modes rather than as a standalone clean Wilson coefficient bound (`aebischer_buras_kumar_2022_arxiv.txt`, `isidori_smith_unterdorfer_2004_arxiv.txt`).

Constraint validity / model dependence

Class: long-distance-limited rare kaon decay. The measured partial branching fraction is robust, but the full-kinematic number is model-dependent because it assumes a vector matrix element and a unit form factor. A model constraint must specify whether it uses the directly measured $m_{ee} > 0.165$ GeV region or the extrapolated total rate, and whether it predicts the chiral long-distance amplitude or only a short-distance $s \to de^+e^-$ Wilson coefficient.

Code coverage in this repo

NO. Focused greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $K_S \to \pi^0e^+e^-$, $S012.10$, a_S, or $s \to de^+e^-$ implementation. The nearest kaon coverage is $\Delta F = 2$: `quarkConstraints/deltaf2.py:209` starts the default neutral-meson-mixing inputs, `quarkConstraints/deltaf2.py:755` evaluates Δm_K, and `quarkConstraints/modern/phenomenology.py:23` lists only ε_K, K, B_d, B_s, and $D0$ policy systems.

Implementation difficulty

HIGH. Production integration would need a new $\Delta S = 1$ semileptonic rare-kaon observable, Wilson normalization for $s \to de^+e^-$, and a documented treatment of the K_S form factor or a conservative experimental-input-only interpretation. The existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ basis does not cover this decay.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:K010`, `NA482003:KSPi0EE`, `IsidoriSmithUnterdorfer2004:KPi0LL`, `AebischerBurasKumar2022:RareKaons`, and `CFW2008:RSFlavor`.

11 $K_S \to \mu^+\mu^-$

Process

Standard notation: $K_S^0 \to \mu^+\mu^-$. This kaon entry covers the short-lived neutral-kaon dimuon decay, the K_S companion of $K_L \to \mu^+\mu^-$. It is a rare $\Delta S = 1$ weak-neutral-current mode, not a neutral-kaon mixing observable.

PDG / equivalent value(s)

The sidecar `pdg_or_equivalent` block, backed by the local `pdg2025_ks_decay_modes.txt` snapshot, records the 2025 PDG limit

$$\mathcal{B}(K_S^0 \to \mu^+\mu^-) < 2.1 \times 10^{-10} \quad (90\% \text{ CL}),$$

from the K_S^0 particle listing, accessed 2026-05-16. Additional sidecar `pdg_or_equivalent` entries from `lhcb_2020_ksmumu_arxiv.txt` show the LHCb standalone 2016–2018 bound, $\mathcal{B} < 2.2 \times 10^{-10}$ at 90% CL, and the combined $\mathcal{B} < 2.1 \times 10^{-10}$ at 90% CL. The Run-1 predecessor, stored in `lhcb_2017_ksmumu_arxiv.txt`, was $\mathcal{B} < 0.8\ (1.0) \times 10^{-9}$ at 90% (95%) CL.

Relevance to the RS / anarchic-flavor pipeline

In warped or anarchic-flavor scenarios this mode probes $s \to d\mu^+\mu^-$ neutral currents rather than the $\Delta F = 2$ kaon-mixing surface already implemented in the repo. Modified Z couplings, heavy neutral-gauge exchange, electroweak penguins, and scalar or pseudoscalar lepton operators can feed the dimuon final state. The sidecar `pdg_or_equivalent` entry from `chobanova_2018_ksmumu_susy_arxiv.txt` gives the Standard Model scale as approximately 5×10^{-12}, so the present limit mainly bounds large new-physics enhancements rather than precision SM-sized effects.

Post-2008 developments

The post-2008 change is mostly experimental: LHCb moved the bound from the Run-1 0.8×10^{-9} result to the combined 2.1×10^{-10} limit, with both values recorded in the sidecar. On the theory side, Chobanova et al. supplied the SM-scale and Higgs-penguin context, while Dery, Ghosh, Grossman, and Schacht showed that time-dependent $K \to \mu^+\mu^-$ interference can isolate the $\ell = 0$ K_S component with hadronic uncertainty below 1%; the latter number is stored in the sidecar entry for `dery_ghosh_grossman_schacht_2021_kmumu_arxiv.txt`.

Constraint validity and model dependence

The experimental upper limit is robust, but a model interpretation is operator-dependent. The total $K_S \to \mu^+\mu^-$ branching fraction includes long-distance physics and is not the same object as a clean short-distance $\ell = 0$ extraction. A catalog-to-code constraint should therefore distinguish vector/axial $s \to d\mu\mu$ contributions from scalar or pseudoscalar operators and should not treat the PDG limit as a pure $\Delta F = 2$ bound.

Code coverage in this repo

NO. Targeted greps across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $K_S \to \mu^+\mu^-$, dimuon rare-kaon, or $s \to d\ell^+\ell^-$ implementation. The nearest kaon hit is `quarkConstraints/deltaf2.py:615`, the $K_L - K_S$ mass difference used for $\Delta F = 2$ mixing, not this decay.

Implementation difficulty

HIGH. K012 needs a new rare-kaon $\Delta S = 1$ semileptonic calculation, Wilson/operator mapping for $s \to d\mu^+\mu^-$, and a documented treatment of long-distance and K_S/K_L interference effects. The existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ basis is not sufficient for this observable.

Key references

- `PDG2025:K012`: PDG K_S^0 particle listing, Γ_{12}, and headline limit.
- `LHCb2020:KSmumu`: standalone and combined LHCb upper limits.
- `LHCb2017:KSmumu`: Run-1 predecessor limit.
- `Chobanova2018:KSmumuSUSY`: SM scale and BSM Higgs-penguin context.
- `DeryGhoshGrossmanSchacht2021:KMuMuClean`: time-dependent clean-probe proposal.

12 K013: $K_L \to \pi^0\gamma\gamma$

Process

Standard notation: $K_L \to \pi^0\gamma\gamma$. This entry covers the long-lived neutral-kaon radiative decay listed by PDG as $\Gamma(\pi^0 2\gamma)/\Gamma_{\text{total}}$, process $K013$ in the kaon family. The orchestrator decision maps the PI's ambiguous $K1 \to \pi\gamma$ seed to this channel; the neighboring $K_S \to \pi^0\gamma\gamma$ entry remains separate.

PDG / equivalent value(s)

The canonical sidecar value is

$$\mathcal{B}(K_L \to \pi^0\gamma\gamma) = (1.273 \pm 0.033) \times 10^{-6},$$

from the PDG K_L^0 listing, 2025 update, accessed 2026-05-16; see `flavor_catalog/references/K013/pdg2025_kl_p` PDG forms this average from the KTeV final result and the NA48 measurement after rescaling both to the current $K_L \to \pi^0\pi^0$ normalization. The sidecar records the PDG-rescaled inputs, $1.28 \pm 0.06 \pm 0.01$ from KTeV and $1.27 \pm 0.04 \pm 0.01$ from NA48, both in units of 10^{-6}, plus the originally reported branching fractions and vector-exchange parameters a_V.

Relevance to the RS / anarchic-flavor pipeline

This is not a clean tree-level KK-gluon constraint like ε_K or Δm_K. It is a $\Delta S = 1$ radiative rare decay dominated by long-distance chiral dynamics and vector-meson-exchange effects. Its practical catalog role is diagnostic: it fixes the size and shape of the two-photon amplitude that feeds the CP-conserving part of $K_L \to \pi^0 \ell^+ \ell^-$, and it checks whether an RS extension with enhanced $s \to d\gamma\gamma$, dipole, or electroweak-penguin structure is already in tension with radiative kaon data.

Post-2008 developments

Relative to the Csaki–Falkowski–Weiler era, the KTeV full-sample analysis is the decisive update: it superseded the earlier KTeV value and is one of the two inputs in the current PDG average. The PKA did not find a newer neutral-mode branching-ratio measurement in the PDG 2025 listing. The main later experimental development is adjacent rather than identical: NA48/2 measured the charged analogue $K^\pm \to \pi^\pm \gamma\gamma$, with the sidecar recording the model-independent $z > 0.2$ and ChPT-extrapolated branching fractions. Cappiello, Cata, and D'Ambrosio 2018 show that radiative-kaon data constrain weak chiral counterterms, which is the right language for this mode rather than a short-distance Wilson-only bound.

Constraint validity and model dependence

Class: long-distance-limited and model-dependent. The measured branching fraction should not be imposed as a hard RS exclusion without a dedicated $\Delta S = 1$ radiative calculation and a prescription for the SM long-distance amplitude. It is safer as a consistency target and as an auxiliary input to $K_L \to \pi^0 \ell^+ \ell^-$ estimates. A new-physics interpretation must avoid double counting two-photon contributions already absorbed into chiral counterterms.

Code coverage in this repo

NO. The PKA reran the plan-required greps and a K013-specific search across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The repository contains kaon $\Delta F = 2$ machinery and unrelated generic γ-symbol occurrences, but no implementation of $K_L \to \pi^0 \gamma\gamma$, no a_V parameter, and no radiative-kaon likelihood.

Implementation difficulty

HIGH. A production constraint would need new $\Delta S = 1$ radiative matching, chiral-amplitude inputs, long-distance SM treatment, and possibly differential-shape information. The existing $\Delta F = 2$ operator basis and running code are not enough.

Key references

Process-local source keys before bibliography consolidation: `PDG2025_KL_pi0gammagamma`, `KTeV2008_KL_pi0gammag`
`NA482002_KL_pi0gammagamma`, `NA482014_Kpm_pigammagamma`, `CappielloCataDAmbrosio2018_radiative_chiral`,

and GabbianiValencia2001_KL_pi0gammagamma.

13 K017: $K^+ \to \ell^+\nu$ and R_K

Process

Standard notation: $R_K = \Gamma(K^+ \to e^+\nu_e(\gamma))/\Gamma(K^+ \to \mu^+\nu_\mu(\gamma))$. This entry covers the charged-kaon leptonic lepton-flavor-universality ratio using the PDG/NA62 convention that includes internal bremsstrahlung and separates direct-emission radiation.

PDG / equivalent value(s)

The PDG API average is

$$R_K = (2.488 \pm 0.009) \times 10^{-5},$$

from PDG API node S010R28, accessed on 2026-05-16 and snapshotted at flavor_catalog/references/K017/pdg2 (PDG2025:K017RK). The same snapshot records the main inputs: NA62 finds $(2.488 \pm 0.007_{\rm stat} \pm 0.007_{\rm syst}) \times 10^{-5}$ (NA62:RK2013), and KLOE finds $(2.493 \pm 0.025_{\rm stat} \pm 0.019_{\rm syst}) \times 10^{-5}$ (KLOE:RK2009). For the SM comparison, Cirigliano–Rosell give

$$R_K^{\rm SM} = (2.477 \pm 0.001) \times 10^{-5},$$

in the local snapshot flavor_catalog/references/K017/cirigliano_rosell_2007_arxiv0707_4464.txt (CiriglianoRosell:RKSM2007).

Relevance to the RS / anarchic-flavor pipeline

This is not a neutral-meson $\Delta F = 2$ observable and does not probe the existing quark scan's KK-gluon mixing lane directly. Its value is that the ratio cancels the common V_{us}, f_K, and much of the kaon-rate normalization, leaving a helicity-suppressed charged-current LFU test (FlaviaNet:Kaon2010). Nonminimal warped models with lepton localization, heavy neutral leptons, nonunitary charged currents, charged scalars, or nonuniversal W-coupling shifts can modify K_{e2} relative to $K_{\mu 2}$. In a quark-only RS interpretation it should therefore be cataloged as a lepton-extension or charged-current diagnostic, not as a hard bound on the current anarchic quark fit.

Post-2008 developments

The 2008 CFW baseline predates the precision R_K consolidation and did not make this charged-current LFU ratio a central RS constraint (CsakiFalkowskiWeiler:RSFlavor2008). Since then, KLOE published a precise inclusive measurement in 2009 (KLOE:RK2009), the FlaviaNet kaon working group consolidated post-2008 kaon-decay inputs in 2010 (FlaviaNet:Kaon2010), and NA62 superseded earlier charged-kaon data with its 2007–2008 data set in 2013 (NA62:RK2013). The modern PDG average is driven by those post-2008 measurements. The theory reference value was already clean before 2008, but the experimental precision now makes R_K a useful null test for scalar or right-handed charged-current effects that evade many hadronic uncertainties.

Constraint validity and model dependence

As a measured ratio, R_K is robust and experimentally clean, provided the radiative convention is kept fixed: the PDG/NA62 ratio includes internal bremsstrahlung, while direct-emission radiation is separated. Its mapping onto this repository's default model is model-dependent. A minimal quark flavor scan with no nonstandard charged-current lepton sector has no direct parameter controlling this observable, whereas a 5D lepton extension or charged-scalar completion could generate visible effects through chirality- enhanced scalar amplitudes or lepton-nonuniversal W couplings.

Code coverage in this repo

NO. The required coverage greps across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` find no $K^+ \to e^+\nu$, $K^+ \to \mu^+\nu$, $K_{\ell 2}$, or R_K implementation. The nearest surfaces are the $\Delta F = 2$ default inputs in `quarkConstraints/deltaf2.py:209` and the modern policy fence `quarkConstraints/modern/phenomenology.py:23`, which enumerates only ε_K, K, B_d, B_s, and $D0$.

Implementation difficulty

MEDIUM. A catalog-to-code implementation would need a new charged-current leptonic-decay observable, including the SM amplitude, radiative-convention bookkeeping, and whatever BSM operator basis the PI wants to expose. The ratio avoids new lattice matrix elements, but it is not covered by the existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR machinery.

Key references

Process-local reference keys before bibliography consolidation: PDG2025:K017RK, NA62:RK2013, KLOE:RK2009, CiriglianoRosell:RKSM2007, FlaviaNet:Kaon2010, and CsakiFalkowskiWeiler:RSFlavor2008.

14 K018: $K_{\ell 3}$ and semileptonic V_{us}

Process

Standard notation: $K_{\ell 3}$, with $K \to \pi e \nu$ and $K \to \pi \mu \nu$. This entry records the kaon semileptonic source input $|V_{us}|f_+(0)$ and its mode-by-mode consistency tests. The first-row CKM unitarity combination is cataloged separately in EW002.

PDG / equivalent value(s)

The sidecar `pdg_or_equivalent` entry `PDG2025:K018:average_fplus_vus`, backed by `pdg2025_vud_vus_kl3.txt`, records

$$|V_{us}|f_+(0) = 0.21656(35),$$

from the 2025 PDG update, accessed 2026-05-16. The same sidecar block records the PDG Table 67.1 products: 0.2169(6) for $K^{\pm}e3$, 0.2168(10) for $K^{\pm}\mu3$, 0.2162(5) for K_Le3, 0.2165(6) for $K_L\mu3$, 0.2169(8) for K_Se3, and 0.2125(47) for $K_S\mu3$. Additional `pdg_or_equivalent` entries record the FLAG $N_f = 2+1+1$ normalization $f_+(0) = 0.9698(17)$ and the PDG-derived $|V_{us}| = 0.22330(53)$; these are normalization inputs, not separate measured $K_{\ell 3}$ rates.

Relevance to the RS / anarchic-flavor pipeline

This is a tree-level charged-current process, not a neutral-meson $\Delta F = 2$ observable. In the current quark scan it is relevant mainly as the semileptonic source behind the fitted $|V_{us}|$ target and the EW002 first-row unitarity test. Nonminimal warped models with modified W-quark couplings, vector-like fermion mixing, lepton nonuniversality, or right-handed charged currents could shift the extracted V_{us} or the $K_{\mu 3}/K_{e3}$ consistency test, but a minimal KK-gluon flavor analysis does not map this measurement into the existing operator basis.

Post-2008 developments

The 2008 RS-flavor baseline `CsakiFalkowskiWeiler:RSFlavor2008` centered on flavor-changing neutral constraints, so the post-2008 K018 story is precision charged-current consolidation. The sidecar entries `FlaviaNet2010:K018:average_fplus_vus` and `FlaviaNet2010:K018:r_mue` record the 2010 FlaviaNet average $|V_{us}|f_+(0) = 0.2163(5)$ and $K_{\mu 3}/K_{e3}$ universality ratio $r_{\mu e} = 1.002(5)$. The 2025 PDG update keeps the same source-level convention while tightening the average to 0.21656(35). FLAG 2024 supplies the auxiliary lattice normalization, $f_+(0) = 0.9698(17)$, and the derived $|V_{us}| = 0.22328(58)$, which feed the CKM-unitarity entry rather than replacing the measured K018 product.

Constraint validity and model dependence

The measured $|V_{us}|f_+(0)$ product is experimentally robust but depends on consistent radiative, isospin, and phase-space conventions. The derived $|V_{us}|$ additionally depends on the lattice $f_+(0)$ average. As an RS constraint, K018 is model-dependent: it is a charged-current CKM and lepton-universality diagnostic, not a standalone FCNC bound unless the model specifies nonstandard W, scalar, tensor, or right-handed-current effects.

Code coverage in this repo

PARTIAL. The repo has a generic CKM target containing $|V_{us}|$: `quarkConstraints/scan.py:43` documents the PDG-derived CKM averages, `quarkConstraints/scan.py:247` labels the reduced CKM observable as `Vus`, and `tests/test_quark_fit.py:587` anchors the target against PDG 2024. The coverage grep found no $K_{\ell 3}$ decay-rate, $|V_{us}|f_+(0)$, $f_+(0)$, K_{e3}, or $K_{\mu 3}$ implementation. The modern flavor surface is still limited to ε_K, K, B_d, B_s, and $D0$ systems at `quarkConstraints/modern/phenomenolo`

Implementation difficulty

HIGH. A live K018 constraint would need a charged-current semileptonic likelihood or EFT wrapper, radiative/isospin convention bookkeeping, lattice $f_+(0)$ provenance, and a decision about whether it is implemented as a direct V_{us} target, an EW002 sub-input, or a BSM charged-current operator constraint. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR machinery does not cover it.

Key references

- `PDG2025:K018Kl3`: PDG 2025 $K_{\ell 3}$ product average and mode values.
- `FLAG2024:K018KaonSemileptonic`: lattice $f_+(0)$ normalization and derived $|V_{us}|$.
- `FlaviaNet:K018Kaon2010`: post-2008 $K_{\ell 3}$ consolidation and $r_{\mu e}$.
- `CsakiFalkowskiWeiler:RSFlavor2008`: 2008 warped-flavor context.

15 Charm

16 C001: $D^0 - \bar{D}^0$ mixing

Process

Standard notation: neutral-charm mixing, parameterized by $x_D = \Delta m_D/\Gamma_D$, $y_D = \Delta\Gamma_D/(2\Gamma_D)$, and $\Delta m_D = |m_1 - m_2|$. This entry covers the CP-even mixing magnitudes used for RS-flavor constraints; the dedicated CP-violation observables $|q/p|$ and ϕ_D belong to C002.

PDG / equivalent value(s)

The headline values are the HFLAV CKM25 all-CPV global-fit results, version through 14 September 2025 (`hflav_ckm25_dmixing_global_fit`):

$$x_D = (0.405 \pm 0.043)\%, \qquad y_D = (0.636 \pm 0.024)\%.$$

The 95% C.L. intervals are $x_D \in [0.320, 0.489]\%$ and $y_D \in [0.590, 0.682]\%$. These values are traced to `flavor_catalog/references/C001/hflav_ckm25_dmixing_global_fit.txt` and to the PDG 2025 Table 70.7 excerpt. PDG Live separately gives the absolute mass difference (`pdg2025_deltam_d_pdgLive`),

$$\Delta m_D = (0.997 \pm 0.116) \times 10^{10} \hbar\, \mathrm{s}^{-1} = (6.56 \pm 0.76) \times 10^{-15}\,\mathrm{GeV},$$

from `flavor_catalog/references/C001/pdg2025_deltam_d_pdgLive.txt`. For the orchestrator-requested δy block, the sidecar records the latest HFLAV/LHCb CPV-sensitive input $\Delta y = (0.031 \pm 0.035 \pm 0.013)\%$ (`hflav_ckm25_delta_y_input`); it also records the derived $\Delta\Gamma_D/\Gamma_D = 2y_D = (1.272 \pm 0.048)\%$ (`hflav_ckm25_dmixing_global_fit`).

Relevance to the RS / anarchic-flavor pipeline

In anarchic warped models, KK-gluon and other heavy neutral exchanges generate $\Delta C = 2$ four-quark operators connecting u and c. The same operator-basis logic used for K, B_d, and B_s mixing applies, but the charm system is up-sector and long-distance dominated. It is therefore a useful complementary test of whether the scan's flavor alignment suppresses both down- and up-sector neutral currents.

Post-2008 developments

The CFW 2008 baseline (`cfw2008_arxiv0804_1954`) already treated warped anarchic flavor as strongly constrained, with generic RS KK-gluon scales around 21 TeV and composite pseudo-Goldstone scenarios around 33 TeV. Since then, the charm-mixing experimental picture has become much sharper. LHCb's 2021 $D^0 \to K_S^0\pi^+\pi^-$ analysis (`lhcb2021_arxiv2106_03744`) observed a nonzero neutral-charm mass difference, and the current HFLAV/PDG global fits decisively exclude the no-mixing point. The same HFLAV fit remains consistent with no indirect CP violation, so the CP-even magnitudes are the clean C001 inputs.

Constraint validity and model dependence

As a new-physics constraint this is robust in the sense that any short-distance $\Delta C = 2$ amplitude contributes to M_{12}^D. It is also SM-theory-limited: the observed mixing is believed to be dominated by long-distance Standard Model amplitudes that are hard to compute. The safe catalog interpretation is therefore conservative, bounding the magnitude of a new-physics contribution rather than subtracting a precise SM prediction.

In down-aligned or kaon-protected RS variants, up-sector observables become leading rather than secondary diagnostics.

Code coverage in this repo

YES. The legacy lane has a default D-mixing input at `quarkConstraints/deltaf2.py:252`, with reject reason `d_mix` at `quarkConstraints/deltaf2.py:256`, D0 hadronic inputs starting at `quarkConstraints/deltaf2.py:655`, and `DELTA_M_D_EXP` at `quarkConstraints/deltaf2.py:665`. The numeric evaluator is `evaluate_d0_mixing` at `quarkConstraints/deltaf2.py:941`. The scan writes `d_mix_ratio` at `quarkConstraints/scan.py:103` and `quarkConstraints/scan.py:439`. The modern policy surface also lists D^0 at `quarkConstraints/modern/phenomenology.py:23` and evaluates the bridge result at `quarkConstraints/modern/phenomenology.py:1356`.

Implementation difficulty

LOW. The existing $\Delta F = 2$ machinery already contains the D0 evaluator and the required VLL/VRR/LR operator structure. A future production update would mainly refresh the experimental value and document the conservative long-distance treatment, not add a new operator basis.

Key references

Process-local source keys before bibliography consolidation: `hflav_ckm25_dmixing_global_fit`, `pdg2025_deltam_d_pdgLive`, `pdg2025_dmixing_review_table70_7`, `lhcb2021_arxiv2106_03744`, and `cfw2008_arxiv0804_1954`.

17 C002: CP violation in $D^0 - \bar{D}^0$ mixing

Process

Standard notation: $|q/p|$ and $\phi_D = \arg(q/p)$ in neutral-charm mixing. This entry covers indirect CP violation in $D^0 - \bar{D}^0$ oscillations; the CP-even mixing magnitudes x_D, y_D, and Δm_D are cataloged separately in C001.

PDG / equivalent value(s)

The canonical values are the HFLAV CKM25 all-CPV global-fit results, version through 14 September 2025, accessed on 2026-05-16:

$$|q/p| = 0.983^{+0.015}_{-0.014}, \qquad \phi_D = -1.51^{+1.03}_{-1.06} \text{ deg.}$$

The corresponding 95% C.L. intervals are $|q/p| \in [0.955, 1.012]$ and $\phi_D \in [-3.63, 0.51]$ degrees. HFLAV also reports that the no-indirect-CPV point $(|q/p|, \phi) = (1, 0)$ has $\Delta\chi^2 = 2.16$, consistent with CP conservation at 0.95σ. These numbers are traced to `flavor_catalog/references/C002/hflav_ckm25_dmi` the PDG 2025 review excerpt in `flavor_catalog/references/C002/pdg2025_dmixing_review_cpv.txt` cross-checks the same central values.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models generate short-distance $\Delta C = 2$ amplitudes through flavor-changing neutral-current exchange. C002 is the CP-odd counterpart of the D-mixing magnitude bound: it tests whether the complex phase of a new $c-u$ mixing amplitude is aligned well enough to avoid visible indirect CP violation. It is especially useful because the repo's current D0 lane constrains $|M_{12}^{D,\mathrm{NP}}|$, while C002 asks about the phase-sensitive part of the same amplitude.

Post-2008 developments

The baseline RS-flavor paper arXiv:0804.1954 predates the modern high-statistics neutral-charm mixing program. Since then, LHCb added CPV-sensitive model-independent $D^0 \to K^0_S\pi^+\pi^-$ inputs, including the post-Run-2 result summarized in `lhcb2023_arxiv2208_06512.txt`. Belle and Belle II have also supplied new large-sample model-independent mixing information (`belle_belleii2025_arxiv241` HFLAV now combines these inputs in a global fit with explicit $|q/p|$ and ϕ_D parameters, and the current result remains compatible with no indirect CP violation.

Constraint validity and model dependence

This is a short-distance-sensitive but global-fit-dependent constraint. The experimental fit is precise enough to bound new CP phases, but the Standard Model charm-mixing amplitude is long-distance dominated and difficult to calculate. A safe RS interpretation should therefore constrain the allowed new-physics phase conservatively, without subtracting a precise SM prediction.

In down-aligned or kaon-protected RS variants, up-sector observables become leading rather than secondary diagnostics.

Code coverage in this repo

PARTIAL. The legacy scan defines a D-mixing input at `quarkConstraints/deltaf2.py:252` and evaluates D0 mixing through `evaluate_d0_mixing` at `quarkConstraints/deltaf2.py:941`, but that result uses $|M_{12}^{\mathrm{NP}}|$ and returns only a magnitude ratio. The scan exports `d_mix_ratio` at `quarkConstraints/scan.py:103` and `quarkConstraints/scan.py:439`. The modern policy labels D0 as `conservative_non_cp_mixing_amplitude_hadronic` at `quarkConstraints/modern/phenomenology.py` A paper-mode helper does retain complex M_{12}^{D0} and exposes `im_M12_D0_NP_GeV` at `quarkConstraints/paper_0710_` but there is no exported $|q/p|$ or ϕ_D evaluator.

Implementation difficulty

LOW. The existing $\Delta F = 2$ operator and RG machinery is the right basis for C002, so no new operator class is required. The missing step is a CPV observable wrapper or likelihood policy that maps the complex D0 mixing amplitude onto a conservative HFLAV $|q/p|, \phi_D$ constraint.

Key references

Process-local source keys before bibliography consolidation: `hflav_ckm25_dmixing_cpv_fit`, `pdg2025_dmixing_rev` `lhcb2023_arxiv2208_06512`, `belle_belleii2025_arxiv2410_22961`, and `cfw2008_arxiv0804_1954`.

18 C003: $\Delta A_{CP}(D^0 \to K^+K^-, \pi^+\pi^-)$

Process

Standard notation: $\Delta A_{CP} \equiv A_{CP}(D^0 \to K^+K^-) - A_{CP}(D^0 \to \pi^+\pi^-)$. This entry covers time-integrated direct CP violation in singly Cabibbo-suppressed neutral-charm decays; indirect CP violation in charm mixing is cataloged separately.

PDG / equivalent value(s)

The canonical equivalent source is HFLAV. The direct-vs-indirect CPV combination updated in 2025 gives

$$\Delta a_{CP}^{\text{dir}} = (-0.159 \pm 0.029)\%,$$

with $a_{CP}^{\text{ind}} = (-0.010 \pm 0.012)\%$ and a no-CPV discrepancy quoted as 5.3σ. The local snapshot is `flavor_catalog/references/C003/hflav2025_direct_indirect_cpv.txt`. The LHCb 2019 discovery anchor gives

$$\Delta A_{CP} = (-15.4 \pm 2.9) \times 10^{-4} = (-0.154 \pm 0.029)\%,$$

as recorded in `lhcb2019_arxiv1903_08726.txt`. HFLAV CKM25 also lists the same measured input $A_{CP}^{K} - A_{CP}^{\pi} = (-0.154 \pm 0.029)\%$ and the all-CPV fit parameters $A_\pi = (0.225 \pm 0.057)\%$ and $A_K = (0.068 \pm 0.051)\%$, traced to `hflav_ckm25_dcpv_inputs.txt`.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped flavor generically carries new CP phases in both up- and down-sector flavor transitions. Unlike neutral-charm mixing, this is not a $\Delta C = 2$ mixing constraint: it probes $\Delta C = 1$ nonleptonic amplitudes, including possible chromomagnetic dipoles and four-quark penguin structures. It is therefore a diagnostic for BSM CP phases in $c \to u$ transitions that can be hidden by a neutral-meson-mixing-only scan.

Post-2008 developments

The CFW 2008 RS-flavor baseline predates the charm direct-CP observation. The important post-2008 change is experimental: LHCb reported the first observation of CP violation in charm decays in 2019, and HFLAV now provides a compact direct-vs-indirect CPV combination with percent-level fit parameters. The 2025 CKM25 charm fit additionally folds the ΔA_{CP} input into a global all-CPV fit together with A_K and A_π.

Constraint validity and model dependence

The experimental observable is robust because the difference cancels many production and detection asymmetries. Interpreting it as a bound on a specific new-physics Wilson coefficient is much less clean: Standard Model long-distance charm amplitudes are difficult to compute, and the HFLAV extraction separates direct and indirect CPV using small mixing corrections. Treat this as a model-dependent $\Delta C = 1$ CP-phase constraint, not as a clean short-distance subtraction.

In down-aligned or kaon-protected RS variants, up-sector observables become leading rather than secondary diagnostics.

Code coverage in this repo

NO. The required grep finds neutral D^0-mixing support but no direct charm CP-asymmetry implementation. The current implemented charm surface is `evaluate_d0_mixing` at `quarkConstraints/deltaf2.py:94` the modern policy labels D^0 as conservative non-CP mixing at `quarkConstraints/modern/phenomenology.py:46`. Focused searches for `DeltaA_CP`, `A_CP`, direct CP, and $D^0 \to K^+K^-, \pi^+\pi^-$ across the required implementation directories found no C003 observable.

Implementation difficulty

HIGH. Production integration would require a new $\Delta C = 1$ operator/matching layer, RG treatment for the relevant four-quark and dipole operators, and a hadronic-amplitude or likelihood prescription for the nonleptonic charm modes. The existing $\Delta F = 2$ basis does not cover it.

Key references

Process-local source keys before bibliography consolidation: `hflav2025_direct_indirect_cpv`, `hflav_ckm25_dcpv_inputs`, `lhcb2019_arxiv1903_08726`, and `cfw2008_arxiv0804_1954`.

19 C004: $D^0 \to \mu^+\mu^-$

Process

Standard notation: $\mathcal{B}(D^0 \to \mu^+\mu^-)$. This entry covers the rare neutral-charm leptonic decay $D^0 \to \mu^+\mu^-$, a $\Delta C = 1$, $c \to u\ell^+\ell^-$ flavor-changing neutral-current test. It is distinct from C001/C002 neutral-charm mixing and from semileptonic rare-charm modes because the final state is purely leptonic and directly probes axial-vector, scalar, and pseudoscalar short-distance operators.

PDG / equivalent value(s)

The current PDG Live/API entry S032.28, accessed on 2026-05-16, gives the headline limit

$$\mathcal{B}(D^0 \to \mu^+\mu^-) < 2.1 \times 10^{-9} \qquad (90\% \text{ C.L.}).$$

The same PDG entry notes that the underlying CMS 2025 analysis reports the 95% C.L. limit

$$\mathcal{B}(D^0 \to \mu^+\mu^-) < 2.4 \times 10^{-9}.$$

These values are traced to `flavor_catalog/references/C004/pdg2026_d0_mumu_pdgLive.txt` and `cms2025_d0_mumu_public.txt`. The requested LHCb predecessor used $9\,\text{fb}^{-1}$ at 7, 8, and 13 TeV and set $\mathcal{B}(D^0 \to \mu^+\mu^-) < 3.1 \times 10^{-9}$ at 90% C.L., with a 95% C.L. companion limit of 3.5×10^{-9} (`lhcb2023_arxiv2212_11203`).

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models can generate flavor-changing $c \to u$ neutral-current couplings through KK gauge exchange, electroweak Z-like effects, Higgs or radion scalar exchange, or lepton-sector extensions. This channel is therefore a useful up-sector $\Delta C = 1$ complement to the implemented $\Delta F = 2$ D-mixing constraint. For the present quark-only scan it is best treated as a catalog and roadmap entry; a live constraint would require lepton-current normalization in addition to the quark flavor spurions.

Post-2008 developments

Since the arXiv:0804.1954 warped-flavor baseline, the experimental bound has moved by orders of magnitude. LHCb's full Run 1+2 search improved the rare charm dimuon limit to 3.1×10^{-9} at 90% C.L. CMS then used the Run 3 inclusive dimuon-trigger program, $64.5\,\mathrm{fb}^{-1}$ from 2022–2023, to set the stronger 95% C.L. limit of 2.4×10^{-9}. On the theory side, the 2024 rare-charm EFT analysis incorporated recent $\Delta C = 1$ data, including $D^0 \to \mu^+\mu^-$, into constraints on $C^{(\prime)}_{7,9,10}$ while emphasizing that null-test observables still leave room for new physics.

Constraint validity and model dependence

The experimental limit is robust, but the Standard Model prediction is not a clean short-distance number. The classic rare-charm calculation estimates the long-distance two-photon contribution as $\mathcal{B}(D^0 \to \mu^+\mu^-)_{\gamma\gamma} \simeq 2.7 \times 10^{-5} \mathcal{B}(D^0 \to \gamma\gamma)$, implying a long-distance floor of at least 3×10^{-13} for the assumptions quoted there. The safe catalog use is therefore an upper bound on extra short-distance contributions, not a precision SM subtraction.

In down-aligned or kaon-protected RS variants, up-sector observables become leading rather than secondary diagnostics.

Code coverage in this repo

NO. The required catalog greps and a focused search for `D0.*mu`, `c.?->.?u`, `rare.?charm`, `C10`, `C_S`, `C_P`, and scalar/pseudoscalar rare-decay terms found no live $D^0 \to \mu^+\mu^-$ implementation in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. Nearby hits are unrelated: D^0 mixing appears at `quarkConstraints/deltaf2.py:256` and `quarkConstraints/deltaf2.py:941`, while $\mu \to e\gamma$ is handled separately at `flavorConstraints/muToEGamma`

Implementation difficulty

HIGH. $D^0 \to \mu^+\mu^-$ is a new $c \to u$ dilepton mode requiring a new $\Delta C = 1$ Wilson-coefficient normalization and a documented long-distance subtraction; it is not covered by the existing $\Delta F = 2$ backend.

Key references

Process-local keys before bibliography consolidation: `PDG2026:D0MuMu`, `CMS2025:D0MuMu`, `LHCb2023:D0MuMu`, `BurdmanGolowichHewettPakvasa2002:RareCharm`, `GisbertHillerSuelmann2024:RareCharmEFT`, and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

20 C005: $D^0 \to e^+e^-$

Process

Standard notation: $\mathcal{B}(D^0 \to e^+e^-)$. This entry covers the rare neutral-charm leptonic decay $D^0 \to e^+e^-$, a $\Delta C = 1$, $c \to ue^+e^-$ flavor-changing neutral-current probe. Charge-conjugate modes are implicit in the experimental listings. It is the electron-mode counterpart to C004, $D^0 \to \mu^+\mu^-$, but has a separate experimental record.

PDG / equivalent value(s)

The current PDG Live/API entry $S032.39$, accessed on 2026-05-16, gives

$$\mathcal{B}(D^0 \to e^+e^-) < 7.9 \times 10^{-8} \qquad (90\%\ \text{C.L.}).$$

The sidecar source key is `PDG2026:D0Ee`, with local snapshot `pdg2026_d0_ee_pdgLive.txt`. The PDG-used input is the Belle 2010 search with $660\,\text{fb}^{-1}$ at KEKB. Belle reports the same 7.9×10^{-8} upper limit and no signal evidence (`Belle2010:D0Ee`; `belle2010_arxiv1003_2345.txt`). The independent BaBar 2012 search used $468\,\text{fb}^{-1}$ and set the weaker 1.7×10^{-7} upper limit at 90% CL (`BaBar2012:D0EeMuMuEMu`; `babar2012_arxiv1206_5419.txt`). The dispatch note called this an LHCb upper limit, but the accessed PDG/API measurement list contains no LHCb $D^0 \to e^+e^-$ entry (`pdg2026_d0_ee_pdgLive.txt`).

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models can induce up-sector flavor-changing neutral currents through nonuniversal KK gauge couplings, flavor-violating Z-like effects, Higgs/radion scalar exchange, or lepton-sector extensions (`CsakiFalkowskiWeiler2008:CompositeFlavor`). C005 therefore tests a pure $c \to u$ dilepton amplitude rather than the four-quark $\Delta F = 2$ operator basis used for neutral D-mixing. For the current quark-only scan it is a catalog and roadmap process; a live veto would need both the $c \to u$ current and electron-current normalization.

Post-2008 developments

The 2008 RS-flavor baseline predates the Belle and BaBar leptonic D^0 searches (`CsakiFalkowskiWeiler2008:Comp` The main post-2008 experimental development is Belle's 2010 improvement to the current PDG limit, followed by BaBar's 2012 cross-check in a separate $e^+e^- \to c\bar{c}$ sample (`Belle2010:D0Ee`, `BaBar2012:D0EeMuMuEMu`). Unlike C004, which has newer hadron-collider dimuon searches, the accessed C005 PDG record still uses the Belle result as the headline input. Rare-charm theory work emphasizes that purely leptonic D^0 limits are clean null tests experimentally but require care when separating short-distance new physics from long-distance charm dynamics (`BurdmanGolowichHewettPakvasa2002:`

Constraint validity and model dependence

The upper limit itself is experimentally robust. Its interpretation as an RS constraint is model dependent because the electron mode is helicity sensitive, bremsstrahlung and electron identification enter the experimental acceptance, and the Standard Model rare-charm amplitude is not a precision short-distance prediction. A conservative use is an upper bound on additional short-distance $c \to ue^+e^-$ amplitudes in a specified operator basis, not a direct subtraction against a sharply predicted Standard Model rate (`BurdmanGolowichHewettPakvasa2002:RareCharm`).

Code coverage in this repo

NO. The PKA greps and a focused search for `D0.*e+e-`, `D0.*ee`, `c.?->.?u.*e`, and `rare.?charm` found no $D^0 \to e^+e^-$ implementation in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. Nearby hits are not C005: D^0 mixing is configured at `quarkConstraints/deltaf2.py:252` and evaluated at `quarkConstraints/deltaf2.py:941`; the modern policy likewise lists only D^0 mixing at `quarkConstraints/modern/phenomenology.py:23` and `quarkConstraints/modern/phenomenology.py:668`. The $\mu \to e\gamma$ dipole check at `flavorConstraints/muToE` is a separate LFV observable.

Implementation difficulty

HIGH. Production use would require a new $\Delta C = 1$ rare leptonic module with vector, axial-vector, scalar, and pseudoscalar normalizations, electron-mode acceptance conventions, and a documented long-distance treatment. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR basis is not enough.

Key references

Process-local keys before bibliography consolidation: `PDG2026:D0Ee`, `Belle2010:D0Ee`, `BaBar2012:D0EeMuMuEMu`, `BurdmanGolowichHewettPakvasa2002:RareCharm`, and `CsakiFalkowskiWeiler2008:CompositeFlavor`. Numeric claims above are tied to these sidecar source keys and local snapshots.

21 C006: $D^0 \to e^\pm\mu^\mp$

Process

Standard notation: $\mathcal{B}(D^0 \to e^\pm\mu^\mp)$. This entry covers the charged-lepton-flavor-violating neutral-charm decay $D^0 \to e^\pm\mu^\mp$. It is a $\Delta C = 1$, $c \to ue\mu$ null test with charge-conjugate final states included in the experimental limit. In flavor structure it is an up-sector analogue of semileptonic LFV probes such as $B^+ \to K^+e^\pm\mu^\mp$, but with a purely leptonic two-body D^0 final state.

PDG / equivalent value(s)

The sidecar source `PDG2026:D0EMu`, PDG Live/API listing S032.40 accessed on 2026-05-16, gives

$$\mathcal{B}(D^0 \to e^\pm\mu^\mp) < 1.3 \times 10^{-8} \qquad (90\% \text{ C.L.}).$$

The PDG-used input is the 2016 LHCb search `LHCb2016:D0EMu`, based on $3.0\,\mathrm{fb}^{-1}$ of pp data at $\sqrt{s} = 7$ and $8\,\mathrm{TeV}$ and normalized with $D^0 \to K^-\pi^+$. The same LHCb source reports no significant excess. Earlier post-2008 B-factory limits were weaker: `Belle2010:D0EMu` reported $\mathcal{B}(D^0 \to e^+\mu^-) + \mathcal{B}(D^0 \to \mu^+e^-) < 2.6 \times 10^{-7}$, and `BaBar2012:D0EeMuMuEMu` reported $\mathcal{B}(D^0 \to e\mu) < 3.3 \times 10^{-7}$, both at 90% C.L. The local snapshots are `flavor_catalog/references/C006/pdg2026_d0_emu_pdgLive.txt` `lhcb2016_arxiv1512_00322.txt`, `belle2010_arxiv1003_2345.txt`, and `babar2012_arxiv1206_5419.txt`.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped flavor already generates nonuniversal c- and u-quark couplings through localization and KK exchange. C006 additionally requires lepton-flavor violation, so it is not a quark-only $\Delta F = 2$ constraint. In a lepton-extended RS pipeline it would constrain four-fermion $(\bar{u}\Gamma c)(\bar{e}\Gamma\mu)$ structures, flavor-changing Z-like couplings combined with lepton flavor violation, scalar exchange, or explicit leptoquark/R-parity-violating UV completions. It is therefore a roadmap process for joint up-sector and charged-lepton flavor alignment.

Post-2008 developments

The arXiv:0804.1954 RS-flavor baseline `CsakiFalkowskiWeiler2008:CompositeFlavor` predates the modern $D^0 \to e\mu$ search sequence. Belle's 2010 result `Belle2010:D0EMu` set the combined charge-mode limit at 2.6×10^{-7}, BaBar's 2012 continuum-charm search `BaBar2012:D0EeMuMuEMu` gave 3.3×10^{-7}, and LHCb's 2016 hadron-collider analysis `LHCb2016:D0EMu` became the current PDG-used input at 1.3×10^{-8}. The LHCb source describes this as an order-of-magnitude improvement over the previous limit and notes sensitivity to leptoquark and R-parity-violating SUSY parameter space.

Constraint validity and model dependence

The experimental upper limit is a clean LFV null result. The RS interpretation is model dependent: the quark-only scan has no charged-LFV $c \to ue\mu$ operator, and different chiral structures map differently onto the two-body D^0 rate. A conservative future use should compare the limit to a defined $\Delta C = 1$ LFV Wilson basis, with the D^0 decay constant and lepton mass effects stated explicitly, rather than reusing the neutral-meson mixing basis.

Code coverage in this repo

NO. The sidecar `code_coverage` block records the required greps across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $D^0 \to e^\pm\mu^\mp$, $c \to ue\mu$, or rare-charm LFV implementation. Nearby hits are different observables: D^0 mixing is configured at `quarkConstraints/deltaf2.py`:252 and evaluated at `quarkConstraints/deltaf2.py`:941; the modern policy surface lists only neutral D^0 mixing at `quarkConstraints/modern/phenomenology.py`:23 and evaluates that bridge at `quarkConstraints/modern/phen`
The $\mu \to e\gamma$ dipole checker at `flavorConstraints/muToEGamma.py`:75, called from `scanParams/scan.py`:524, is a separate charged-LFV observable.

Implementation difficulty

MEDIUM. A first live constraint would need a new $\Delta C = 1$ LFV operator normalization and a $D^0 \to e\mu$ branching-ratio wrapper. It should not require a global fit or the existing $\Delta F = 2$ RG machinery, but the current neutral-meson SLL/SLR/VLL/VRR/LR basis is not sufficient.

Key references

Process-local keys before bibliography consolidation: `PDG2026:D0EMu`, `LHCb2016:D0EMu`, `Belle2010:D0EMu`, `BaBar2012:D0EeMuMuEMu`, and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

22 C007: $D^+ \to \pi^+\mu^+\mu^-$

Process

Standard notation: $\mathcal{B}(D^+ \to \pi^+\mu^+\mu^-)$. This entry covers the rare charged-charm semileptonic decay $D^+ \to \pi^+\mu^+\mu^-$, a $\Delta C = 1$, $c \to u\mu^+\mu^-$ flavor-changing neutral-current probe. Charge-conjugate modes are implied by the experimental sources.

PDG / equivalent value(s)

The current PDG Live/API entry S031.42, accessed on 2026-05-16, lists

$$\mathcal{B}(D^+ \to \pi^+\mu^+\mu^-) < 6.7 \times 10^{-8} \qquad (90\% \text{ C.L.})$$

from the LHCb 2021 search in $1.6\,\text{fb}^{-1}$ of 2016 pp data. The LHCb source table gives the companion 95% C.L. limit 7.4×10^{-8}. The predecessor LHCb 2013 nonresonant search, using $1.0\,\text{fb}^{-1}$ at $\sqrt{s} = 7\,\text{TeV}$, reported $7.3(8.3) \times 10^{-8}$ at 90% (95%) C.L. These numbers are traced to `flavor_catalog/references/C007/pdg2026_dplus_piplus_mumu_pdgLive.txt`, `lhcb2021_arxiv2011_00217.txt`, and `lhcb2013_arxiv1304_6365.txt`.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models can induce up-sector flavor-changing neutral currents through nonuniversal KK gauge couplings, Z-like flavor violation, Higgs or radion scalar effects, and lepton-current extensions. This process is the charged semileptonic $\Delta C = 1$ analogue of the implemented neutral D^0-mixing lane: it probes $c \to u$ vector, axial-vector, and possible scalar semileptonic operators rather than four-quark $\Delta F = 2$ operators. It is therefore a roadmap constraint for a broader flavor backend, not a constraint currently applied by the quark scan.

Post-2008 developments

The arXiv:0804.1954 RS-flavor baseline predates the modern LHCb rare-charm program. Since then, LHCb first set nonresonant $D^+ \to \pi^+\mu^+\mu^-$ limits at the 10^{-8} level in 2013 and then folded the channel into a 2021 search for 25 rare or forbidden D^+ and D_s^+ decays. The 2021 analysis found no evidence in any of the investigated modes and tightened the PDG-listed C007 limit to 6.7×10^{-8} at 90% C.L. On the theory side, post-2008 rare-charm EFT work, especially de Boer and Hiller 2015, clarified that $D_{(s)} \to P\ell^+\ell^-$ branching-ratio recasts must separate short-distance Wilson coefficients from resonance-dominated long-distance physics; the LHCb 2021 introduction quotes short-distance SM branching fractions of order 10^{-12}, with dilepton resonances dominating the physical spectrum.

Constraint validity and model dependence

The experimental upper limit is robust as a counting result, but its use as an RS short-distance bound is model dependent. Long-distance hadronic resonances, weak-annihilation effects, and the exact phase-space definition of the "nonresonant" signal matter. A conservative catalog use is an upper bound on additional short-distance $c \to u\mu^+\mu^-$ contributions in the same experimental region, with an explicit recast needed before treating it as a hard scan veto.

Code coverage in this repo

NO. The required catalog greps and a focused search for `D+.*pi+.*mu`, `c.?->.?u`, `rare.?charm`, `semileptonic.*charm`, `C9`, and `C10` found no $D^+ \to \pi^+\mu^+\mu^-$ implementation in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. Nearby hits are unrelated: D^0 mixing appears at `quarkConstraints/deltaf2.py:252` and `quarkConstraints/deltaf2.py:941`, the modern policy lists only D^0 mixing at `quarkConstraints/modern/pheno` and `quarkConstraints/modern/phenomenology.py:668`, and $\mu \to e\gamma$ is separate at `flavorConstraints/muToEGa`

Implementation difficulty

HIGH. A production implementation needs a new $\Delta C = 1$ semileptonic operator basis and matching for $C_{9,10}^{(\prime)}$ and possible scalar/tensor terms, plus a $D \to \pi\mu^+\mu^-$ mode calculation with form factors, phase-space cuts, and a documented treatment of resonant long-distance contributions.

Key references

Process-local keys before bibliography consolidation: `PDG2026:DPlusPiMuMu`, `LHCb2021:DPlusPiMuMu`, `LHCb2013:DPlusPiMuMu`, `DeBoerHiller2015:RareCharmLeptons`, and `CsakiFalkowskiWeiler2008:CompositeFl`

23 C008: $D^+ \to \pi^+ e^\pm \mu^\mp$

Process

Standard notation: $\mathcal{B}(D^+ \to \pi^+ e^+ \mu^-)$ and $\mathcal{B}(D^+ \to \pi^+ e^- \mu^+)$. This entry covers the two opposite-sign charged-lepton-flavor-violating semileptonic charm decays usually summarized as $D^+ \to \pi^+ e^\pm \mu^\mp$. Charge-conjugate decays are implied by the experimental sources.

PDG / equivalent value(s)

The sidecar sources `PDG2026:DPlusPiEMuPlus` and `PDG2026:DPlusPiEMuMinus`, PDG Live/API entries S031.110 and S031.111 accessed on 2026-05-16, list separate 90% C.L. upper limits:

$$\mathcal{B}(D^+ \to \pi^+ e^+ \mu^-) < 2.1 \times 10^{-7}, \qquad \mathcal{B}(D^+ \to \pi^+ e^- \mu^+) < 2.2 \times 10^{-7}.$$

Both current limits come from `LHCb2021:DPlusPiEMu`, a search in $1.6\,\mathrm{fb}^{-1}$ of 2016 pp data. The LHCb table gives companion 95% C.L. limits of 2.3×10^{-7} and 2.2×10^{-7}, respectively. The predecessor search `BaBar2011:RareForbiddenCharm`, using $384\,\mathrm{fb}^{-1}$, reported 90% C.L. limits of 2.9×10^{-6} and 3.6×10^{-6}. These values are traced to `flavor_catalog/references/C008/pdg2026_dplus_piplus_ep_mum_pdg` `pdg2026_dplus_piplus_em_mup_pdgLive.txt`, `lhcb2021_arxiv2011_00217.txt`, and `babar2011_arxiv1107_4465`

Relevance to the RS / anarchic-flavor pipeline

This process tests a product of up-sector flavor violation and charged-lepton flavor violation: schematically, short-distance operators such as $(\bar{u}\,\Gamma c)(\bar{e}\,\Gamma'\mu)$. Minimal quark-only RS scans do not generate a live constraint for this mode, but lepton-sector extensions with nonuniversal KK gauge couplings, scalar exchange, or flavor-violating brane terms can generate it. It is therefore a useful catalog marker for a future combined quark-plus-lepton flavor backend rather than a current veto.

Post-2008 developments

The arXiv:0804.1954 RS-flavor baseline `CsakiFalkowskiWeiler2008:CompositeFlavor` predates the modern rare/forbidden semileptonic charm searches. BaBar `BaBar2011:RareForbiddenCharm` subsequently set or improved limits for charged-charm LFV modes in $384\,\mathrm{fb}^{-1}$. LHCb `LHCb2021:DPlusPiEMu` searched for 25 rare and forbidden D^+ and D_s^+ modes in 2016 data, found no significant deviation from background, and tightened the two C008 charge channels to the present 2×10^{-7}-level limits.

Constraint validity and model dependence

The experimental limits are robust upper bounds on the listed exclusive branching fractions. Their interpretation as RS constraints is model dependent because the mapping requires a $\Delta C = 1$, LFV semileptonic operator basis, $D \to \pi$ form factors, lepton-chirality assumptions, and a phase-space treatment matching the LHCb analysis. The Standard Model background for charged-LFV final states is negligible for catalog purposes.

Code coverage in this repo

NO. The sidecar `code_coverage` block records the required coverage greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`, plus a focused search for `D+.*pi+.*e.*mu`, `pi+.*e.*mu`, rare-charm, and `c->u` LFV patterns, found no $D^+ \to \pi^+ e^\pm \mu^\mp$ implementation. Nearby hits are unrelated: D^0 mixing appears at `quarkConstraints/deltaf2.py:252` and `quarkConstraints/deltaf2.py:941`, the modern policy

lists only D^0 mixing at `quarkConstraints/modern/phenomenology.py:23` and `quarkConstraints/modern/pheno`
and the implemented LFV observable is $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. A production implementation needs a new $\Delta C = 1$ LFV semileptonic operator basis and a $D^+ \to \pi^+ e\mu$ branching-fraction calculation with form factors, phase-space acceptance, and chirality conventions. This is beyond adding an input to the existing $\Delta F = 2$ or $\mu \to e\gamma$ paths.

Key references

Process-local keys before bibliography consolidation: `PDG2026:DPlusPiEMuPlus`, `PDG2026:DPlusPiEMuMinus`, `LHCb2021:DPlusPiEMu`, `BaBar2011:RareForbiddenCharm`, and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

24 Beauty

25 B001: Δm_d in B_d mixing

Process

Standard notation: Δm_d. This entry covers the mass splitting of the neutral B_d eigenstates, equivalently the oscillation frequency in $B_d^0 - \bar{B}_d^0$ mixing. It is the beauty-sector down-type $\Delta F = 2$ magnitude constraint already used by the quark-scan code.

PDG or equivalent value

The canonical measured value is the HFLAV/PDG 2025 average recorded in `B001.yaml` under `pdg_or_equivalent`, source key `hflav_pdg2025_bd_mixing`. HFLAV gives

$$\Delta m_d = 0.5069 \pm 0.0019 \text{ ps}^{-1},$$

using time-dependent measurements from ALEPH, DELPHI, L3, OPAL, CDF, D0, BABAR, Belle, and LHCb. With the HFLAV assumptions of no CP violation in mixing and no B^0 width difference, the same snapshot records the companion world averages $x_d = 0.7697 \pm 0.0035$ and $\chi_d = 0.1860 \pm 0.0011$. The local source snapshot is `flavor_catalog/references/B001/hflav_pdg2025_bd_mixing.txt`, accessed on 2026-05-16.

Relevance to RS / anarchic flavor

Anarchic warped models generate tree-level flavor-changing neutral currents from KK gluon and electroweak exchange after rotating to the quark-mass basis. B_d mixing probes the $b \leftrightarrow d$ down-sector alignment and is a direct test of the same Wilson-coefficient pipeline used for kaon and

B_s mixing. Compared with ε_K, it is less chirally enhanced but constrains a different row-column structure of the anarchic profiles.

Post-2008 developments

The arXiv:0804.1954 RS-flavor baseline, source key `cfw_2008_rs_flavor`, highlighted KK-gluon flavor bounds near 21 TeV in standard RS scenarios and 33 TeV in the pseudo-Goldstone setup, with boundary kinetic terms and flavor assumptions as caveats. Since then, Δm_d has become a precision-average observable. Belle II has repeated the benchmark measurement with early SuperKEKB data, reporting

$$\Delta m_d = 0.516 \pm 0.008_{\text{stat}} \pm 0.005_{\text{syst}}\ \text{ps}^{-1}$$

from $190\,\text{fb}^{-1}$, with both numbers recorded under source key `belleii_2023_bd_mixing`. On the theory side, source key `flag2024_b_mixing` documents the FLAG Review 2024 heavy-quark inputs relevant for B-mixing matrix elements and global unitarity-triangle fits; those inputs are theory normalization data, not the measured observable.

Constraint validity / model dependence

This is a robust neutral-meson-mixing magnitude constraint. The experimental average is precise, but turning it into a new-physics exclusion requires a policy for the SM contribution and CKM inputs. The current code uses an allowed NP budget rather than a full correlated CKM fit, so catalog users should treat it as a conservative magnitude screen, not as a global-fit replacement.

Code coverage in this repo

YES-D2-BD. The legacy $\Delta F = 2$ backend defines the B_d input at `quarkConstraints/deltaf2.py:225` and notes the hadronic matrix-element treatment at `quarkConstraints/deltaf2.py:236`. The in-code B_d hadronic and experimental constants are at `quarkConstraints/deltaf2.py:631–638`. The scan dispatches `b_d` through `evaluate_bd_mixing` at `quarkConstraints/deltaf2.py:383–391`; the evaluator itself is `quarkConstraints/deltaf2.py:903–915`, with the running wrapper at `quarkConstraints/deltaf2.py:960–966`. The modern metadata layer also carries B_d as a policy system at `quarkConstraints/modern/phenomenology.py:23` and mirrors the input at `quarkConstraints/modern/inputs.py:967–976`.

Implementation difficulty

LOW. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR machinery and BMU-style running cover the B_d mixing magnitude observable. A future production refresh would mainly improve provenance, unit conversion, and likelihood policy rather than adding new operators or RG evolution.

Key references

Process-local reference keys before bibliography consolidation: `hflav_pdg2025_bd_mixing`, `belleii_2023_bd_mixin` `flag2024_b_mixing`, and `cfw_2008_rs_flavor`.

26 B002: $S_{\psi K_S}$ and $\sin 2\beta$

Process

Standard notation: $S_{\psi K_S}$, equivalently $\sin 2\beta$ or $\sin 2\phi_1$. This entry covers the mixing-induced CP asymmetry in $B_d^0 \to J/\psi K_S^0$, conventionally reported as $S_{\psi K_S} \simeq \sin 2\beta$. It is the beauty-sector benchmark CP phase of the unitarity triangle and is dominated by interference between $B_d - \bar{B}_d$ mixing and the $b \to c\bar{c}s$ tree amplitude.

PDG or equivalent value

The canonical values are the `pdg_or_equivalent` entries in `B002.yaml`. HFLAV Summer 2025 (`hflav_triangle_summer2025_sin2beta`) gives the current all-charmonium average

$$\sin(2\beta) = \sin(2\phi_1) = 0.710 \pm 0.011,$$

with a stat-only uncertainty of 0.010 and fit CL 0.34, accessed on 2026-05-16 in `references/B002/hflav_triangle_`
The same HFLAV block gives the direct $J/\psi K_S$ mode average

$$S_{\psi K_S} = 0.712 \pm 0.011.$$

The corresponding physical CKM-angle solution is $\beta = (22.63^{+0.45}_{-0.44})^\circ$, with the trigonometric alternate solution also stored in the sidecar.

Relevance to the RS / anarchic-flavor pipeline

In anarchic warped models, tree-level KK-gluon and electroweak exchange can generate a new complex contribution to M_{12}^d. $S_{\psi K_S}$ is therefore the phase counterpart of the existing B_d mass-difference constraint: a scan point can satisfy Δm_d while still shifting $\arg(M_{12}^d)$. This observable is especially useful because the decay amplitude is CKM-favored and theoretically clean compared with many hadronic beauty modes.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline, the central experimental story is precision rather than discovery. Belle and BaBar final samples established the B-factory anchor, and the post-2023 update added two important inputs: LHCb measured

$$S_{\psi K_S} = 0.717 \pm 0.013_{\text{stat}} \pm 0.008_{\text{syst}}$$

using $6\,\text{fb}^{-1}$ (`lhcb_2024_spsiks`), and Belle II measured

$$S = 0.724 \pm 0.035_{\text{stat}} \pm 0.009_{\text{syst}}$$

with $362\,\text{fb}^{-1}$ (`belleii_2024_sin2phi1`). These inputs moved the HFLAV average to the percent-level regime. On the theory side, the main development is sharper accounting of doubly Cabibbo-suppressed penguin pollution; the `frings_nierste_wiebusch_2015` snapshot quotes $|\Delta\phi_d| \leq 0.68^\circ$.

Constraint validity and model dependence

This is a robust neutral-meson-mixing phase constraint, but a catalog number is not a standalone exclusion without a CKM convention. A live implementation must decide whether the SM β input is fixed from a global fit or fitted simultaneously, and how to budget the small penguin phase. New physics in the decay amplitude would make the usual $S_{\psi K_S} \simeq \sin 2\beta$ identification model-dependent, but for RS quark-flavor scans the dominant target is the new phase in B_d mixing.

Code coverage in this repo

PARTIAL. No implementation named `S_psi`, `J/psi K_S`, `sin2beta`, or equivalent was found in the required greps. However, `quarkConstraints/deltaf2.py:225` defines the existing B_d mixing input, `quarkConstraints/deltaf2.py:631` and `quarkConstraints/deltaf2.py:638` define its hadronic and Δm_d inputs, and `quarkConstraints/deltaf2.py:903` evaluates $M_{12}^{d,\mathrm{NP}}$. That evaluator reduces the result to $|M_{12}^{d,\mathrm{NP}}|$ at `quarkConstraints/deltaf2.py:910`, so the phase needed for $S_{\psi K_S}$ is not exposed. The shared $\Delta F = 2$ running basis is documented at `quarkConstraints/qcd_running.py:7`.

Implementation difficulty

LOW. The existing $\Delta F = 2$ Wilson and hadronic machinery covers the needed operator basis. The missing work is to retain the complex B_d mixing amplitude, define the SM phase convention and allowed penguin uncertainty, and add an observable policy for the measured $S_{\psi K_S}$.

Key references

Process-local raw reference keys before bibliography consolidation: `hflav_triangle_summer2025_sin2beta`, `lhcb_2024_spsiks`, `belleii_2024_sin2phi1`, `frings_nierste_wiebusch_2015`, `faller_fleischer_jung_mannel_2` and `csaki_falkowski_weiler_2008`.

27 B003: Δm_s in B_s mixing

Process

Standard notation: Δm_s. This entry covers the $B_s^0 - \bar{B}_s^0$ mass splitting, equivalently the oscillation frequency measured in time-dependent flavor-tagged B_s decays. It is the magnitude observable in the $\Delta B = 2$, $b \leftrightarrow s$ neutral-meson mixing system.

PDG or equivalent value

The canonical equivalent value used here is the HFLAV Fall 2024 recommended lifetime-and-oscillation average recorded in `B003.yaml` under `pdg_or_equivalent`, source key `hflav_2024_dms`, with local snapshot `flavor_catalog/references/B003/hflav_2024_dms.txt`:

$$\Delta m_s = 17.766 \pm 0.006 \text{ ps}^{-1}$$

from CDF, LHCb, and CMS. The same HFLAV page gives the derived companion quantities $x_s = 26.94 \pm 0.10$ and $\chi_s = 0.499314 \pm 0.000005$, using $1/\Gamma_s = 1.516 \pm 0.006$ ps, $\Delta\Gamma_s = +0.0781 \pm 0.0035$ ps^{-1}, and no CP violation in mixing. The published-only HFLAV/PDG 2025 preparation page, source key `hflav_pdg2025_dms`, gives the nearly identical $\Delta m_s = 17.765 \pm 0.006$ ps^{-1}, snapshotted at `flavor_catalog/references/B003/hflav_pdg2025_dms.txt`.

Relevance to RS / anarchic flavor

In anarchic Randall–Sundrum flavor, tree-level KK-gluon and electroweak exchange generate $\Delta B = 2$ four-quark operators. B_s mixing is the down-sector $s-b$ analogue of B_d mixing and complements the stronger kaon constraints by testing a different localization and CKM structure. The existing code treats Δm_s as a direct bound on the new-physics contribution to M_{12}^s, with separate weights for current-current and left-right operators.

Post-2008 developments

The 2008 RS-flavor baseline in `cfw_2008_rs_flavor` predates the precision LHCb B_s oscillation program. The key post-2008 experimental step for this catalog is the LHCb measurement in `lhcb_2021_deltams_arxiv`: using 6 fb^{-1} of Run-2 $B_s \to D_s\pi$ data, LHCb finds

$$\Delta m_s = 17.7683 \pm 0.0051_{\text{stat}} \pm 0.0032_{\text{syst}} \text{ ps}^{-1},$$

and a LHCb-only combination

$$\Delta m_s = 17.7656 \pm 0.0057 \text{ ps}^{-1}.$$

On the theory side, `flag_2024_bmixing` records the FLAG Review 2024 $N_f = 2+1+1$ auxiliary inputs

$$f_{B_s}\sqrt{\hat{B}_{B_s}} = 256.1 \pm 5.7 \text{ MeV}, \qquad \xi = 1.216 \pm 0.016,$$

with a warning that correlated decay-constant and mixing inputs should not be reconstructed from separate averages.

Constraint validity / model dependence

The experimental oscillation frequency is very clean. The model-dependent step is translating it into a new-physics allowance: a conservative catalog can bound $|M_{12}^{s,\text{NP}}|$ directly from the measured splitting, while a sharper SM-subtracted interpretation depends on CKM inputs and FLAG lattice matrix elements. The constraint is therefore a robust $\Delta F = 2$ magnitude test, but cancellations between SM and new physics or correlated CKM-fit choices should be made explicit in any live likelihood.

Code coverage in this repo

YES-D2-BS. The v1 $\Delta F = 2$ input bundle defines the B_s system at `quarkConstraints/deltaf2.py:238` with key `b_s` at `quarkConstraints/deltaf2.py:239` and down-sector generations `(1, 2)` at `quarkConstraints/d`

The hadronic constants include the B_s decay constant at `quarkConstraints/deltaf2.py:645` and Δm_s input at `quarkConstraints/deltaf2.py:651`. The budget helper is at `quarkConstraints/deltaf2.py:893;` the evaluator is `evaluate_bs_mixing` at `quarkConstraints/deltaf2.py:922`, with the QCD-running wrapper at `quarkConstraints/deltaf2.py:969`. The modern policy also carries B_s in `quarkConstraints/modern/phenomenology.py:23`, defines the policy entry at `quarkConstraints/modern/pheno` and routes it through the bridge evaluator at `quarkConstraints/modern/phenomenology.py:657` and `quarkConstraints/modern/phenomenology.py:1346`. The shared $\Delta F = 2$ operator basis is documented at `quarkConstraints/qcd_running.py:7`.

Implementation difficulty

LOW. The existing $\Delta F = 2$ basis and RG path already cover the needed VLL, VRR, scalar, and left-right structures for B_s mixing. Future production work is mostly input hygiene: decide the HFLAV/PDG value policy, refresh FLAG normalization constants, and document whether the live constraint uses a conservative absolute M_{12}^{NP} budget or an SM-subtracted one.

Key references

Process-local reference keys before bibliography consolidation: `hflav_2024_dms`, `hflav_pdg2025_dms`, `lhcb_2021_deltams_arxiv`, `flag_2024_bmixing`, and `cfw_2008_rs_flavor`.

28 B004: ϕ_s in $B_s^0 \to J/\psi\phi$

Process

Standard notation: $\phi_s^{c\bar{c}s}$, often shortened to ϕ_s. This entry covers the mixing-induced CP phase extracted from time-dependent angular analyses of $B_s^0 \to J/\psi K^+K^-$ near the ϕ resonance, conventionally reported as $B_s^0 \to J/\psi\phi$. The vector–vector final state requires the experimental fit to separate CP-even and CP-odd angular components.

PDG or equivalent value

The canonical equivalent source is the HFLAV PDG 2025 ϕ_s input table, snapshotted as `references/B004/hflav_p` with source key `hflav_pdg2025_phis_inputs`. HFLAV gives the all-combined average

$$\phi_s^{c\bar{c}s} = -0.041 \pm 0.016 \text{ rad}, \qquad \Delta\Gamma_s = +0.079 \pm 0.006 \text{ ps}^{-1}.$$

For the strict $B_s^0 \to J/\psi K^+K^-$ channel combination, HFLAV gives

$$\phi_s^{c\bar{c}s} = -0.050 \pm 0.017 \text{ rad}, \qquad \Delta\Gamma_s = +0.077 \pm 0.006 \text{ ps}^{-1}.$$

The dominant recent LHCb Run-2 input, snapshotted as `references/B004/lhcb_2024_jpsikk_phis_arxiv.txt` with source key `lhcb_2024_jpsikk_phis_arxiv`, reports

$$\phi_s = -0.039 \pm 0.022_{\text{stat}} \pm 0.006_{\text{syst}} \text{ rad}$$

from 6 fb^{-1} and approximately 349000 signal decays. The same source quotes the CKM-fit Standard Model expectation $-2\beta_s = -0.0368^{+0.0009}_{-0.0006}$ rad.

Relevance to RS / anarchic flavor

In anarchic warped models, tree-level KK-gluon and electroweak exchange can generate a complex new contribution to $B_s - \bar{B}_s$ mixing. ϕ_s is therefore the phase complement to the Δm_s magnitude constraint: a point can satisfy the mass splitting while still shifting $\arg M_{12}^s$. The process is especially clean for RS quark scans because the dominant decay amplitude is the CKM-favored $b \to c\bar{c}s$ transition, while new physics mainly enters through $\Delta B = 2$ mixing.

Post-2008 developments

The 2008 RS-flavor baseline in `cfw_2008_rs_flavor` predates the LHCb era and treats neutral-meson mixing as a generic $\Delta F = 2$ stress test rather than as a precision ϕ_s input. Since then, Tevatron, ATLAS, CMS, and especially LHCb measurements have turned ϕ_s into a precision phase test. The 2024 LHCb Run-2 analysis superseded the earlier Run-2 result in the same channel, added improved flavor tagging and decay-time calibration, and found no evidence for polarization dependence (`lhcb_2024_jpsikk_phis_arxiv`). HFLAV's PDG 2025 combination now reaches 0.016 rad precision and is consistent with the small SM phase (`hflav_pdg2025_phis_inputs`).

Constraint validity / model dependence

This is a robust neutral-meson-mixing phase constraint, but it is not identical to a pure $\arg M_{12}^s$ observable without assumptions. The interpretation uses a CKM phase convention, assumes no large new physics in the tree-dominated decay amplitude, and must budget subleading penguin effects. The vector–vector final state also requires an angular analysis and experimental correlations with $\Delta\Gamma_s$. For the present catalog, it should be classified as a robust $\Delta F = 2$ phase observable with small decay-amplitude systematics.

Code coverage in this repo

PARTIAL. A focused grep for `phi_s`, `phis`, `J/psi`, `Jpsi`, `psi.*phi`, `B_s.*phi`, and `beta_s` found no direct observable implementation in the required implementation and test directories. The underlying B_s mixing lane exists: `quarkConstraints/deltaf2.py:239` defines the `b_s` input, `quarkConstraints/deltaf2.py:651` stores Δm_s, and `quarkConstraints/deltaf2.py:922` evaluates the B_s mixing contribution. However, `quarkConstraints/deltaf2.py:929` reduces the computed complex amplitude to $|M_{12}^{s,\mathrm{NP}}|$, so the phase needed for ϕ_s is not exposed. The modern bridge also evaluates only the magnitude-style B_s mixing contribution at `quarkConstraints/modern/phenomenology.py:` the shared $\Delta F = 2$ operator basis is documented at `quarkConstraints/qcd_running.py:7`.

Implementation difficulty

LOW. No new operator basis is required: the existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ machinery covers the B_s mixing amplitude. The missing work is to retain the complex M_{12}^s, combine it with the SM $-2\beta_s$ convention, and add a likelihood or tolerance policy including the HFLAV uncertainty and small penguin-systematic allowance.

Key references

Process-local reference keys before bibliography consolidation: `hflav_pdg2025_phis_inputs`, `lhcb_2024_jpsikk_ph` and `cfw_2008_rs_flavor`.

29 B005: $B_s^0 \to \mu^+\mu^-$

Process

Standard notation: $\mathcal{B}(B_s^0 \to \mu^+\mu^-)$. This entry covers the time-integrated rare leptonic decay $B_s^0 \to \mu^+\mu^-$, a helicity-suppressed $b \to s\ell^+\ell^-$ flavor-changing neutral-current channel. It is distinct from the inclusive and exclusive semileptonic $b \to s\ell^+\ell^-$ entries because the final state is purely leptonic and the leading hadronic input is the B_s decay constant.

PDG or equivalent value

The current PDG live/API value, accessed on 2026-05-16, is

$$\mathcal{B}(B_s^0 \to \mu^+\mu^-)_{\rm exp} = (3.34 \pm 0.27) \times 10^{-9}.$$

The source key is `PDG2026:BsMuMu`, with local snapshot `flavor_catalog/references/B005/pdg_2026_bsmumu.txt` The 2024 Czaja–Misiak SM-status review quotes the same weighted experimental average and gives the Standard Model prediction

$$\mathcal{B}(B_s^0 \to \mu^+\mu^-)_{\rm SM} = (3.64 \pm 0.12) \times 10^{-9}.$$

Its source key is `CzajaMisiak2024:BsMuMuSM`. The sidecar retains the older HFLAV rare-decay average $(3.45\pm0.29)\times10^{-9}$ as auxiliary provenance, together with CMS $(3.83^{+0.38\,+0.19\,+0.14}_{-0.36\,-0.16\,-0.13})\times10^{-9}$, LHCb $(3.09^{+0.46+0.15}_{-0.43-0.11}) \times 10^{-9}$, and ATLAS $(2.8^{+0.8}_{-0.7}) \times 10^{-9}$ inputs under the `HFLAV2023:BsMuMu`, `CMS2023:BsMuMu`, `LHCb2022:BsMuMu`, and `ATLAS2019:BsMuMu` source keys.

Relevance to the RS / anarchic-flavor pipeline

This mode probes the short-distance $b \to s\mu^+\mu^-$ Hamiltonian in a clean single-number observable. In anarchic RS scenarios, flavor-changing neutral gauge couplings, KK electroweak exchange, Higgs/radion scalar exchange, or leptoquark-like completions can shift axial-vector, scalar, and pseudoscalar Wilson coefficients. The scalar and pseudoscalar terms are especially important because they can remove the Standard Model helicity suppression and are prominent in large-tan β supersymmetric or extended-Higgs interpretations.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline, this channel changed from an upper-limit target into a precision observable. LHCb and CMS first established the decay, and the present average combines modern LHCb, CMS, and ATLAS inputs. CMS now reports the most precise single-experiment branching-fraction measurement, while LHCb provides a simultaneous branching-fraction and effective-lifetime analysis. Theory has also improved: modern SM predictions include precise lattice f_{B_s}, CKM inputs, electroweak and QCD corrections, and the time-integration correction associated with the B_s width difference.

Constraint validity and model dependence

This is a robust short-distance rare-decay constraint, but it is not covered by the repo's neutral-meson-mixing machinery. The SM comparison is clean relative to angular $b \to s\ell\ell$ observables, yet it still depends on the definition of the time-integrated branching fraction, f_{B_s}, CKM inputs, and the treatment of scalar and pseudoscalar operators. It is directly useful for models that predict $C_{10}^{(\prime)}$, $C_S^{(\prime)}$, or $C_P^{(\prime)}$, including scalar leptoquarks, Z' models, and tan-beta-enhanced Higgs sectors.

Code coverage in this repo

NO. The required catalog greps and a focused search for `B_s.*mu`, `mumu`, `C10`, scalar/pseudoscalar rare-decay terms, and $b \to s\mu\mu$ found no live $B_s^0 \to \mu^+\mu^-$ implementation in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. The nearest related code is B_s mixing, e.g. `quarkConstraints/deltaf2.py:239` and `quarkConstraints/deltaf2.py:922`, which is a $\Delta F = 2$ observable rather than this $\Delta B = 1$ decay.

Implementation difficulty

MEDIUM. A catalog-to-code implementation needs a new $\Delta B = 1$ leptonic rare-decay observable, Wilson conventions for $C_{10}^{(\prime)}, C_S^{(\prime)}, C_P^{(\prime)}$, and a branching-ratio formula with the B_s lifetime and width-difference convention. It does not need an angular likelihood or exclusive form factors, but a production-grade RS matching path would need carefully documented scalar and electroweak operator normalization.

Key references

Process-local raw reference keys before bibliography consolidation: `PDG2026:BsMuMu`, `HFLAV2023:BsMuMu`, `CMS2023:BsMuMu`, `LHCb2022:BsMuMu`, `ATLAS2019:BsMuMu`, `CzajaMisiak2024:BsMuMuSM`, and `CsakiFalkowskiWeil`

30 B006: $B^0 \to \mu^+\mu^-$

Process

Standard notation: $\mathcal{B}(B^0 \to \mu^+\mu^-)$. This entry covers the rare leptonic $b \to d\mu^+\mu^-$ flavor-changing neutral-current decay of the neutral B_d meson. It is the $b \to d$ analogue of $B_s^0 \to \mu^+\mu^-$, but is CKM-suppressed and currently enters the catalog as an upper limit rather than an observed branching fraction.

PDG or equivalent value

The current PDG live/API listing `S042.7`, accessed on 2026-05-16 and stored under source key `PDG2026:BdMuMu`, gives

$$\mathcal{B}(B^0 \to \mu^+\mu^-) < 1.5 \times 10^{-10} \quad (90\% \text{ CL}).$$

The PDG input used for this limit is the CMS Run 2 result (`CMS2023:BdMuMu`), which also reports $< 1.9 \times 10^{-10}$ at 95% CL. The local canonical snapshot is `flavor_catalog/references/B006/pdg_2026_bdmumu.txt`. As an equivalent HFLAV rare-decay ratio constraint, the April 2023 table gives

$$\mathcal{B}(B^0 \to \mu^+\mu^-)/\mathcal{B}(B_s^0 \to \mu^+\mu^-) < 8.1 \times 10^{-2} \quad (90\% \text{ CL}),$$

using the LHCb input (`HFLAV2023:BdOverBsMuMu`). For scale, the reduced-uncertainty Standard Model prediction recorded in `BobethEtAl2013:BdMuMuSM` is

$$\mathcal{B}(B_d \to \mu^+\mu^-)_{\text{SM}} = (1.06 \pm 0.09) \times 10^{-10}.$$

Relevance to RS / anarchic flavor

This mode is a clean $\Delta B = 1$ leptonic rare decay. In warped/anarchic flavor models, flavor-changing Z, KK electroweak gauge, Higgs, radion, or other scalar exchanges can contribute to axial-vector, scalar, and pseudoscalar $b \to d\mu\mu$ Wilson coefficients. Scalar and pseudoscalar terms are especially diagnostic because they can compete with the helicity-suppressed Standard Model amplitude.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline (`CsakiFalkowskiWeiler2008:CompositeFlavor`), this channel moved from pre-LHC upper limits to a mature LHC rare-decay search. ATLAS combined Run 1 and 2015–2016 data to set 2.1×10^{-10} at 95% CL (`ATLAS2019:BdMuMu`). LHCb later set 2.6×10^{-10} at 95% CL and constrained the B^0/B_s^0 dimuon branching-ratio ratio (`LHCb2022:BdMuMu`). CMS Run 2, with $140\,\text{fb}^{-1}$ at $\sqrt{s} = 13$ TeV, now supplies the PDG canonical 90%-CL limit (`CMS2023:BdMuMu`; `PDG2026:BdMuMu`). On the theory side, $B_{s,d} \to \ell^+\ell^-$ Standard Model predictions now include reduced perturbative electroweak and QCD uncertainties (`BobethEtAl2013:BdMuMuSM`).

Constraint validity / model dependence

The experimental object is robust as an upper limit on a short-distance branching fraction, but the catalog interpretation is model-dependent. A live constraint needs a consistent $\Delta B = 1$ Hamiltonian, CKM convention, normalization to f_{B_d}, and treatment of scalar/pseudoscalar operators. Unlike angular $b \to d\ell\ell$ modes it does not require a multi-bin form-factor likelihood.

Code coverage in this repo

NO. The required catalog greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $B^0 \to \mu^+\mu^-$ or $b \to d\mu\mu$ observable implementation. Nearby hits such as `quarkConstraints/deltaf2.py:225` and `quarkConstraints/deltaf2.py:903` are B_d $\Delta F = 2$ mixing entries, not this rare leptonic $\Delta B = 1$ decay.

Implementation difficulty

MEDIUM. This requires a new $\Delta B = 1$ leptonic rare-decay observable and Wilson normalization for $C_{10}^{(\prime)}$, $C_S^{(\prime)}$, and $C_P^{(\prime)}$. It should not need new lattice matrix-element machinery beyond decay-constant inputs, RG for a full angular analysis, or exclusive semileptonic form factors.

Key references

Process-local raw reference keys before bibliography consolidation: `PDG2026:BdMuMu`, `HFLAV2023:BdOverBsMuMu`, `CMS2023:BdMuMu`, `LHCb2022:BdMuMu`, `ATLAS2019:BdMuMu`, `BobethEtAl2013:BdMuMuSM`, and `CsakiFalkowskiWeile`

31 B009: $B^+ \to \tau^+\nu_\tau$

Process

Standard notation: $\mathcal{B}(B^+ \to \tau^+\nu_\tau)$. This entry covers the purely leptonic charged-B branching fraction. It is a beauty-family charged-current observable, not a neutral-meson mixing constraint.

PDG / equivalent value(s)

The canonical sidecar value is the HFLAV End-of-December-2025 average

$$\mathcal{B}(B^+ \to \tau^+\nu_\tau) = (1.12 \pm 0.19) \times 10^{-4},$$

from `flavor_catalog/references/B009/hflav_dec2025_btaunu.txt`. The PDG 2025 REST entry gives the compatible average $(1.09^{+0.25}_{-0.24}) \times 10^{-4}$, with the PDG uncertainty scaled by the factor recorded in `pdg2025_btaunu_api.txt`. HFLAV lists the post-2008 inputs as Belle hadronic tag $(0.72^{+0.27}_{-0.25} \pm 0.11) \times 10^{-4}$, Belle semileptonic tag $(1.25 \pm 0.28 \pm 0.27) \times 10^{-4}$, Belle II hadronic tag $(1.24 \pm 0.41 \pm 0.19) \times 10^{-4}$, BaBar hadronic tag $(1.83^{+0.53}_{-0.49} \pm 0.24) \times 10^{-4}$, and BaBar semileptonic tag $(1.70 \pm 0.80 \pm 0.20) \times 10^{-4}$. For comparison to the $|V_{ub}|f_B$ Standard Model prediction, the sidecar records the UTfit Summer 2024 SM prediction $(0.865 \pm 0.041) \times 10^{-4}$.

Relevance to the RS / anarchic-flavor pipeline

This channel probes $b \to u\tau\nu$ charged currents. In the SM it is fixed by $f_B|V_{ub}|$, the charged-B lifetime, and the tau mass; therefore it is a clean place where a new charged current can interfere with the SM amplitude. It is directly relevant to charged-Higgs, W', and leptoquark benchmarks. For warped or anarchic-flavor extensions, it is not part of the current quark $\Delta F = 2$ lane, but it would test KK charged-gauge-boson or non-minimal charged-scalar effects in the b-to-u current and in the tau-neutrino current.

Post-2008 developments

Relative to the Csaki–Falkowski–Weiler 2008 flavor baseline (`cfw2008_arxiv0804_1954.txt`), the measurement landscape changed substantially. BaBar published semileptonic-tag and hadronic-tag results in 2010 and 2013; Belle followed with hadronic-tag and semileptonic-tag measurements in 2013 and 2015. Belle II added a hadronic-FEI measurement in 2025, which HFLAV includes in the average while noting that the red Belle II input is not part of the PDG average. On the theory/global-fit side, the sidecar records the UTfit Summer 2024 SM prediction, so the comparison is now driven mainly by $|V_{ub}|$ and global-fit conventions rather than by an unavailable leptonic decay constant.

Constraint validity and model dependence

The experimental average is straightforward, but the constraint is model-dependent. It is robust as a charged-current amplitude test; it is not a generic RS bound unless the model supplies a calculable $b \to u\tau\nu$ matching contribution. A charged scalar, W', or leptoquark can interfere constructively or destructively with the SM, so a one-number exclusion requires the operator normalization and neutrino-flavor assumptions. The SM comparison is global-fit dependent through $|V_{ub}|$, even though the leptonic decay constant input itself is now standard.

Code coverage in this repo

NO. I reran the required coverage greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The broad plan greps find the existing neutral $\Delta F = 2$ and $\mu \to e\gamma$ surfaces, but no charged-current leptonic-B observable. Targeted searches for `B^+ -> tau`, `B+ --> tau`, `tau.*nu`, `b.*->.*u`, `charged Higgs`, `leptoquark`, and W' returned no code hits. Docs-only catalog-planning rows were not counted as implementation evidence.

Implementation difficulty

MEDIUM. The catalog-level formula is simple and uses standard experimental and lattice/CKM inputs, but production use would require a new charged-current observable and model-specific matching for W', charged-scalar, or leptoquark amplitudes. It should not extend the legacy `deltaf2.py` path; if promoted, it belongs in the modern constraint surface.

Key references

Process-local snapshots before bibliography consolidation: `hflav_dec2025_btaunu.txt`, `pdg2025_btaunu_api.txt`, `utfit_summer2024_btaunu.txt`, `belle2013_arxiv1208_4678.txt`, `belle2015_arxiv1503_05613.txt`, `belleii2025_arxiv2502_04885.txt`, `babar2013_arxiv1207_0698.txt`, `babar2010_arxiv0912_2453.txt`, and `cfw2008_arxiv0804_1954.txt`.

32 B011: Inclusive $B \to X_s\gamma$

Process

Standard notation: $\mathcal{B}(\bar{B} \to X_s\gamma)$. This entry covers the CP- and isospin-averaged inclusive radiative decay $\bar{B} \to X_s\gamma$, equivalently the partonic FCNC transition $b \to s\gamma$, at the canonical photon-energy threshold $E_\gamma > 1.6\,\text{GeV}$. It is separated from the exclusive $B \to K^*\gamma$ and $B_s \to \phi\gamma$ modes.

PDG / equivalent value(s)

The canonical sidecar value is the HFLAV inclusive average

$$\mathcal{B}(\bar{B} \to X_s\gamma)_{\text{exp}} = (3.49 \pm 0.19) \times 10^{-4}, \qquad E_\gamma > 1.6\,\text{GeV},$$

accessed on 2026-05-16, with local snapshots `references/B011/hflav_results_2026-05-16.txt` and `references/B011/hflav_rare_decays_dec2024_table67.txt`. The HFLAV table reports the same average as 349 ± 19 in units of 10^{-6} after extrapolating the Belle, Belle II, BaBar, and CLEO inputs to the common threshold. For theory comparison, the sidecar records the Misiak–Rehman–Steinhauser NNLO SM prediction

$$\mathcal{B}(\bar{B} \to X_s\gamma)_{\text{SM}} = (3.40 \pm 0.17) \times 10^{-4}, \qquad E_\gamma > 1.6\,\text{GeV},$$

from `references/B011/misiak_rehman_steinhauser_2020_arxiv.txt` (arXiv:2002.01548). It also records the PDG-reviewed Misiak et al. number $(3.36 \pm 0.23) \times 10^{-4}$ from `references/B011/misiak_et_al_2015_ar` (arXiv:1503.01789).

Relevance to the RS / anarchic-flavor pipeline

This observable tests the electromagnetic and chromomagnetic dipole sector, usually parameterized by C_7 and C_8. In anarchic RS models, KK fermion, gauge, Higgs, or charged-scalar loops can generate flavor-changing dipoles even when the tree-level neutral-meson mixing constraints are passed. The inclusive rate is therefore a sharp complement to the existing $\Delta F = 2$ lane: it probes loop-level chirality-flipping flavor structure rather than the four-quark operators currently audited in the quark scan.

Post-2008 developments

The main post-CFW updates are experimental averaging and NNLO theory. Belle published a full-data inclusive $B \to X_{s+d}\gamma$ photon-spectrum analysis, Belle II added a hadronic-tag photon-spectrum measurement, and HFLAV now averages Belle, Belle II, BaBar, and CLEO inputs at the common $E_\gamma > 1.6\,\text{GeV}$ threshold. On the SM side, Misiak et al. updated the NNLO prediction after the original paper-era calculations, and Misiak, Rehman, and Steinhauser subsequently incorporated improved treatment of nonperturbative effects and progress toward removing the charm-mass interpolation.

Constraint validity and model dependence

This is a robust loop/dipole constraint but not a plug-in likelihood without model work. The inclusive observable is cleaner than exclusive radiative modes, yet its comparison still depends on the photon-energy cut, extrapolation to the common threshold, perturbative matching, resolved-photon and other nonperturbative uncertainties, and the convention used for C_7, C_8. A catalog-level entry is unambiguous; using it as a hard RS constraint requires a dedicated $b \to s\gamma$ Wilson-coefficient pipeline.

Inclusive $B \to X_s\gamma$ constrains $|C_7, C_8|$ dipoles; the RS right-handed-dipole signature requires exclusive polarization and time-dependent CP observables sensitive to C_7' (see B012 once available).

Code coverage in this repo

NO. The required catalog greps were rerun. A focused implementation grep for `X_s`, `b->s gamma`, `C7`, and `C8` across `quarkConstraints/`, `flavorConstraints/`, `scanParams/`, `neutrinos/`, `yukawa/`, `qcd/`, `warpConfig/`, `solvers/`, and `tests/` returned no hits. The modern quark surface enumerates only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`, and says it is policy-only at `quarkConstraints/modern/phenomenology.py:165`. The only dipole implementation found is the lepton-sector $\mu \to e\gamma$ module at `flavorConstraints/muToEGamma.py:1`.

Implementation difficulty

HIGH. Production use would need a new $b \to s\gamma$ observable module, matching onto C_7 and C_8, QCD running in the dipole basis, normalization to the inclusive rate, and a documented treatment of the SM prediction and experimental photon-energy threshold. None of that machinery is present in the audited $\Delta F = 2$ code.

Key references

Process-local raw reference keys before bibliography consolidation: `hflav_rare_decays_dec2024`, `hflav_results_2026`, `pdg2024_b_meson_prod_decay`, `misiak_rehman_steinhauser_2020`, `misiak_et_al_2015`, `belleii_2022`, and `belle_2016`.

33 B012: Exclusive $B \to K^*\gamma$

Process

Standard notation: $B \to K^*(892)\gamma$. This entry covers the exclusive radiative $b \to s\gamma$ channels $B^0 \to K^{*0}\gamma$ and $B^+ \to K^{*+}\gamma$, with emphasis on the time-dependent $S_{K^*\gamma}$ photon-helicity observable.

PDG value

The HFLAV Dec. 2024 averages give

$$\mathcal{B}(B^0 \to K^{*0}\gamma) = (41.63 \pm 0.92) \times 10^{-6}, \qquad \mathcal{B}(B^+ \to K^{*+}\gamma) = (39.5 \pm 1.0) \times 10^{-6},$$

from `references/B012/hflav_rare_decays_dec2024_btokstargamma.txt` (`hflav_rare_decays_dec2024_btokst`
The same HFLAV snapshot gives $\Delta_{0+}(B \to K^*\gamma) = 0.059 \pm 0.014$ and $A_{CP}(B \to K^*\gamma) = -0.004 \pm 0.011$ (`hflav_rare_decays_dec2024_btokstargamma`); the HFLAV note marks the isospin averages as naive and to be treated with caution. The PDG 2025 pdgLive time-dependent average is

$$S_{K^{*0}\gamma}(B^0 \to K^{*0}\gamma) = -0.08 \pm 0.17, \qquad C_{K^{*0}\gamma} = 0.03 \pm 0.10,$$

from `references/B012/pdglive_2025_kstargamma_cp.txt` (`pdglive_2025_kstargamma_cp`).

Relevance to RS / anarchic flavor

This is the exclusive radiative complement to the inclusive $B \to X_s\gamma$ entry. The inclusive rate mostly constrains the size of the electromagnetic dipole combination, while $S_{K^*\gamma}$ is a chirality test: a large right-handed dipole coefficient C_7' can enhance the mixing-induced asymmetry even when the total rate remains near the SM value. Anarchic RS models can generate chirality-flipped dipoles through KK-fermion, KK-gauge, or Higgs-sector loops, so this process directly probes a direction not covered by the present $\Delta F = 2$ scan.

Post-2008 developments

Since the arXiv:0804.1954 flavor-baseline era (`cfw_2008_rs_flavor`), BaBar produced a high-statistics branching-fraction, direct-CP, and isospin-asymmetry measurement in 2009 (`babar_2009_btokstargamma`); Belle reported evidence for isospin violation in 2017 (`belle_2017_isospin_cp`); LHCb observed photon polarization in $b \to s\gamma$ using $B^\pm \to K^\pm\pi^\mp\pi^\pm\gamma$ (`lhcb_2014_photon_polarization`); and Belle II added $B \to K^*\gamma$ measurements now entering HFLAV (`belleii_2024_btokstargamma`). PDG 2025 also includes a new Belle/Belle II input in the $S_{K^*\gamma}$ average (`pdglive_2025_kstargamma_cp`).

Constraint validity / model dependence

The branching fractions are measured cleanly but are less theory-clean than the inclusive rate because exclusive form factors and spectator effects enter the SM comparison. The time-dependent asymmetry is a sharper null-test of photon helicity, but its present uncertainty is still statistics-limited. Translating the entry into an RS bound requires a convention-matched C_7, C_7' and C_8, C_8' pipeline plus exclusive-amplitude inputs; the measured averages alone should not be treated as a standalone likelihood.

Code coverage in this repo

NO. The required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $B \to K^*\gamma$, C_7', photon-polarization, or right-handed $b \to s\gamma$ implementation. The modern quark surface is limited to neutral-meson mixing systems at `quarkConstraints/modern/phenomenology.py:23`; the only dipole code found is the unrelated lepton-sector $\mu \to e\gamma$ check at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. Production use needs new dipole matching, RG evolution for C_7, C_7' and C_8, C_8', exclusive $B \to K^*$ form-factor handling, and a time-dependent CP/helicity observable. None of that is supplied by the current $\Delta F = 2$ operator basis.

Key references

Process-local keys before bibliography consolidation: `hflav_rare_decays_dec2024_btokstargamma`, `pdglive_2025_kstargamma_cp`, `belleii_2024_btokstargamma`, `belle_2017_isospin_cp`, `babar_2009_btokstarga`
`lhcb_2014_photon_polarization`, and `cfw_2008_rs_flavor`.

34 B015: Inclusive $B \to X_s\ell^+\ell^-$

Process

Standard notation: $\mathcal{B}(B \to X_s\ell^+\ell^-)$. This entry covers the inclusive rare electroweak-penguin process $B \to X_s\ell^+\ell^-$, equivalently inclusive $b \to s\ell^+\ell^-$, with $\ell = e, \mu$. It excludes the exclusive modes $B \to K\ell\ell$ and $B \to K^*\ell\ell$, and the LFU ratios R_K and R_{K^*}, which are separate catalog entries B016–B019.

PDG / equivalent value(s)

The canonical sidecar value is the HFLAV December 2024 inclusive average,

$$\mathcal{B}(B \to X_s\ell^+\ell^-) = (5.84 \pm 0.69) \times 10^{-6},$$

with the same table listing the PDG value $(5.84^{+1.31}_{-1.23}) \times 10^{-6}$. HFLAV combines the BaBar input $(6.73^{+0.70+0.60}_{-0.64-0.56}) \times 10^{-6}$ and the Belle input $(4.11 \pm 0.83^{+0.85}_{-0.81}) \times 10^{-6}$; the source notes different low-dilepton-mass thresholds and different charmonium treatments in `references/B015/hflav_dec2024_B_to_Xsll.tx`
For the standard perturbative bins, Huber–Hurth–Lunghi quote weighted experimental averages of $(1.58 \pm 0.37) \times 10^{-6}$ for $1 < q^2 < 6\,\mathrm{GeV}^2$ and $(0.48 \pm 0.10) \times 10^{-6}$ at high q^2, as recorded in `references/B015/huber_hurth_lunghi_1503_04849_arxiv.txt`.

Relevance to the RS / anarchic-flavor pipeline

This is a $\Delta B = 1$ semileptonic FCNC channel. It probes flavor-changing Z couplings, electroweak-penguin matching, and dipole correlations with $b \to s\gamma$, rather than the $\Delta F = 2$ four-quark operators already used by the repo's neutral-meson constraints. In RS/anarchic flavor it is most useful as a diagnostic of semileptonic Wilson coefficients such as C_7, C_9, C_{10} and their chirality-flipped partners.

Post-2008 developments

Since the CFW 2008 RS-flavor baseline, the inclusive channel has gained a BaBar measurement and a modern HFLAV average, but it remains statistics-limited and experimentally based on sum-of-exclusive reconstruction. Theory has also matured: the 2015 Huber–Hurth–Lunghi analysis includes NNLO QCD, NLO QED, power corrections, collinear-photon effects, and Belle II projections with $50\,\mathrm{ab}^{-1}$. Current inclusive branching fractions are compatible with the SM within uncertainties; the sharper exclusive and LFU anomaly discussions belong in B016–B019.

Constraint validity and model dependence

The inclusive perturbative windows are cleaner than exclusive form-factor observables, but a catalog single number is not a plug-in hard bound. The interpretation depends on q^2 binning, charm-resonance vetoes, photon treatment, and the mapping from RS flavor parameters to the $\Delta B = 1$ weak Hamiltonian. It is best classified as a theory-controlled but mode-calculation-limited rare-decay constraint.

In custodial RS, reduced Zb_Lb_L pressure can make this observable relatively more discriminating; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms.

Code coverage in this repo

NO. The modern phenomenology policy enumerates only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/ph` and states that this is policy-only rather than a numeric backend at `quarkConstraints/modern/phenomenology.py`. The only live lepton flavor routine found is $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $b \to s\ell\ell$, $X_s\ell\ell$, C_9, or C_{10} implementation.

Implementation difficulty

HIGH. Production use would require a new $\Delta B = 1$ semileptonic operator basis, matching and running for C_7, C_9, C_{10} plus primed operators, bin-aware SM predictions, charm/QED treatment, and an experimental likelihood or covariance model.

Key references

Process-local snapshots: HFLAV2024Dec:BtoXsll, BaBar2014:BtoXsll, Belle2005:BtoXsll, Huber-HurthLunghi2015:BtoXsll, and CsakiFalkowskiWeiler2008:CompositeFlavor.

35 B016: $B \to K\ell^+\ell^-$ exclusive branching fractions

Process

Standard notation: $B \to K\ell^+\ell^-$, with $\ell = e, \mu$. This entry covers the exclusive pseudoscalar-kaon $b \to s\ell^+\ell^-$ branching fractions, centered on $B^+ \to K^+\ell^+\ell^-$ and $B^0 \to K^0\ell^+\ell^-$. Lepton-universality ratios are cataloged in the companion $B018$ entry, and vector-kaon angular observables in $B017$.

PDG or equivalent value

The canonical measured values are the HFLAV rare-B averages from the end of December 2025, accessed on 2026-05-16 and snapshotted locally under `flavor_catalog/references/B016/`. For source key `HFLAV2025Dec:BplusToKplusll`, HFLAV quotes

$$\mathcal{B}(B^+ \to K^+\ell^+\ell^-) = (5.76 \pm 0.40) \times 10^{-7},$$

and for source key `HFLAV2025Dec:B0ToK0ll`,

$$\mathcal{B}(B^0 \to K^0\ell^+\ell^-) = (3.28 \pm 0.32) \times 10^{-7}.$$

Both values are recorded in `B016.yaml` under `pdg_or_equivalent`. HFLAV notes that treatments of charmonium intermediate components differ among inputs.

Relevance to RS / anarchic flavor

This is a $\Delta B = 1$ flavor-changing neutral-current probe of the same down-sector flavor structure that appears in B_s mixing, but it tests semileptonic and dipole operators rather than four-quark $\Delta F = 2$ operators. In warped or anarchic-flavor models, flavor-changing Z or KK electroweak-gauge couplings and loop dipoles can feed C_7, C_9, C_{10}, and primed semileptonic coefficients. The branching fractions are therefore useful diagnostics of $b \to s\ell\ell$ structure, while the LFU ratios and K^* angular program carry related but distinct information.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline, source key `CsakiFalkowskiWeiler2008:CompositeFlavor`, this channel moved from sparse B-factory evidence into precision rare-B phenomenology. BaBar provided early post-baseline $B \to K^{(*)}\ell^+\ell^-$ asymmetry inputs (`BaBar2009:BtoKllAsymmetries`). LHCb measured differential $B \to K^{(*)}\mu^+\mu^-$ branching fractions and isospin asymmetries with hadron-collider data (`LHCb2014:BtoKmumuDifferential`), and Belle measured both electron and muon $B \to K\ell\ell$ modes together with R_K (`Belle2021:BtoKllLFU`). HFLAV now provides current charged and neutral exclusive-K branching-fraction averages, while the separate R_K updates shifted the clean LFU-anomaly interpretation toward correlated semileptonic-Wilson-coefficient and hadronic-systematics questions.

Constraint validity / model dependence

The HFLAV branching fractions are measured observables and are appropriate for cataloging, but using them as hard RS cuts is model-dependent. A faithful constraint must specify the dilepton q^2 regions, charmonium vetoes, form-factor inputs, nonlocal charm treatment, correlations with C_7 from $b \to s\gamma$, and correlations with R_K and $B \to K^*\ell\ell$. The entry is therefore a semileptonic $\Delta B = 1$ constraint seed, not a drop-in replacement for a global $b \to s\ell\ell$ fit.

Code coverage in this repo

NO. The modern phenomenology surface lists only ϵ_K, K, B_d, B_s, and D^0 systems at `quarkConstraints/modern/p`
The only live lepton-flavor module is $\mu \to e\gamma$, implemented at `flavorConstraints/muToEGamma.py:75`.
Required greps over the implementation directories found no $B \to K\ell\ell$, $b \to s\ell\ell$, C_9, C_{10}, or R_K backend.

Implementation difficulty

HIGH. Integration would require a new $\Delta B = 1$ semileptonic Hamiltonian, matching to C_7, C_9, C_{10}, and primed operators, q^2-bin handling, form factors, nonlocal-charm uncertainties, and experimental covariance support. The existing $\Delta F = 2$ basis is not enough.

Key references

Process-local snapshots: `HFLAV2025Dec:BplusToKplusll`, `HFLAV2025Dec:B0ToK0ll`, `Belle2021:BtoKllLFU`, `LHCb2014:BtoKmumuDifferential`, `BaBar2009:BtoKllAsymmetries`, and `CsakiFalkowskiWeiler2008:Composit`

36 B017: $B^0 \to K^{*0}\ell^+\ell^-$ and linked $b \to s\ell^+\ell^-$ decays

Process

Standard notation: $B^0 \to K^*(892)^0\ell^+\ell^-$, with linked $b \to s\ell^+\ell^-$ observables. The plan row $B017$ is the exclusive $B \to K^*\ell^+\ell^-$ channel. The PKA sidecar also preserves nearby inclusive, exclusive-K, and LFU-ratio context from rows $B015$, $B016$, $B018$, and $B019$. This TeX entry foregrounds the $K^*\ell\ell$ channel and leaves the final split of those linked rows to CA/orchestrator review.

PDG or equivalent value

The YAML sidecar is the numerical source of truth. In HFLAV's December 2025 rare-decay tables, the exclusive lepton-summed K^* average is

$$\mathcal{B}(B^0 \to K^*(892)^0\ell^+\ell^-) = (9.9 \pm 1.2) \times 10^{-7},$$

under source key `HFLAV2025Dec:B0ToKst0ll`. The same sidecar records linked HFLAV averages $\mathcal{B}(B \to X_s\ell^+\ell^-) = (5.84 \pm 0.69) \times 10^{-6}$ and $\mathcal{B}(B^+ \to K^+\ell^+\ell^-) = (5.76 \pm 0.40) \times 10^{-7}$ under `HFLAV2025Dec:BtoXsll` and `HFLAV2025Dec:BplusToKplusll`. For the K^* LFU ratio, the LHCb Run-1/Run-2 analysis (`LHCb2023:RKstar`) reports

$$R_{K^*} = 0.927^{+0.093+0.036}_{-0.087-0.035} \quad (0.1 < q^2 < 1.1\,\mathrm{GeV}^2/c^4),$$

and

$$R_{K^*} = 1.027^{+0.072+0.027}_{-0.068-0.026} \quad (1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4).$$

Relevance to the RS / anarchic-flavor pipeline

This is a $\Delta B = 1$ FCNC probe of the same flavor structure that drives the repo's neutral-meson constraints, but it tests semileptonic operators rather than $\Delta F = 2$ four-quark operators. KK electroweak gauge exchange, flavor-changing Z couplings, and loop-induced dipoles can feed C_9, C_{10}, C_9', and C_{10}', with C_7 correlations from the radiative $b \to s\gamma$ lane. Non-universal lepton localization would show up most cleanly in R_K and R_{K^*}, while the branching fractions and angular observables probe a broader combination of short-distance coefficients and hadronic inputs.

Post-2008 developments

Relative to the Csaki–Falkowski–Weiler 2008 RS-flavor baseline, this channel has become data-rich. HFLAV now maintains current rare-B averages through the end of 2025. LHCb's 2023 simultaneous R_K/R_{K^*} update superseded the earlier LHCb LFU results and moved the clean LFU ratios into agreement with the Standard Model. The $B^0 \to K^{*0}\mu^+\mu^-$ angular program remains important: LHCb's 2020 CP-averaged analysis states that the earlier tension with SM predictions persists, with significance depending on theory nuisance choices. Post-update global fits therefore no longer look like a simple LFU violation story; they are mainly a correlated C_9-type and hadronic-systematics problem.

Constraint validity and model dependence

The LFU ratios are comparatively clean null tests. Branching fractions and angular observables are more model- and theory-limited because they require form factors, nonlocal charm treatment, bin definitions, and correlated experimental likelihoods. A catalog value is useful, but using this channel as a hard RS bound requires a global-fit interface rather than a single-number cut.

In custodial RS, reduced Zb_Lb_L pressure can make semileptonic $b \to s$ channels relatively more discriminating; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms. Use R_K/R_{K^*} as precision LFU null tests; do not impose the pre-2023 anomaly narrative as a prior.

Code coverage in this repo

NO. The modern phenomenology surface lists only ϵ_K, K, B_d, B_s, and D^0 systems at `quarkConstraints/modern/p`
The lepton flavor module implements $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`. Re-

quired plan greps plus targeted searches for C_9, C_{10}, R_K, X_s, K^*, and P_5' found no live $b \to s\ell^+\ell^-$ backend.

Implementation difficulty

HIGH. A production implementation would need a new $\Delta B = 1$ semileptonic Hamiltonian, matching onto C_7, C_9, C_{10} and primed operators, RG conventions, q^2-bin-aware SM predictions, form factors, nonlocal charm uncertainties, and experimental covariance handling.

Key references

Process-local snapshots: HFLAV2025Dec:BtoXsll, HFLAV2025Dec:BplusToKplusll, HFLAV2025Dec:B0ToKst0ll, LHCb2023:RKstar, LHCb2020:KstAngular, WenXu2023:BsllGlobalFit, and CsakiFalkowskiWeiler2008:Composite

37 B018: R_K in $B^+ \to K^+\ell^+\ell^-$

Process

Standard notation: $R_K \equiv \mathcal{B}(B^+ \to K^+\mu^+\mu^-)/\mathcal{B}(B^+ \to K^+e^+e^-)$. This beauty-sector entry covers the charged $B \to K\ell\ell$ lepton-flavor universality ratio, not the exclusive K^* counterpart R_{K^*} or the absolute $B \to K\ell\ell$ branching fractions.

PDG or equivalent value

The current HFLAV-equivalent source is the Dec. 2025 rare-decay table. For the low-q^2 bin $0.1 < q^2 < 1.1\,\mathrm{GeV}^2/c^4$, HFLAV lists

$$R_K = 0.994 \pm 0.090,$$

under `HFLAV2025Dec:RKBplusKLowQ2`, from LHCb's $0.994^{+0.090+0.029}_{-0.082-0.027}$ measurement. For the central bin $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$, HFLAV quotes

$$R_K = 0.947 \pm 0.047,$$

under `HFLAV2025Dec:RKBplusKCentralQ2`, combining LHCb $0.949^{+0.042}_{-0.041} \pm 0.022$ with CMS $0.78^{+0.46+0.09}_{-0.23-0.05}$. HFLAV also records a Belle-only full-q^2 ratio 1.08 ± 0.16 under `HFLAV2025Dec:RKBplusKFullQ2Belle`. Local snapshots are `references/B018/hflav_dec2025_rk_lowq2.txt`, `references/B018/hflav_dec2025_rk_cent` and `references/B018/hflav_dec2025_rk_fullq2_belle.txt`, accessed 2026-05-16.

Relevance to the RS / anarchic-flavor pipeline

R_K is a clean $\Delta B = 1$ semileptonic FCNC ratio. In an RS or anarchic-flavor interpretation it probes nonuniversal contributions to the $b \to s\ell^+\ell^-$ weak Hamiltonian, especially flavor-changing Z or KK-gauge effects and semileptonic Wilson coefficients C_9^ℓ and C_{10}^ℓ. It is therefore complementary to the repo's neutral-meson $\Delta F = 2$ constraints.

Post-2008 developments

The CFW 2008 baseline predates the modern R_K anomaly program. The key post-2008 developments are experimental: LHCb's 2021 update measured $R_K = 0.846^{+0.042+0.013}_{-0.039-0.012}$ in the central bin and reported a 3.1σ tension with the SM (`LHCb2021:UpdatedRK`). The later LHCb simultaneous R_K/R_{K^*} analysis using $9\,\mathrm{fb}^{-1}$ superseded that result and found the four LFU measurements compatible with the SM (`LHCb2023:RKRKstarDetailed`). HFLAV Dec. 2025 therefore treats the charged R_K bins as SM-consistent averages. On the theory side, Bordone–Isidori–Pattori established the ratio as a clean probe once radiative corrections and experimental cuts are handled consistently.

Constraint validity and model dependence

This is a robust LFU null test, but not a standalone bound on the current quark-scan model. It requires a $\Delta B = 1$ semileptonic matching calculation, bin definitions, QED/radiative treatment, and experimental correlations. A universal shift in electron and muon coefficients cancels strongly in the ratio, while lepton-nonuniversal new physics can be visible.

In custodial RS, reduced Zb_Lb_L pressure can make R_K relatively more discriminating as an LFU cross-check; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms. Use R_K/R_{K^*} as precision LFU null tests; do not impose the pre-2023 anomaly narrative as a prior.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no R_K, $b \to s\ell\ell$, C_9, C_{10}, or LFU implementation. The implemented flavor surface is neutral-meson mixing: `quarkConstraints/modern/phenomenology.` enumerates only ϵ_K, K, B_d, B_s, and D^0, and `quarkConstraints/deltaf2.py:209` defines the legacy $\Delta F = 2$ inputs. The only live lepton-flavor routine found is `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. Production use would require a new $\Delta B = 1$ semileptonic operator basis, matching and running for lepton-flavor-dependent C_9, C_{10}, and primed coefficients, SM bin predictions including radiative corrections, and likelihood handling for HFLAV/LHCb/CMS inputs.

Key references

Process-local snapshots: HFLAV2025Dec:RKBplusKLowQ2, HFLAV2025Dec:RKBplusKCentralQ2, HFLAV2025Dec:RKBplusKFullQ2Belle, LHCb2023:RKRKstarDetailed, LHCb2021:UpdatedRK, BordoneIsidoriPattori2016:RKRKstarSM, and CsakiFalkowskiWeiler2008:CompositeFlavor.

38 B019: R_{K^*} in $B^0 \to K^{*0}\ell^+\ell^-$

Process

Standard notation: $R_{K^*} \equiv \mathcal{B}(B^0 \to K^{*0}\mu^+\mu^-)/\mathcal{B}(B^0 \to K^{*0}e^+e^-)$, with K^{*0} the neutral vector kaon. This beauty-sector entry covers the neutral exclusive lepton-universality ratio; R_K is B018 and absolute $B \to K^*\ell\ell$ rates and angular observables are B017.

PDG or equivalent value

The current HFLAV-equivalent Dec. 2025 neutral-K^{*0} entries give

$$R_{K^*} = 0.927 \pm 0.097, \qquad 0.1 < q^2 < 1.1\ \mathrm{GeV}^2/c^4,$$

from the LHCb-only Run-1+Run-2 input $0.927^{+0.093+0.036}_{-0.087-0.035}$ (`HFLAV2025Dec:RKstarLowQ2LHCb`, `LHCb2023:RKRKstar`
In the central bin,

$$R_{K^*} = 1.028 \pm 0.074, \qquad 1.1 < q^2 < 6.0\ \mathrm{GeV}^2/c^4,$$

HFLAV combines the LHCb value $1.027^{+0.072+0.027}_{-0.068-0.026}$ with Belle $1.06^{+0.63}_{-0.38}\pm 0.14$ (`HFLAV2025Dec:RKstarCentralQ2`). Local snapshots are `references/B019/hflav_dec2025_rkstar_lowq2_lhcb.txt` and `references/B019/hflav_dec`
accessed 2026-05-16.

Relevance to RS / anarchic flavor

R_{K^*} probes $\Delta B = 1$ semileptonic FCNC amplitudes in $b \to s\ell^+\ell^-$. In RS/anarchic-flavor language it is sensitive to nonuniversal lepton couplings and flavor-changing b-s currents from KK gauge, Z, or other semileptonic effects. It is therefore complementary to the repo's existing neutral-meson $\Delta F = 2$ constraints and to B018's pseudoscalar R_K ratio.

Post-2008 developments

The CFW 2008 baseline (`CsakiFalkowskiWeiler2008:CompositeFlavor`) predates the R_{K^*} anomaly program. The key post-2008 milestone was the LHCb 2017 measurement (`LHCb2017:RKstar`): $R_{K^*} = 0.66^{+0.11}_{-0.07}\pm 0.03$ for $0.045 < q^2 < 1.1\ \mathrm{GeV}^2/c^4$ and $0.69^{+0.11}_{-0.07}\pm 0.05$ for $1.1 < q^2 < 6.0\ \mathrm{GeV}^2/c^4$, corresponding to roughly 2.1–2.5σ tensions depending on bin and SM prediction. The LHCb 2023 simultaneous R_K/R_{K^*} analysis superseded that result; its R_{K^*} central values moved upward and the four LFU measurements have combined SM compatibility of 0.2σ (`LHCb2023:RKRKstarDetailed`). Theory work by Bordone–Isidori–Pattori clarified that the central bin is SM-unity to about a 0.01 QED uncertainty, while low-q^2 R_{K^*} needs explicit photon-pole and radiative treatment (`BordoneIsidoriPattori2C`

Constraint validity / model dependence

This is a clean LFU null test, but it is not a drop-in bound for the current quark scan. The ratio cancels many hadronic uncertainties, yet a usable constraint still depends on bin definitions, electron bremsstrahlung, radiative corrections, correlations, and a model prediction for lepton-specific C_9^ℓ, C_{10}^ℓ, and primed semileptonic coefficients.

In custodial RS, reduced Zb_Lb_L pressure can make R_{K^*} relatively more discriminating as an LFU cross-check; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms. Use R_K/R_{K^*} as precision LFU null tests; do not impose the pre-2023 anomaly narrative as a prior.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no R_{K^*}, $K^{*0}ee$, $K^{*0}\mu\mu$, C_9, C_{10}, LFU, or $b \to s\ell\ell$ implementation. The modern phenomenology surface enumerates only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`, and it is policy-only at `quarkConstraints/modern/phe` The legacy inputs are $\Delta F = 2$ at `quarkConstraints/deltaf2.py:209`, with B_d and B_s mixing evaluators at `quarkConstraints/deltaf2.py:903` and `quarkConstraints/deltaf2.py:922`. The only live lepton-flavor routine found is $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. Integration requires a new $\Delta B = 1$ semileptonic Hamiltonian, lepton-specific C_9/C_{10} and primed coefficients, bin-aware SM/QED predictions, and HFLAV/LHCb likelihood or covariance handling.

Key references

Process-local reference keys before bibliography consolidation: `HFLAV2025Dec:RKstarLowQ2LHCb`, `HFLAV2025Dec:RKstarCentralQ2`, `LHCb2023:RKRKstarDetailed`, `LHCb2017:RKstar`, `BordoneIsidoriPattori20` and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

39 B021: $\Lambda_b^0 \to \Lambda\ell^+\ell^-$

Process

Standard notation: $\Lambda_b^0 \to \Lambda\ell^+\ell^-$. This is the baryonic exclusive $b \to s\ell^+\ell^-$ channel. The current canonical measured mode is $\Lambda_b^0 \to \Lambda\mu^+\mu^-$; no comparable PDG electron-mode average was found in this PKA pass.

PDG or equivalent value

The sidecar is the numerical source of truth. PDG Live's 2025 Λ_b^0 listing gives

$$\mathcal{B}(\Lambda_b^0 \to \Lambda\mu^+\mu^-) = (1.08 \pm 0.28) \times 10^{-6},$$

from the decay-mode table under source key `PDG2025:LambdaBToLambdaMuMu`. The LHCb angular analysis under `LHCb2015:LambdaBToLambdaMuMu` reports, for $15 < q^2 < 20\,\mathrm{GeV}^2/c^4$,

$$\frac{d\mathcal{B}}{dq^2} = (1.18^{+0.09}_{-0.08} \pm 0.03 \pm 0.27) \times 10^{-7}(\mathrm{GeV}^2/c^4)^{-1},$$

and $A_{\mathrm{FB}}^{\ell} = -0.05\pm0.09\pm0.03$, $A_{\mathrm{FB}}^{h} = -0.29\pm0.07\pm0.03$. Local snapshots are in `flavor_catalog/references/B02`

Relevance to RS / anarchic flavor

This mode probes $\Delta B = 1$ semileptonic FCNC operators rather than the $\Delta F = 2$ four-quark basis implemented for neutral meson mixing. Warped flavor models can generate flavor-changing Z couplings, KK electroweak exchange, and loop-induced dipoles feeding C_7, C_9, C_{10}, and primed semileptonic coefficients. The baryon final state adds spin and polarization information unavailable in $B \to K^{(*)}\ell^+\ell^-$, so it is a complementary diagnostic of chirality and hadronic systematics.

Post-2008 developments

The arXiv:0804.1954 RS-flavor baseline predates this channel as an observed rare decay (`CsakiFalkowskiWeiler20`
CDF reported the first observation in 2011 with 24 signal events, 5.8 standard-deviation significance, and $\mathcal{B} = (1.73\pm0.42\pm0.55)\times10^{-6}$ (`CDF2011:LambdaBToLambdaMuMu`). LHCb later used $3.0\,\mathrm{fb}^{-1}$ to publish the q^2-binned branching-fraction and angular analysis quoted above (`LHCb2015:LambdaBToLambdaMuMu`). Lattice-QCD work by Detmold, Lin, Meinel, and Wingate supplied post-2008 $\Lambda_b \to \Lambda$ form factors, so this baryonic mode should not be folded into the mesonic $B \to K^{(*)}\ell\ell$ treatment (`DetmoldLinMeinelWingate2013:LambdaBToLambdaFormFactors`).

Constraint validity / model dependence

This is a useful catalog constraint but not a single-number hard bound. The interpretation depends on q^2 binning, charmonium vetoes, normalization to $\Lambda_b^0 \to J/\psi\Lambda$, baryonic form factors, and nonlocal charm effects. Its most robust role is as a $\Delta B = 1$ semileptonic cross-check with different hadronic systematics from mesonic rare B decays.

Code coverage in this repo

NO. The modern phenomenology surface lists only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenome`
The only live lepton-flavor module is the $\mu \to e\gamma$ dipole check at `flavorConstraints/muToEGamma.py:75`. Required plan greps and targeted searches for Λ_b, $b \to s$, C_9, C_{10}, and semileptonic rare decays found no B021 implementation.

Implementation difficulty

HIGH. A production constraint would need a new $\Delta B = 1$ semileptonic Hamiltonian, RS matching onto C_7, C_9, C_{10} and primed operators, baryonic form factors, q^2-bin predictions, angular likelihoods, and experimental covariance/normalization treatment.

Key references

PDG2025:LambdaBToLambdaMuMu, LHCb2015:LambdaBToLambdaMuMu, CDF2011:LambdaBToLambdaMuMu, DetmoldLinMeinelWingate2013:LambdaBToLambdaFormFactors, and CsakiFalkowskiWeiler2008:CompositeFl

40 B022: $B^+ \to K^+\nu\bar{\nu}$

Process

Standard notation: $\mathcal{B}(B^+ \to K^+\nu\bar{\nu})$. This beauty-sector entry covers the leading exclusive invisible $b \to s\nu\bar{\nu}$ mode, with emphasis on the charged channel now measured by Belle II. The related vector mode $B \to K^*\nu\bar{\nu}$ is B023.

PDG or equivalent value

The current HFLAV-equivalent Dec. 2025 table gives

$$\mathcal{B}(B^+ \to K^+\nu\bar{\nu}) = (1.38 \pm 0.35) \times 10^{-5},$$

with local snapshot flavor_catalog/references/B022/hflav_dec2025_bplus_kplus_nunu.txt accessed 2026-05-16 (HFLAV2025Dec:BplusKplusNuNu). The same HFLAV table displays the PDG value $(2.30^{+0.71}_{-0.64}) \times 10^{-5}$, and the PDG live/API listing records the Belle II evidence input $(2.3 \pm 0.5_{\text{stat}}{}^{+0.5}_{-0.4\text{syst}}) \times 10^{-5}$ (PDG2026:BplusKplusNuNu, BelleII2024:BplusKplusNuNu). Belle II used 362 fb^{-1} at the $\Upsilon(4S)$, reported 3.5σ evidence for the decay, and found the combined result 2.7σ above the Standard Model expectation (BelleII2024:BplusKplusNuNu). For theory orientation, the HPQCD 2023 SM prediction is $(5.58\pm0.37)\times10^{-6}$, summed over neutrino flavours and including the charged-mode long-distance term (HPQCD2023:BtoKNuNuSM).

Relevance to RS / anarchic flavor

This mode probes a $\Delta B = 1$ neutral-current transition with missing energy. In RS/anarchic-flavor models it can receive contributions from flavor-changing b-s currents coupled to Z, KK electroweak gauge bosons, or other neutral invisible states. It is complementary to $\Delta F = 2$ B_s mixing because it tests a semileptonic/invisible operator, not a neutral-meson mixing amplitude.

Post-2008 developments

The CFW 2008 RS-flavor baseline predates the modern $B \to K\nu\bar{\nu}$ program (CsakiFalkowskiWeiler2008:Composi BaBar later set hadronic-tag limits, including $\mathcal{B}(B^+ \to K^+\nu\bar{\nu}) < 3.7\times10^{-5}$ at 90% CL (BaBar2013:BKstarNuNu). Belle II changed the status of the mode with the 2024 evidence measurement quoted above (BelleII2024:BplusKpl On the theory side, modern lattice form factors support the SM prediction quoted above (HPQCD2023:BtoKNuNuSM), and Belle II has released a 2025 model-agnostic likelihood package to reinterpret the measurement under non-SM invisible kinematics or WET coefficient choices (BelleII2025:BKnnLikelihood).

Constraint validity / model dependence

The SM prediction is comparatively clean: there are no charmonium veto regions as in $b \to s\ell^+\ell^-$, and the dominant hadronic input is the $B \to K$ vector form factor. The experimental extraction is less universal: Belle II's headline branching fraction assumes SM kinematics, while invisible new particles can change the q^2 spectrum and selection efficiency. A live constraint should therefore use either the published likelihood or a validated reweighting prescription rather than only a one-bin branching-fraction pull.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $B \to K\nu\bar{\nu}$, $b \to s\nu\bar{\nu}$, or rare invisible decay implementation. The modern phenomenology surface lists only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`; its note says this is policy-only at `quarkConstraints/modern/phenomenology.py:166`. The legacy $\Delta F = 2$ inputs are at `quarkConstraints/delta` with B_d and B_s mixing evaluators at `quarkConstraints/deltaf2.py:903` and `quarkConstraints/deltaf2.py:92` The only live lepton-flavor routine seen is $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. This would require a new $\Delta B = 1$ $b \to s\nu\bar{\nu}$ Hamiltonian, exclusive $B \to K$ form-factor integration, invisible-final-state acceptance treatment, and Belle II likelihood or kinematic-reweighting support. The existing neutral-meson $\Delta F = 2$ SLL/SLR/VLL/VRR/LR machinery does not cover this observable.

Key references

`HFLAV2025Dec:BplusKplusNuNu`, `PDG2026:BplusKplusNuNu`, `BelleII2024:BplusKplusNuNu`, `BelleII2025:BKnn` `BaBar2013:BKstarNuNu`, `HPQCD2023:BtoKNuNuSM`, and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

41 B023: $B \to K^*\nu\bar{\nu}$

Process

Standard notation: $\mathcal{B}(B \to K^*\nu\bar{\nu})$. This beauty-sector entry covers the invisible vector-kaon modes $B^0 \to K^{*0}(892)\nu\bar{\nu}$ and $B^+ \to K^{*+}(892)\nu\bar{\nu}$. The pseudoscalar companion $B \to K\nu\bar{\nu}$ is B022.

PDG or equivalent value

The canonical current PDG/HFLAV values are upper limits. HFLAV Dec. 2025, cross-linked to PDG Live 2025, lists

$$\mathcal{B}(B^0 \to K^{*0}(892)\nu\bar{\nu}) < 1.8 \times 10^{-5}, \qquad 90\%\ \text{CL},$$

and

$$\mathcal{B}(B^+ \to K^{*+}(892)\nu\bar{\nu}) < 4.0 \times 10^{-5}, \qquad 90\%\ \text{CL}.$$

The neutral limit is from Belle's semileptonic-tag search (`Belle2017:BHnunuSemileptonic`); the charged limit is from Belle's full-sample hadronic-tag search (`Belle2013:BHnunuHadronic`). BaBar's 2013 hadronic-tag result is superseded but remains part of the HFLAV input history (`BaBar2013:BKstarNunu`). Local snapshots are under `references/B023/`, accessed 2026-05-16.

Relevance to RS / anarchic flavor

These modes probe the $\Delta B = 1$ FCNC transition $b \to s\nu\bar{\nu}$. In RS/anarchic-flavor models they are sensitive to flavor-changing Z, heavy electroweak-gauge, or other neutral-current couplings generated after fermion localization and mass-basis rotation. They are complementary to B_s mixing because the final state tests a semileptonic neutral current, not a $\Delta F = 2$ four-quark amplitude, and the K^* final state can also separate left- and right-handed short-distance structures through polarization.

Post-2008 developments

The CFW 2008 baseline predates all current $B \to K^*\nu\bar{\nu}$ limits. (`CsakiFalkowskiWeiler2008:CompositeFlavor`). Belle's 2013 full-$\Upsilon(4S)$ hadronic-tag analysis set the charged-mode limit $B^+ \to K^{*+}\nu\bar{\nu} < 4.0\times10^{-5}$ (`Belle2013:BHnunuHadronic`). BaBar's 2013 hadronic-tag result found no significant excess and remains useful as superseded input history (`BaBar2013:BKstarNunu`). Belle's 2017 semileptonic-tag search improved the neutral mode to $< 1.8 \times 10^{-5}$ and quoted a combined vector limit of $< 2.7 \times 10^{-5}$ (`Belle2017:BHnunuSemileptonic`). Buras–Girrbach-Noe–Niehoff–Straub 2015 updated the SM treatment using form factors and electroweak corrections, quoting $\mathcal{B}(B^0 \to K^{*0}\nu\bar{\nu}) = (9.2\pm1.0)\times10^{-6}$, and framed the right-handed-current observables used in later new-physics studies (`BurasGirrbachNoeNiehoffStraub2015:BKstarNunu`).

Constraint validity / model dependence

Experimentally this is a clean missing-energy upper limit, but it is not a drop-in bound for the present quark scan. A production constraint needs the $b \to s\nu\bar{\nu}$ weak Hamiltonian, $B \to K^*$ form factors, treatment of right-handed currents and polarization, and either an experimental likelihood or a conservative reinterpretation of the published upper limits.

Code coverage in this repo

NO. Targeted greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $B \to K^* \nu\bar{\nu}$, $b \to s\nu\bar{\nu}$, or invisible-B implementation. The modern phenomenology surface is limited to ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`. The legacy default inputs are $\Delta F = 2$ at `quarkConstraints/deltaf2.py:209`, with B_d and B_s mixing evaluators at `quarkConstraints/deltaf2.py:903` and `quarkConstraints/deltaf2.py:922`. The only live lepton-flavor routine found is $\mu \to e\gamma$ at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. Implementing B023 requires a new $\Delta B = 1$ $b \to s\nu\bar{\nu}$ operator basis, RS matching for neutral-current left- and right-handed coefficients, $B \to K^*$ form-factor and polarization treatment, and an upper-limit likelihood or validated CL reinterpretation.

Key references

`PDG2025:B0KstarNunu`, `PDG2025:BpKstarNunu`, `HFLAV2025Dec:B0KstarNunu`, `HFLAV2025Dec:BpKstarNunu`, `Belle2017:BHnunuSemileptonic`, `Belle2013:BHnunuHadronic`, `BaBar2013:BKstarNunu`, `BurasGirrbachNoeNiel` and `CsakiFalkowskiWeiler2008:CompositeFlavor`.

42 B025: R_D in $B \to D\tau\nu$

Process

Standard notation: $R_D \equiv \mathcal{B}(B \to D\tau\nu_\tau)/\mathcal{B}(B \to D\ell\nu_\ell)$, with $\ell = e, \mu$. This beauty-family entry covers the tau-versus-light-lepton charged-current LFU ratio in exclusive $b \to c\ell\nu$ semileptonic decays.

PDG or equivalent value

The canonical current equivalent value is the HFLAV CKM 2025 joint average, updated on 2025-09-28 and snapshotted in `flavor_catalog/references/B025/hflav_ckm2025_rdrds.txt`:

$$R_D = 0.358 \pm 0.024.$$

The source key is `hflav_ckm2025_rdrds`. The same fit gives $R_{D^*} = 0.281 \pm 0.011$, correlation $\rho(R_D, R_{D^*}) = -0.374$, and $\chi^2/\text{dof} = 16.683/14$ with confidence level 0.27. For comparison, HFLAV quotes an arithmetic SM reference $R_D = 0.296 \pm 0.004$, $R_{D^*} = 0.254 \pm 0.005$. Against that reference the combined R_D-R_{D^*} discrepancy is quoted as about 3.8σ, with $p = 1.48 \times 10^{-4}$. The sidecar records the individual R_D inputs from BaBar, Belle, LHCb, and Belle II.

Relevance to the RS / anarchic-flavor pipeline

R_D probes $b \to c\tau\nu$ charged currents rather than neutral $\Delta F = 2$ mixing. It is therefore not a direct constraint on the current quark-scan lane, but it is important for any warped/anarchic extension with non-universal charged currents, third-generation enhanced lepton couplings, charged scalar exchange, W'-like vectors, or leptoquark-motivated matching. The ratio cancels many normalization inputs, so a persistent excess is a clean diagnostic of LFU violation once the model supplies the relevant semileptonic Wilson coefficients and neutrino-flavor assumptions.

Post-2008 developments

The Csaki–Falkowski–Weiler 2008 baseline predates the $R_D^{(*)}$ anomaly program. The key post-2008 shift was BaBar's 2012 and 2013 evidence for an excess in $B \to D^{(*)}\tau\nu$, followed by Belle hadronic-tag and semileptonic-tag measurements, LHCb measurements with muonic tau decays, and Belle II semileptonic-tag and CKM-2025 hadronic-tag inputs. HFLAV now uses a joint correlated average rather than a single-experiment comparison. On the SM side, lattice and form-factor work matured substantially: HFLAV references FNAL/MILC, HPQCD, JLQCD, FLAG 2024, and later phenomenological combinations, so the tension is now limited by correlated experimental averaging and form factor conventions rather than by the absence of a basic SM prediction.

Constraint validity and model dependence

The experimental ratio is robust, but interpreting it as an RS bound is model-dependent. A live constraint needs the operator basis for $(\bar{c}\,\Gamma b)(\bar{\tau}\,\Gamma\nu)$, the assumed neutrino species, and the integrated $B \to D$ form-factor response to vector, scalar, and tensor operators. Because HFLAV fits R_D together with R_{D^*}, a serious implementation should use the two-dimensional likelihood or covariance rather than a standalone Gaussian for R_D.

Code coverage in this repo

NO. The PKA reran the required coverage greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The broad plan greps recover the existing neutral-meson $\Delta F = 2$ and $\mu \to e\gamma$ surfaces, but no R_D, $B \to D\tau\nu$, $b \to c$ charged-current, charged-Higgs, or leptoquark observable implementation.

Implementation difficulty

HIGH. Promotion to a live constraint requires a new semileptonic charged-current observable, form-factor/mode integration for at least the $B \to D$ final state, and a correlated treatment with R_{D^*}. This belongs in the modern constraint surface if it is later implemented.

Key references

Process-local snapshots before bibliography consolidation: `hflav_ckm2025_rdrds.txt`, `hflav_ckm2025_logfile.t` `babar2012_arxiv1205_5442.txt`, `babar2013_arxiv1303_0571.txt`, `belle2015_arxiv1507_03233.txt`, `belle2019_arxiv1910_05864.txt`, `lhcb2023_arxiv2302_02886.txt`, `lhcb2025_arxiv2406_03387.txt`, `belleii2025_arxiv2504_11220.txt`, `flag2024_arxiv2411_04268.txt`, and `cfw2008_arxiv0804_1954.txt`.

43 B026: R_{D^*} in $B \to D^* \tau\nu$

Process

Standard notation: $R_{D^*} \equiv \mathcal{B}(B \to D^*\tau\nu_\tau)/\mathcal{B}(B \to D^*\ell\nu_\ell)$, with $\ell = e, \mu$. This beauty-family entry covers charged-current lepton-flavor-universality tests in exclusive $b \to c\ell\nu$ transitions with a vector charm meson in the final state.

PDG or equivalent value

The canonical current equivalent value is the HFLAV CKM 2025 joint R_D-R_{D^*} average, updated on 2025-09-28 and snapshotted in `flavor_catalog/references/B026/hflav_ckm2025_rdrds.txt` with source key `hflav_ckm2025_rdrds`:

$$R_{D^*} = 0.281 \pm 0.011.$$

The same HFLAV table gives $R_D = 0.358 \pm 0.024$, correlation $\rho(R_D, R_{D^*}) = -0.374$, and $\chi^2/\text{dof} = 16.683/14$ for the experimental combination. HFLAV's displayed arithmetic SM reference is $R_D = 0.296 \pm 0.004$ and $R_{D^*} = 0.254 \pm 0.005$. With the quoted covariance, HFLAV reports the current combined R_D-R_{D^*} discrepancy as 3.8σ ($p = 1.48 \times 10^{-4}$); the R_{D^*}-only pull against that reference is 2.3σ. All numerical values in this paragraph use source key `hflav_ckm2025_rdrds`.

Relevance to RS / anarchic flavor

R_{D^*} is not a neutral-meson $\Delta F = 2$ bound and does not constrain the current quark-scan backend directly. It becomes relevant in warped or anarchic extensions with non-universal charged currents, third-generation enhanced lepton couplings, charged-scalar exchange, W'-like vectors, or leptoquark-motivated matching. Because the observable is a ratio, many CKM and normalization uncertainties cancel, making it a sensitive diagnostic once a model predicts the $b \to c\tau\nu$ Wilson coefficients.

Post-2008 developments

The 2008 CFW/Perez-Randall-era baseline (`cfw2008_arxiv0804_1954`) predates the $R_D^{(*)}$ anomaly program. The post-2008 experimental sequence starts with BaBar's 2012 and 2013 excess in $B \to$

$D^{(*)}\tau\nu$ (babar2012_arxiv1205_5442; babar2013_arxiv1303_0571), followed by Belle hadronic-tag, tau-polarization, and semileptonic-tag measurements, LHCb muonic-tau and three-prong-tau measurements, and Belle II inputs recorded in the HFLAV CKM 2025 average. HFLAV now performs a correlated average rather than comparing one experiment to one SM number (hflav_ckm2025_rdrds; hflav_ckm2025_logfile). On the SM side, $B \to D^*\ell\nu$ form-factor theory changed materially after 2008: HFLAV cites benchmark, FNAL/MILC, HPQCD, JLQCD, and FLAG inputs, with the local FLAG snapshot stored as flag2024_arxiv2411_04268. The residual tension is therefore tied to correlated experimental averaging and form-factor conventions, not to a missing basic SM prediction.

Constraint validity / model dependence

The HFLAV average is a robust experimental summary, but interpreting it as an RS constraint is highly model-dependent. A live likelihood needs the $(\bar{c}\,\Gamma b)(\bar{\tau}\,\Gamma\nu)$ charged-current operator basis, neutrino-flavor assumptions, and the integrated $B \to D^*$ response to vector, scalar, pseudoscalar, and tensor operators. A standalone Gaussian for R_{D^*} would discard the documented correlation with R_D; any serious implementation should use the two-dimensional HFLAV covariance or a dedicated likelihood.

Code coverage in this repo

NO. The PKA reran the required greps over quarkConstraints/, qcd/, flavorConstraints/, neutrinos/, yukawa/, warpConfig/, solvers/, scanParams/, and tests/. The broad searches recover existing neutral-meson $\Delta F = 2$ and $\mu \to e\gamma$ code, but targeted searches for $R(D^*)$, $D^*\tau$, $B \to D^*$, $b \to c$, charged-current, charged-Higgs, and leptoquark observable code returned no implementation hits.

Implementation difficulty

HIGH. Promotion to a live constraint requires new charged-current semileptonic matching, $B \to D^*$ form-factor integration, and a correlated R_D-R_{D^*} likelihood. This is beyond the existing $\Delta F = 2$ and $\mu \to e\gamma$ surfaces and should target quarkConstraints/modern/ only if the PI later approves implementation.

Key references

Process-local reference keys before bibliography consolidation: hflav_ckm2025_rdrds, hflav_ckm2025_logfile, babar2012_arxiv1205_5442, babar2013_arxiv1303_0571, belle2015_arxiv1507_03233, belle2017_arxiv1612_0 belle2018_arxiv1709_00129, belle2019_arxiv1910_05864, lhcb2023_arxiv2302_02886, lhcb2023_arxiv2305_0 lhcb2025_arxiv2406_03387, belleii2024_arxiv2401_02840, belleii2025_arxiv2504_11220, flag2024_arxiv24 and cfw2008_arxiv0804_1954.

44 B032: $\bar{B} \to \pi\bar{K}$

Process

Standard notation: $\bar{B} \to \pi\bar{K}$, equivalently the four two-body charmless $B \to K\pi$ modes with charge conjugates implied. This entry covers branching fractions, direct CP asymmetries, and the time-dependent $C_{K^0\pi^0}$ and $S_{K^0\pi^0}$ coefficients.

PDG or equivalent value

The canonical branching-fraction source is the HFLAV End-of-December 2025 rare/charmless update, locally snapshot in `references/B032/hflav_dec2025_btopik_values.txt` under source key `hflav_dec2025_btopik`. In HFLAV units of 10^{-5},

mode	$\mathcal{B}$
$B^+ \to K^0\pi^+$	2.392 ± 0.062
$B^+ \to K^+\pi^0$	1.322 ± 0.043
$B^0 \to K^+\pi^-$	2.007 ± 0.040
$B^0 \to K^0\pi^0$	1.012 ± 0.043

The same HFLAV snapshot gives direct asymmetries in %:

$$A_{CP}(K^0_S\pi^+) = (-2.67 \pm 0.87)\%, \quad A_{CP}(K^+\pi^0) = (+2.7 \pm 1.2)\%,$$

$$A_{CP}(K^+\pi^-) = (-8.31 \pm 0.31)\%, \quad A_{CP}(K^0\pi^0) = (-1 \pm 13)\%.$$

For the neutral time-dependent mode, the PDG 2025 API snapshot `references/B032/pdg2025_k0pi0_time_depen` gives

$$C_{K^0\pi^0} = 0.00 \pm 0.08, \qquad S_{K^0\pi^0} = 0.64 \pm 0.13.$$

The sign convention should be checked before combining with Belle-style $A = -C$ inputs.

Relevance to the RS / anarchic-flavor pipeline

$\bar{B} \to \pi\bar{K}$ is a $b \to s$ charmless penguin system with tree, QCD-penguin, and electroweak-penguin amplitudes. Anarchic RS models can induce nonstandard $b \to s$ flavor-changing Z, dipole, or four-quark operators after KK matching. This process is therefore a diagnostic of penguin flavor structure that is complementary to the existing neutral-meson $\Delta F = 2$ constraints, but it is not a clean tree-level bound by itself.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor benchmark era, the main changes are experimental. Belle and BaBar final samples were folded into HFLAV, LHCb measured $A_{CP}(B^+ \to K^+\pi^0) = 0.025 \pm 0.015 \pm 0.006 \pm 0.003$ and explicitly sharpened the charged–neutral $K\pi$ direct-asymmetry puzzle, under source key `lhcb2021_bplus_kplus_pi0_acp`, and Belle II added 2024 measurements of all four $K\pi$ branching fractions and direct asymmetries under `belleii2024_btokpi`. Belle II also measured $B^0 \to K_S^0\pi^0$ time-dependent CP in 2023, providing the largest modern input to the PDG/HFLAV $S_{K^0\pi^0}$ average. The Belle II $K\pi$ sum-rule test, $-0.03 \pm 0.13 \pm 0.04$, is currently consistent with the SM expectation (`belleii2024_btokpi`).

Constraint validity and model dependence

This is an experimentally mature but SM-theory-limited nonleptonic constraint. Branching fractions and CP asymmetries are easy to catalog, while interpreting them as RS exclusions requires a hadronic amplitude framework, electroweak penguin conventions, strong phases, and correlated treatment of the four isospin-related modes. It should be treated as a model-dependent penguin diagnostic until a dedicated likelihood or conservative amplitude scan exists.

Code coverage in this repo

NO. A focused grep for $B \to K\pi$, charmless, nonleptonic, and electroweak-penguin strings across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no implementation. The modern policy surface lists only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`; its notes say this is a policy-only fence with no numeric EFT backend at `quarkConstraints/modern/phenomenology.py:165`. The implemented default inputs in `quarkConstraints/deltaf2.py:209` are neutral-meson $\Delta F = 2$ systems only, and the dipole code found at `flavorConstraints/muToEGamma.py:75` is the lepton $\mu \to e\gamma$ constraint.

Implementation difficulty

HIGH. A production constraint would need a new nonleptonic $b \to s\bar{q}q$ operator/amplitude layer, electroweak-penguin matching, hadronic strong-phase treatment, and a correlated four-mode likelihood or conservative theory envelope. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR basis does not cover this calculation.

Key references

Process-local raw reference keys before bibliography consolidation: `hflav_dec2025_btopik`, `pdg2025_k0pi0_time_d` `belleii2024_btokpi`, `lhcb2021_bplus_kplus_pi0_acp`, `belleii2023_b0_ks_pi0_cp`, and `cfw2008_rs_flavor`.

45 B033: $B^0 \to \phi K_S^0$ Penguin CP Asymmetry

Process

Standard notation: $S_{\phi K_S}$ and $C_{\phi K_S}$, equivalently $\sin(2\beta_{\rm eff})_{\phi K^0}$ and $C_{\phi K^0}$ in the HFLAV convention. This entry covers the time-dependent CP asymmetry in the penguin-dominated beauty decay $B^0 \to \phi K_S^0$, usually reported through $\sin(2\beta_{\rm eff})$ and $C_{\phi K^0}$. It probes the $b \to s\bar{s}s$ loop amplitude and is interpreted by comparing its weak phase with the reference $b \to c\bar{c}s$ value from $B^0 \to J/\psi K_S^0$.

PDG or equivalent value

The source-of-truth numerical block is `pdg_or_equivalent` in `B033.yaml`. HFLAV Summer 2025 (`hflav_triangle_summer2025_phiK0`) gives, for the ϕK^0 average,

$$\sin(2\beta_{\text{eff}})_{\phi K^0} = 0.74 \pm 0.12, \qquad C_{\phi K^0} = -0.09 \pm 0.12.$$

HFLAV uses $S = -\eta_{CP} \sin(2\beta_{\text{eff}})$ and quotes $\eta_{CP} = -1$ for ϕK_S, so this $\sin(2\beta_{\text{eff}})$ is the usual $S_{\phi K_S}$ coefficient in this convention. The comparison benchmark on the same HFLAV page is the all-charmonium

$$\sin(2\beta) = 0.710 \pm 0.011,$$

with the strict $J/\psi K_S$ mode average 0.712 ± 0.011. The local snapshot is `references/B033/hflav_triangle_summe` accessed on 2026-05-16.

Relevance to the RS / anarchic-flavor pipeline

In anarchic warped models, KK gluons, electroweak KK modes, or flavor-changing Z/H-like effects can add new weak phases to $b \to s\bar{s}s$ $\Delta B = 1$ amplitudes. $B^0 \to \phi K_S^0$ is therefore a diagnostic of new penguin phases that would not be captured by a pure Δm_{B_s} or Δm_{B_d} magnitude constraint. A large departure of $S_{\phi K_S}$ from $S_{\psi K_S}$ would point to decay-amplitude new physics rather than just the CKM phase measured in the golden charmonium mode.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline, the old B-factory tension in penguin CP phases has been replaced by a stable world average compatible with $S_{\psi K_S}$. HFLAV Summer 2025 combines BaBar's 2012 Dalitz analysis, Belle's 2010 time-dependent Dalitz analysis, and Belle II's 2023 time-dependent $B^0 \to \phi K_S^0$ measurement. Belle II is the important new post-B-factory input: the `belleii_2023_phik0` snapshot records $S = 0.54 \pm 0.26^{+0.06}_{-0.08}$ and, with HFLAV's $C = -A$ convention, $C = -0.31 \pm 0.20 \pm 0.05$. The present HFLAV average differs from the charmonium benchmark by only 0.03, far below the current ϕK^0 uncertainty; this difference is recorded explicitly in `B033.yaml`.

Constraint validity and model dependence

This is a useful but theory-limited nonleptonic penguin observable. It is cleaner as a null test than as a standalone hard bound: the SM prediction is close to $\sin(2\beta)$, but subleading amplitudes and hadronic strong phases are mode dependent. A production implementation would need a $\Delta B = 1$ operator basis, RG evolution, and a hadronic amplitude or phenomenological likelihood. Treat it as charmless-penguin contextual input unless the PI chooses a specific hadronic framework.

Code coverage in this repo

NO. Focused greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no implementation of $B \to \phi K$, $b \to s\bar{s}s$ penguin CP asymmetries, $\sin(2\beta_{\rm eff})$, or time-dependent charmless B CP observables. The nearest adjacent code is neutral-meson mixing: `quarkConstraints/deltaf2.py:209` defines the $\Delta F = 2$ input registry, `quarkConstraints/deltaf2.py:903` evaluates B_d mixing, and `quarkConstraints/deltaf2.py:92` evaluates B_s mixing. Those do not compute $\Delta B = 1$ penguin decay amplitudes.

Implementation difficulty

HIGH. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR1/LR2 machinery is not enough. A live constraint would require $\Delta B = 1$ four-quark penguin operators, matching from RS sources, QCD running to the b scale, and a hadronic treatment of the $B \to \phi K$ amplitude and strong phases.

Key references

Process-local raw reference keys before bibliography consolidation: `hflav_triangle_summer2025_phiK0`, `babar_2012_b0_kkks`, `belle_2010_b0_kkks`, `belleii_2023_phik0`, and `csaki_falkowski_weiler_2008`.

46 B034: $B_s^0 \to \phi\phi$ Penguin CP Phase

Process

Standard notation: $\phi_s^{s\bar{s}s}$ and $|\lambda|$ in $B_s^0 \to \phi(1020)\phi(1020)$. This entry covers the flavour-tagged, time-dependent angular CP analysis of the charmless penguin decay $B_s^0 \to \phi\phi$, a $b \to s\bar{s}s$ transition. The process is a counterpart to the cleaner $b \to c\bar{c}s$ phase measurements, but with the weak phase attached to a loop-induced decay amplitude.

PDG or equivalent value

The current source-of-truth numerical block is `B034.yaml`. The PDG B_s^0 listing records the $b \to s\bar{s}s$ convention

$$\beta_s = (3.7 \pm 3.5) \times 10^{-2} \text{ rad},$$

and its AAIJ 23AT note gives the LHCb combined phase and direct-CP parameter

$$\phi_s^{s\bar{s}s} = -0.074 \pm 0.069 \text{ rad}, \qquad |\lambda| = 1.009 \pm 0.030.$$

The PDG convention is recorded under `pdg2025_bs_phiphi_cp`; the same combined LHCb values appear in `lhcb2023_bs_phiphi_cp_cds`. As contextual rate information for the channel, HFLAV May 2024 quotes

$$\mathcal{B}(B_s^0 \to \phi\phi) = (1.85 \pm 0.14) \times 10^{-5}.$$

This rate is tracked under `hflav2024_bs_phiphi_branching`. Local snapshots are in `flavor_catalog/references/` accessed on 2026-05-16.

Relevance to RS / anarchic flavor

In warped anarchic flavour models, flavour-changing KK gluon, electroweak-KK, or misaligned Z/H-like effects can induce new $\Delta B = 1$ penguin operators with phases not fixed by the Standard Model CKM fit. $B_s^0 \to \phi\phi$ is therefore a useful null test for new weak phases in $b \to s\bar{s}s$ amplitudes. It is complementary to the existing repo's B_s-mixing magnitude lane: a new phase could appear in the decay amplitude or in mixing-decay interference even when a simple Δm_s bound is satisfied.

Post-2008 developments

The process matured after the arXiv:0804.1954 era. LHCb's 2014 analysis used $3.0\,\text{fb}^{-1}$, about 4000 signal decays, and found $\phi_s = -0.17 \pm 0.15 \pm 0.03$ rad (`lhcb2014_bs_phiphi_cp_arxiv`). The 2019 update used $5.0\,\text{fb}^{-1}$, about 9000 signal decays, and reported $\phi_s^{s\bar{s}s} = -0.073 \pm 0.115 \pm 0.027$ rad with $|\lambda| = 0.99 \pm 0.05 \pm 0.01$ (`lhcb2019_bs_phiphi_cp_cds`). The 2023 precision measurement used $6\,\text{fb}^{-1}$ at 13 TeV, giving $\phi_s^{s\bar{s}s} = -0.042 \pm 0.075 \pm 0.009$ rad and $|\lambda| = 1.004 \pm 0.030 \pm 0.009$ before combination (`lhcb2023_bs_phiphi_cp_cds`). The combined LHCb result remains consistent with CP conservation and Standard Model expectations.

Constraint validity / model dependence

This is a strong experimental null test but a model-dependent hard constraint. The Standard Model expectation for the penguin phase is close to zero, while $|\lambda| = 1$ corresponds to no direct CP violation, but a production bound requires a $\Delta B = 1$ operator normalization, QCD running, polarization treatment, hadronic amplitudes, and strong-phase assumptions. It should be cataloged as a high-value contextual penguin-phase observable until a specific nonleptonic framework is chosen.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $B_s^0 \to \phi\phi$, $b \to s\bar{s}s$ penguin, time-dependent charmless CP, or angular-likelihood implementation. Adjacent code only covers neutral B_s mixing: `quarkConstraints/deltaf2.py:239` defines the B_s input, `quarkConstraints/deltaf2.py:922` evaluates B_s mixing, and `quarkConstraints/modern/phenomenology.py:657` bridges the modern B_s-mixing evaluator.

Implementation difficulty

HIGH. The existing $\Delta F = 2$ basis does not compute $\Delta B = 1$ penguin amplitudes. A live constraint needs new nonleptonic operators, matching, running, angular/polarization likelihood handling, and a hadronic-amplitude prescription.

Key references

`pdg2025_bs_phiphi_cp`, `lhcb2023_bs_phiphi_cp_cds`, `lhcb2019_bs_phiphi_cp_cds`, `lhcb2014_bs_phiphi_cp_arxi`
`hflav2024_bs_phiphi_branching`, and `cfw2008_arxiv0804_1954`.

47 Top/Higgs/EW

48 S, T, U Oblique Electroweak Parameters

Process

This entry covers the Peskin–Takeuchi oblique parameters S, T, and U, treated as universal new-physics corrections to electroweak gauge-boson self energies. The process is not flavor changing, but it is part of the `top_higgs_ew` family because it is a leading precision bound on warped, vector-like, composite, and custodial electroweak sectors.

PDG / equivalent value(s)

The canonical values are the PDG 2025 electroweak-review fit in `flavor_catalog/references/EW001/pdg_2025_el`
accessed on 2026-05-16. With $U = 0$, PDG quotes $S = 0.026 \pm 0.075$, $T = 0.047 \pm 0.066$, and $\rho(S, T) = 0.90$. Letting all three float gives $S = 0.021 \pm 0.096$, $T = 0.04 \pm 0.12$, and $U = 0.008 \pm 0.092$, with correlations recorded in the sidecar. For comparison, the public Gfitter oblique page gives $S = 0.05 \pm 0.11$, $T = 0.09 \pm 0.13$, and $U = 0.01 \pm 0.11$. The CDF-aware HEPfit/de Blas 2022 study quotes $M_W = 80.4335 \pm 0.0094\,\mathrm{GeV}$; in its standard-average $U = 0$ oblique fit, $S = 0.100 \pm 0.073$ and $T = 0.202 \pm 0.056$, while its floating-U fit has $U = 0.134 \pm 0.087$.

Relevance to the RS / anarchic-flavor pipeline

Bulk RS models generically shift S through mixing of the SM electroweak bosons with KK gauge modes, and shift T when custodial symmetry is broken by gauge, Higgs, or fermion embeddings. PDG summarizes the warped contribution as $S \simeq 30v^2/M_{\mathrm{KK}}^2$ and quotes $S \leq 0.18$ at 95% probability as implying $M_{\mathrm{KK}} \gtrsim 3.2\,\mathrm{TeV}$. These bounds are orthogonal to the catalog's $\Delta F = 2$ constraints, but they test the same KK spectrum and localization choices before flavor observables are even evaluated.

Post-2008 developments

Relative to the CFW baseline, arXiv:0804.1954, the important change is not a new LEP run but a much more mature global-fit environment: Higgs and top inputs are measured directly, full two-loop and selected higher-order SM calculations are included, and PDG now presents both $U = 0$ and floating-U oblique fits. The 2022 CDF W-mass result created a distinct scenario in which T or U can be pulled positive; the 2025 PDG review treats that impact as a global-fit issue rather than replacing the main STU table with the CDF-only preference.

Constraint validity and model dependence

The STU language is cleanest for heavy, approximately universal new physics whose leading effects are vacuum-polarization corrections. It is less complete when vertex corrections, non-universal fermion couplings, light new states, or SMEFT flat directions are numerically important. In RS applications, custodial protection, brane kinetic terms, Higgs localization, and vector-like fermion embeddings can change the translation from a fitted S, T, U point to a KK mass bound.

In custodial RS, reduced Zb_Lb_L pressure can reshuffle which EW observable is most discriminating; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms.

Code coverage in this repo

NO. Greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no S/T/U, oblique, EWPO, or W-mass-fit implementation. Only generic electroweak constants and VEV comments surfaced, for example `qcd/constants.py:11`, `qcd/running.py:3`, `quarkConstraints/qcd_running.py:99`, `yukawa/neutrino.py:14`, and `yukawa/charged_lep`

Implementation difficulty

HIGH. A production constraint would need a new electroweak precision likelihood, correlation handling for the PDG/GFitter/HEPfit fit conventions, and RS matching from the gauge and fermion KK spectra to vacuum-polarization self energies or equivalent SMEFT operators. This is a new electroweak mode/matching calculation, not a reuse of the existing $\Delta F = 2$ basis.

Key references

`pdg_2025_electroweak_stu`; `gfitter_oblique_parameters`; `hepfit_deblas_2022_cdf_stu`; `hepfit_precision_ew`
`peskin_takeuchi_1992_oblique`; `cfw_2008_rs_flavor`.

49 First-Row CKM Unitarity and the Cabibbo Anomaly

Process

This entry covers the first-row CKM test $\Delta^{(1)}_{\text{CKM}} \equiv |V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 - 1$. It combines superallowed nuclear beta-decay, neutron/pion beta-decay cross checks, kaon $K_{\ell 3}$ and $K_{\mu 2}/\pi_{\mu 2}$ inputs, and lattice normalizations of $f_+(0)$ and f_K/f_π. The catalog scope is the Cabibbo-angle anomaly as a precision charged-current constraint, not a single kaon branching fraction.

PDG / equivalent value(s)

The source-of-truth numerical entries are in `EW002.yaml`. The current PDG V_{ud}, V_{us} review, published with the 2024 Review of Particle Physics and updated in 2025, uses the superallowed-beta value $|V_{ud}| = 0.97367(32)$ and a kaon-sector average $|V_{us}| = 0.22431(85)$, with scale factor $S = 2.5$. Its quoted first-row sum is

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 0.9983(6)(4),$$

an overall 2σ deficit, rising to about 3σ if nuclear-structure uncertainties are not included. FLAG 2024 gives the sharper semileptonic-kaon comparison: $|V_{us}|f_+(0) = 0.21654(41)$, $f_+(0) = 0.9698(17)$ for $N_f = 2+1+1$, hence $|V_{us}| = 0.22328(58)$. Combining this with the PDG beta-decay V_{ud} gives $|V_u|^2 = 0.99802(66)$, a roughly 3.0σ deviation. The same FLAG review notes that the f_K/f_π route gives a weaker $|V_u|^2 = 0.99888(67)$, about 1.7σ.

Relevance to the RS / anarchic-flavor pipeline

Warped and partially composite models can perturb the apparent CKM matrix through mixing with KK or vector-like quarks, non-universal W-quark couplings, and correlated shifts in Z-quark couplings. Thus EW002 probes a charged-current direction complementary to the existing $\Delta F = 2$ neutral meson constraints. In lepton-extended variants, a new contribution to muon decay can also change the inferred G_F, feeding directly into beta- and kaon-decay CKM extractions.

Post-2008 developments

Relative to arXiv:0804.1954, which emphasized RS-GIM and neutral-current flavor bounds with quoted KK-gluon scales near 21 and 33 TeV, the post-2008 story is a new charged-current precision problem. Hardy–Towner 2020 shifted the superallowed result to $|V_{ud}| = 0.97373(31)$ after updated radiative-correction theory. PDG now quotes a closely related $|V_{ud}| = 0.97367(32)$ and explicitly frames the first-row test as a possible 2–3σ inconsistency. FLAG 2024 supplies sub-percent lattice inputs for $f_+(0)$ and f_K/f_π, with QED and strong-isospin corrections now central to the comparison. The modern SMEFT literature treats the anomaly as a modified W-coupling problem correlated with electroweak precision and FCNC observables.

Constraint validity and model dependence

The unitarity sum is experimentally clean but theory-limited. The limiting issues are the inner electroweak radiative correction and nuclear-structure terms in V_{ud}, lattice $f_+(0)$ and f_K/f_π, kaon radiative and isospin-breaking corrections, and correlations between $K_{\ell 3}$ and $K_{\mu 2}$. Its new-physics interpretation is global-fit-dependent: a shift can live in semileptonic charged currents, G_F, right-handed W couplings, or correlated Z and $\Delta F = 2$ effects.

In custodial RS, reduced Zb_Lb_L pressure can make this observable relatively more or less discriminating inside a combined EW fit; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms.

Code coverage in this repo

PARTIAL. The repo fits and gates a unitary CKM target: PDG-derived per-element tolerances are documented at `quarkConstraints/scan.py:42`, the reduced observable set ($|V_{us}|, |V_{cb}|, |V_{ub}|, J$)

is implemented at `quarkConstraints/fit.py:47`, and the modern default CKM target is stored at `quarkConstraints/modern/inputs.py:631`. However, greps found no V_{ud}, superallowed-beta, $K_{\ell 3}$, $f_+(0)$, or $\Delta_{\rm CKM}$ likelihood implementing EW002 as a constraint.

Implementation difficulty

HIGH. A production constraint needs new charged-current SMEFT or direct RS W-coupling matching, optional G_F shifts for lepton extensions, and a correlated low-energy likelihood using beta, kaon, radiative-correction, and lattice inputs. This is outside the existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR1/LR2 machinery.

Key references

`pdg_2025_vud_vus_ckm_unitarity`; `flag_2024_vus_ckm_unitarity`; `hardy_towner_2020_superallowed_vud`; `bryman_cirigliano_crivellin_inguglia_2022_lfu_ckm`; `crivellin_kirk_kitahara_mescia_2022_vlq_caa`; `cfw_2008_rs_flavor`.

50 $|V_{cb}|$ and $|V_{ub}|$ Inclusive–Exclusive Tensions

Process

This entry covers the inclusive versus exclusive determinations of $|V_{cb}|$ and $|V_{ub}|$ from semileptonic b-hadron decays. Although the underlying modes are beauty charged currents, the NIT-1 process id places the entry in `top_higgs_ew`; it is an electroweak CKM-input diagnostic with direct ties to $b \to c\tau\nu$ and $b \to u\tau\nu$ BSM fits.

PDG / equivalent value(s)

The canonical PDG 2024 snapshot is `flavor_catalog/references/EW003/pdg_2024_vcb_vub.txt`. PDG quotes $|V_{cb}|_{\rm incl} = (42.2 \pm 0.5) \times 10^{-3}$ and $|V_{cb}|_{\rm excl} = (39.8 \pm 0.6) \times 10^{-3}$, with the scaled average $(41.1 \pm 1.2) \times 10^{-3}$ to be treated with caution because the two inputs are only marginally consistent, about 3.0σ. For $|V_{ub}|$, PDG quotes $(4.13 \pm 0.12_{\rm exp}{}^{+0.13}_{-0.14}{}_{\rm theo} \pm 0.18_{\Delta\rm model}) \times 10^{-3}$ inclusive and $(3.70 \pm 0.10 \pm 0.12) \times 10^{-3}$ exclusive, giving $(3.82 \pm 0.20) \times 10^{-3}$ after error scaling. HFLAV's 2023 semileptonic report refines the split: the preferred inclusive $|V_{cb}|$ input is $(41.97 \pm 0.48) \times 10^{-3}$, while the combined exclusive fit gives $(|V_{ub}|, |V_{cb}|) = (3.43 \pm 0.12, 39.77 \pm 0.46) \times 10^{-3}$. FLAG 2024 gives lattice-anchored exclusive values $|V_{cb}| = 39.23(65) \times 10^{-3}$ for $B \to D^*\ell\nu$ with $N_f = 2 + 1$, and $|V_{ub}| = (3.61 \pm 0.16) \times 10^{-3}$ for $B \to \pi\ell\nu$.

Relevance to the RS / anarchic-flavor pipeline

The present scan fits CKM observables as output of anarchic quark masses, so the inclusive–exclusive split is a diagnostic for how much CKM target choice can move flavor predictions. In an RS extension with non-SM charged-current operators, the same observables also set the normalization for $b \to c$ and $b \to u$ semitauonic Wilson-coefficient fits.

Post-2008 developments

Relative to arXiv:0804.1954, the main change is not a new RS formula but a new input ecosystem. Belle, Belle II, BaBar, and LHCb now feed HFLAV averages; inclusive $|V_{cb}|$ uses mature kinetic and q^2-moment heavy-quark expansion fits; and FLAG 2024 provides full-recoil lattice-plus-experiment fits for $B \to D^{(*)}\ell\nu$ and $B \to \pi\ell\nu$. The same period also turned $R(D^{(*)})$, $B_c \to \tau\nu$, and $B \to (\pi, \rho, \omega)\tau\nu$ into standard charged-current EFT interpretation channels.

Constraint validity and model dependence

For e, μ final states these are CKM extraction inputs, not clean standalone BSM exclusions. Inclusive values depend on heavy-quark expansion schemes and shape-function/model choices, while exclusive values depend on lattice form factors, parameterization choices, and experimental correlations. Tau-mode translations are more model dependent because vector, scalar, and tensor operators can change both rates and templates.

Code coverage in this repo

PARTIAL. CKM target handling exists, including $|V_{cb}|$ and $|V_{ub}|$ tolerances at `quarkConstraints/scan.py:42`, `quarkConstraints/scan.py:46`, `quarkConstraints/scan.py:47`, `quarkConstraints/scan.py:52`, and the regression check at `tests/test_quark_fit.py:575`. No grep hit implements inclusive or exclusive semileptonic B rates, $B \to D^{(*)}\ell\nu$, $B \to \pi\ell\nu$, or $b \to c/u\,\tau\nu$ WET operators in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`.

Implementation difficulty

HIGH. A production constraint would need a new charged-current semileptonic likelihood, scheme handling for inclusive extractions, lattice form-factor inputs for exclusive modes, and optional $b \to c/u\,\tau\nu$ Wilson operators. This is outside the existing $\Delta F = 2$ basis.

Key references

`pdg_2024_vcb_vub`; `hflav_2023_semileptonic_report`; `flag_2024_b_semileptonic_lattice`; `iguro_2024_bc_tau` `bhatta_2021_bu_tau_np`; `cfw_2008_rs_flavor_baseline`.

51 $t \to cZ$ FCNC Top Decay

Process

The observable is the flavor-changing neutral-current branching fraction $\mathcal{B}(t \to cZ)$, catalog id T001 in the top/Higgs/electroweak family. It covers direct LHC searches for a top quark coupled to a charm quark and a Z boson, including analyses that combine $t\bar{t}$ events with one FCNC top decay and single-top production in association with a Z boson.

PDG-or-equivalent value

The source-of-truth numerical entries are the `pdg_or_equivalent` block in `T001.yaml`. The PDG 2025 top-quark listing quotes the ATLAS Run-2 limits in units of 10^{-3} (`PDG2025TopFCNCTZQ`):

$$\mathcal{B}(t \to Zc) < 1.3 \times 10^{-4}$$

for the left-handed tcZ benchmark (`PDG2025:T001:tZc_left`) and

$$\mathcal{B}(t \to Zc) < 1.2 \times 10^{-4}$$

for the right-handed benchmark (`PDG2025:T001:tZc_right`), both at 95% CL. The same PDG table records the CMS comparison limit $\mathcal{B}(t \to Zc) < 0.049\%$, also at 95% CL (`CMS2017:T001:tZc`). The local PDG snapshot is `flavor_catalog/references/T001/pdg_2025_top_fcnc_tzq.txt`.

Relevance to the RS / anarchic-flavor pipeline

In warped or partially composite flavor models, non-universal fermion profiles and mass-basis rotations can produce flavor off-diagonal neutral-current couplings. The $t \to cZ$ channel is especially diagnostic of the top sector: it probes the t-c rotation and the degree of right-handed top compositeness more directly than light-meson $\Delta F = 2$ observables. A direct top FCNC limit is therefore complementary to the existing kaon, B, and D-mixing lane rather than a substitute for it.

Post-2008 developments

The CFW-era reference point is arXiv:0804.1954, which framed warped flavor constraints before the full LHC top-FCNC program. Since then, the key experimental change is the move from Tevatron and early-LHC inclusive $t \to Zq$ limits to flavor-separated LHC constraints. CMS used 19.7 fb^{-1} at $\sqrt{s} = 8$ TeV (`CMS2017:T001:dataset`) and found $\mathcal{B}(t \to Zc) < 0.049\%$ (`CMS2017:T001:tZc`). ATLAS later used 139 fb^{-1} at $\sqrt{s} = 13$ TeV (`ATLAS2023:T001:dataset`), combining single-top tZ production and $t\bar{t}$ FCNC-decay topologies, and improved the $t \to Zc$ reach to the 10^{-4} level (`ATLAS2023:T001:tZc_left`, `ATLAS2023:T001:tZc_right`). A 2026 RS-specific study, arXiv:2601.14966, recasts post-LHC constraints on Ztc and notes that HL-LHC direct searches may reach an equivalent $\mathcal{B}(t \to cZ)$ level of about 10^{-6} (`AirenFranceschini2026:T001:HL_LHC_projection`).

Constraint validity and model dependence

Experimentally this is a clean upper limit; there is no SM-sized background from true $t \to cZ$ decays. The SM branching estimate is about 10^{-14}, so any observed signal would be unambiguous new physics (`AguilarSaavedra2004:T001:SM`; `PDG2025:T001:SM`). The interpretation is not fully model independent: ATLAS quotes separate left- and right-handed tensor-SMEFT benchmarks, while an RS model may generate vector-like Ztc couplings, heavy-vector exchange, or correlated contact interactions. A scan backend would need to map the RS parameters into a chosen operator convention before applying the PDG-number as a hard cut.

In down-aligned or kaon-protected RS variants, top FCNCs become leading rather than secondary diagnostics.

Code coverage in this repo

Coverage is **NO**. The PKA reran the required plan greps and a targeted search for $t \to cZ$, tZc, tcZ, and generic FCNC text across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only non-doc hits were generic FCNC prose/comments, not an observable, likelihood, or branching-ratio check.

Implementation difficulty

The cataloging difficulty is medium, but production integration is **HIGH**. The repo has no electroweak top-FCNC operator basis, no tcZ width or single-production reinterpretation, and no top-sector flavor-rotation interface. Implementing it honestly would require a new RS-to-SMEFT or direct Ztc matching convention before this limit can be compared to scan points.

Key references

`PDG2025TopFCNCTZQ`; `ATLAS2023TopFCNCTZQ`; `CMS2017TZQFCNC`; `AguilarSaavedra2004TopFCNC`; `CsakiFalkowskiW`
`AirenFranceschini2026RSTopFCNC`.

52 $t \to uZ$ FCNC Top Decay

Process

Standard notation: $\mathcal{B}(t \to uZ)$. This entry catalogs the flavor-changing neutral-current top decay $t \to uZ$ in the top/Higgs/electroweak family. It covers direct LHC searches for a top quark coupled to an up quark and a Z boson, including interpretations that combine $t\bar{t}$ events with one FCNC top decay and single-top production in association with a Z boson.

PDG-or-equivalent value

The source-of-truth numerical entries are the `pdg_or_equivalent` block in `T002.yaml`. The current PDG 2025 top-quark listing quotes the ATLAS AAD 23AS limits in units of 10^{-3} (`PDG2025TopFCNCTZQ`):

$$\mathcal{B}(t \to Zu) < 6.2 \times 10^{-5}$$

for the left-handed tuZ benchmark (`PDG2025:T002:tZu_left`) and

$$\mathcal{B}(t \to Zu) < 6.6 \times 10^{-5}$$

for the right-handed benchmark (`PDG2025:T002:tZu_right`), both at 95% CL. The same PDG table records the CMS SIRUNYAN 17E comparison limit $\mathcal{B}(t \to Zu) < 0.022\%$, also at 95% CL (`CMS2017:T002:tZu`). The local PDG snapshot is `flavor_catalog/references/T002/pdg_2025_top_fcnc_tzq.tx`

Relevance to the RS / anarchic-flavor pipeline

In warped models with bulk fermions, non-universal zero-mode profiles and mass-basis rotations can induce flavor off-diagonal neutral-current Z couplings. The $t \to uZ$ channel probes the first–third generation up-sector rotation and is therefore complementary to charm and neutral-meson constraints. In anarchic RS language it is sensitive to the same profile and Yukawa structures that control u-c-t mixing, while remaining a direct top observable rather than a low-energy hadronic matrix-element constraint.

Post-2008 developments

The CFW reference point is arXiv:0804.1954, which framed generic warped-flavor bounds before the full LHC top-FCNC program (`CsakiFalkowskiWeiler2008WarpedFlavor`). Shortly after it, the RS-specific Casagrande–Goertz–Haisch–Neubert–Pfoh analysis included $t \to c(u)Z$, noting that $t \to uZ$ is typically about two orders of magnitude below $t \to cZ$ in the minimal RS model (`Casagrande2008:T002:RS_suppression`). Since then the key experimental development is flavor-separated LHC sensitivity. CMS used 19.7 fb^{-1} at $\sqrt{s} = 8$ TeV (`CMS2017:T002:dataset`) and found $\mathcal{B}(t \to Zu) < 0.022\%$ (`CMS2017:T002:tZu`). ATLAS later used 139 fb^{-1} at $\sqrt{s} = 13$ TeV (`ATLAS2023:T002:dataset`), combining single-top tZ production and $t\bar{t}$ FCNC-decay topologies, and improved the $t \to Zu$ reach to the 6×10^{-5} level (`ATLAS2023:T002:tZu_left`, `ATLAS2023:T002:tZu_right`).

Constraint validity and model dependence

Experimentally this is a clean 95% CL upper limit; there is no Standard Model signal at the observable level. Aguilar-Saavedra's benchmark SM estimate is $\mathcal{B}(t \to uZ) \simeq 8\times10^{-17}$ (`AguilarSaavedra2004:T002:SM`) The interpretation is operator-dependent: ATLAS quotes left- and right-handed tensor-SMEFT benchmarks, while an RS model can generate vector-like Ztu couplings, KK-mediated contact effects, or correlated shifts in other electroweak vertices. A live constraint therefore needs a matching convention before the branching-ratio limit can be used as a hard scan cut.

In down-aligned or kaon-protected RS variants, top FCNCs become leading rather than secondary diagnostics.

Code coverage in this repo

Coverage is **NO**. The PKA reran the required plan greps and a targeted search for $t \to uZ$, tZu, tuZ, tZq, and generic FCNC text across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only targeted code-directory hits were generic FCNC prose/comments, not an observable implementation, likelihood, or branching-ratio check.

Implementation difficulty

The cataloging difficulty is medium, but production integration is **HIGH**. The repo has no electroweak top-FCNC operator basis, no tuZ decay-width or single-production reinterpretation, and no top-sector flavor-rotation interface. Implementing it honestly would require new RS-to-SMEFT or direct Ztu matching before comparing this limit to scan points.

Key references

PDG2025TopFCNCTZQ; ATLAS2023TopFCNCTZQ; CMS2017TZQFCNC; AguilarSaavedra2004TopFCNC; CsakiFalkowskiW
CasagrandeGoertzHaischNeubertPfoh2008RSFlavor.

53 $t \to c\gamma$ FCNC Top-Photon Decay

Process

Standard notation: $\mathcal{B}(t \to c\gamma)$. This entry catalogs the flavor-changing neutral-current top decay $t \to c\gamma$, process id T003 in the top/Higgs/electroweak family. It covers LHC searches for a top quark coupled to a charm quark and a photon, including single-top-plus-photon production and $t\bar{t}$ events with one FCNC top decay.

PDG value

The sidecar headline uses the CMS-TOP-21-013 Run-2 charm-specific result, based on 138 fb^{-1} at $\sqrt{s} = 13$ TeV (`CMS2024:T003:dataset`):

$$\mathcal{B}(t \to c\gamma) < 1.51 \times 10^{-5}$$

at 95% CL, with an expected limit of 1.54×10^{-5} (`CMS2024:T003:tcgamma`). The ATLAS full-Run-2 comparison result used 139 fb^{-1} at $\sqrt{s} = 13$ TeV (`ATLAS2023:T003:dataset`) and gave

$$\mathcal{B}(t \to c\gamma) < 4.2 \times 10^{-5}$$

for a left-handed $tq\gamma$ coupling and $< 4.5 \times 10^{-5}$ for a right-handed coupling (`ATLAS2023:T003:tcgamma_left`, `ATLAS2023:T003:tcgamma_right`). The PDG live top summary separately lists a generic $t \to \gamma q$, $q = u, c$, row $< 9.5 \times 10^{-6}$ at 95% CL, defined as $\Gamma(t \to \gamma q)/\Gamma(t \to Wb)$; that row is not charm separated (`PDG2026:T003:gammaq_summary`).

Relevance to RS / anarchic flavor

In anarchic warped flavor, the IR-localized top and non-universal fermion profiles can generate mass-basis flavor violation in the up sector. The top-photon channel probes an electromagnetic dipole or heavy-vector matching direction that is absent from the existing neutral-meson $\Delta F = 2$ lane. It is therefore complementary to T001 ($t \to cZ$) and T005 ($t \to cg$): the same top-charm rotations can appear with different gauge quantum numbers.

Post-2008 developments

Relative to the arXiv:0804.1954 RS-flavor baseline (`CsakiFalkowskiWeiler2008WarpedFlavor`), the main change is experimental. The LHC program moved this channel from indirect expectations to direct full-Run-2 searches with separate $tu\gamma$ and $tc\gamma$ hypotheses. ATLAS reached the 4.2×10^{-5} left-handed charm benchmark in its 2023 result (`ATLAS2023:T003:tcgamma_left`). CMS then used 138 fb^{-1} and improved the charm-specific observed limit to 1.51×10^{-5} (`CMS2024:T003:dataset`; `CMS2024:T003:tcgamma`). The Standard Model expectation is below current limits by many orders of magnitude, so present bounds are new-physics searches rather than SM precision tests.

Constraint validity / model dependence

The upper limit is experimentally clean because the SM decay rate is negligible. Its interpretation is model dependent: CMS quotes a single nonzero anomalous coupling at a time, while ATLAS separates left- and right-handed $tq\gamma$ benchmarks. A warped-model scan would need to map its top-sector rotations and electromagnetic dipoles into the same operator normalization before using the number as a hard exclusion.

Code coverage in this repo

NO. The required greps across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $t \to c\gamma$, $tc\gamma$, $tq\gamma$, or top-photon FCNC implementation. Existing coverage is limited to neutral-meson $\Delta F = 2$ inputs at `quarkConstraints/deltaf2.py:209`, the modern policy list $\epsilon_K, K, B_d, B_s, D^0$ at `quarkConstraints/modern/phen`
and a lepton-sector $\mu \to e\gamma$ dipole check at `flavorConstraints/muToEGamma.py:75`.

Implementation difficulty

HIGH. Cataloging the external upper limit is straightforward, but a live RS constraint would require a top electromagnetic FCNC operator convention, RS-to-SMEFT or direct dipole matching, a $t \to c\gamma$ width normalization, and a choice about how to reinterpret production-plus-decay LHC limits when more than one chirality or flavor coupling is present.

Key references

`CMS2024TopFCNCTQGamma`; `ATLAS2023TopFCNCTQGamma`; `PDG2026TopGammaQSummary`; `AguilarSaavedra2004TopFC`
`CsakiFalkowskiWeiler2008WarpedFlavor`.

54 $t \to u\gamma$ FCNC Top Decay

Process

Standard notation: $\mathcal{B}(t \to u\gamma)$. This entry catalogs the flavor-changing neutral-current top decay $t \to u\gamma$, process id T004 in the top/Higgs/electroweak family. It covers direct searches for an anomalous photon coupling between a top quark and an up quark, including FCNC single-top-plus-photon production and $t\bar{t}$ events with one FCNC top decay.

PDG value

The `pdg_or_equivalent` block in `T004.yaml` records both the PDG combined row and flavor-separated primary limits. The PDG 2026 top-quark listing gives a combined $t \to \gamma q$, $q = u, c$, branching-fraction limit

$$\mathcal{B}(t \to \gamma q) < 9.5 \times 10^{-6}$$

at 95% CL (`PDG2026:T004:tgammaq_combined`). For the flavor-separated $t \to u\gamma$ mode, the tightest primary limit found is the ATLAS result:

$$\mathcal{B}(t \to u\gamma) < 0.85 \times 10^{-5}$$

for the left-handed $tq\gamma$ benchmark and $\mathcal{B}(t \to u\gamma) < 1.2 \times 10^{-5}$ for the right-handed benchmark, both at 95% CL (`ATLAS2023:T004:tugamma_left`, `ATLAS2023:T004:tugamma_right`). CMS later reported $\mathcal{B}(t \to u\gamma) < 0.95\times10^{-5}$, with 1.20×10^{-5} expected, at 95% CL (`CMS2024:T004:tugamma_observed`). Local snapshots are under `flavor_catalog/references/T004/`.

Relevance to RS / anarchic flavor

In warped or partially composite flavor models, non-universal fermion profiles and mass-basis rotations can induce top FCNC couplings. The photon channel is a dipole-like probe, complementary to $t \to uZ$, $t \to ug$, and $t \to uh$. Compared with charm final states, T004 isolates the top-up flavor rotation and therefore probes a different corner of the anarchic up-sector structure.

Post-2008 developments

The comparison baseline is arXiv:0804.1954 (`CFW2008:T004:rs_flavor_context`). Since that reference epoch, the important change is experimental: ATLAS and CMS set direct mode-resolved limits (`ATLAS2023:T004:dataset`; `CMS2024:T004:dataset`). The high-energy single-production strategy also became a standard way to discuss future sensitivity for $t \to uZ/u\gamma$-equivalent interactions.

Constraint validity / model dependence

Experimentally this is a clean null-search upper limit; the SM rate is far below current reach and is lower than the corresponding $t \to c\gamma$ rate because of CKM suppression (`AguilarSaavedra2017:T004:SM_and_proje`
The interpretation is not model-independent: ATLAS quotes separate chiral $tq\gamma$ tensor benchmarks, and an RS backend must choose an operator normalization before comparing scan points to the branching-fraction limit.

Code coverage in this repo

Coverage is **NO**. The PKA reran the required catalog greps across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The targeted $t \to u\gamma$, $tu\gamma$, $tq\gamma$, and top-photon FCNC search returned no observable, Wilson coefficient, or branching-ratio implementation.

Implementation difficulty

Implementation difficulty is **HIGH**. The repo has no top-FCNC photon dipole basis, no RS-to-$tu\gamma$ matching convention, and no top radiative decay-width or production-recast module. A production backend would need those ingredients before this limit could become a scan cut.

Key references

`PDG2026TopTGammaQ`; `ATLAS2023TopFCNCTQGamma`; `CMS2024TopFCNCTQGamma`; `CsakiFalkowskiWeiler2008Warped`
`AguilarSaavedra2017UltraBoostedTopFCN`.

55 $t \to cg$ Top-Gluon FCNC

Process

The observable is the flavor-changing neutral-current top decay $\mathcal{B}(t \to cg)$, catalog id T005 in the top/Higgs/electroweak family. Experimentally it is constrained mainly through anomalous single-top production, $cg \to t$, then translated into an equivalent top-decay branching fraction under a specified chromomagnetic-operator convention.

PDG / equivalent value(s)

The source-of-truth values are in the `pdg_or_equivalent` block of `T005.yaml`. The canonical experimental source is the PDG/pdgLive Q007TUG top-FCNC gluon datablock, accessed on 2026-05-16 (`PDG2025TopFCNCTUG`; `references/T005/pdg_2025_top_tug_tcg.txt`). Its ATLAS Run-2 entry, AAD 2022T, uses 139 fb^{-1} of 13 TeV pp data (`ATLAS2022TopFCNCTUG`). For the charm-gluon production mode the PDG block gives

$$\sigma(cg \to t)\,\mathcal{B}(t \to bW)\,\mathcal{B}(W \to \ell\nu) < 4.7 \text{ pb}$$

at 95% CL, interpreted in the EFT basis as

$$|C_{uG}^{ct}|/\Lambda^2 < 0.14 \text{ TeV}^{-2}, \qquad \mathcal{B}(t \to cg) < 3.7 \times 10^{-4}.$$

(`PDG2025TopFCNCTUG`; `ATLAS2022TopFCNCTUG`). The same PDG block records the CMS 7 and 8 TeV comparison result based on 5.0 fb^{-1} and 19.7 fb^{-1}: $|\kappa_{tcg}|/\Lambda < 1.8 \times 10^{-2}$ TeV^{-1}, corresponding to $\mathcal{B}(t \to cg) < 4.1 \times 10^{-4}$ (`CMS2017TopFCNCTUG`). For SM context, not as a measured PDG value, Aguilar-Saavedra gives $\mathcal{B}(t \to cg) = 4.6 \times 10^{-12}$ as the central SM benchmark (`AguilarSaavedra2004TopFCNC`).

Relevance to the RS / anarchic-flavor pipeline

This channel probes color-carrying top-charm flavor violation rather than the neutral-meson $\Delta F = 2$ four-quark operators already audited in the quark scan. In anarchic warped flavor, the heavy color sector and the IR-localized top can generate non-universal mass-basis couplings and loop-level chromomagnetic dipoles. A $t \to cg$ limit is therefore a direct up-sector top-flavor diagnostic, complementary to D^0 mixing and to electroweak channels such as $t \to cZ$ or $t \to cH$.

Post-2008 developments

The 2008 CFW reference point emphasized KK-gluon exchange and generic anarchic-flavor bounds near 21 TeV for ordinary RS and 33 TeV in the composite pseudo-Goldstone Higgs setup, but it did not provide a direct $t \to cg$ catalog limit (`CsakiFalkowskiWeiler2008WarpedFlavor`). Since then the important change is experimental: LHC searches now constrain the top-gluon FCNC operator directly. CMS used 7 and 8 TeV t-channel single-top data (`CMS2017TopFCNCTUG`), while ATLAS later used the full Run-2 139 fb^{-1} dataset at 13 TeV with a neural-network separation of $ug \to t$, $cg \to t$, and backgrounds (`ATLAS2022TopFCNCTUG`). The resulting limits remain many orders of magnitude above the SM prediction, so they are new-physics limits rather than SM precision tests.

Constraint validity and model dependence

The experimental upper bound is clean in the sense that the SM rate is negligible. Its use as an RS scan cut is model dependent: ATLAS quotes a left-handed SMEFT C_{uG}^{ct} interpretation, while CMS uses an older κ_{tcg} convention. The conversion from single-top production to $\mathcal{B}(t \to cg)$ assumes one FCNC operator at a time, standard top decays, and a specific normalization of the chromomagnetic interaction. A model producing both chiralities, contact terms, or correlated heavy-gluon resonance effects needs a dedicated reinterpretation.

In down-aligned or kaon-protected RS variants, top FCNCs become leading rather than secondary diagnostics.

Code coverage in this repo

NO. The required plan greps were rerun on 2026-05-16, followed by a targeted search for $t \to cg$, tcg, tqg, C_{uG}, and $cg \to t$ across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only implementation-directory hits were generic FCNC prose/comments at `quarkConstraints/PAPER_0710_1869.md:35` and `tests/test_paper_c`
The modern phenomenology surface enumerates only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenom`
and states that it is policy-only at `quarkConstraints/modern/phenomenology.py:165`.

Implementation difficulty

HIGH. Cataloging the external limit is straightforward, but live integration would need a top chromomagnetic FCNC operator convention, RS-to-SMEFT matching, decay-width and $cg \to t$ production normalization, PDF/collider acceptance assumptions or a validated recast, and QCD-running choices not present in the existing $\Delta F = 2$ code.

Key references

`PDG2025TopFCNCTUG`; `ATLAS2022TopFCNCTUG`; `CMS2017TopFCNCTUG`; `AguilarSaavedra2004TopFCNC`; `CsakiFalkowskiWeiler2008WarpedFlavor`.

56 $t \to ug$ Top-Gluon FCNC

Process

Standard notation: $\mathcal{B}(t \to ug)$. This entry covers the flavor-changing neutral-current top decay $t \to ug$, catalog id T006 in the top/Higgs/electroweak family. Experimentally it is constrained mainly through anomalous single-top production, $ug \to t$, then translated into an equivalent top-decay branching fraction under a specified chromomagnetic-operator convention.

PDG / equivalent value(s)

The canonical experimental source is the PDG/pdgLive Q007TUG top-FCNC gluon datablock, accessed on 2026-05-16 (`PDG2025:T006:ATLAS2022:t_ug`; `references/T006/pdg_2025_top_tug.txt`). Its current ATLAS entry, AAD 2022T, uses 139 fb^{-1} of 13 TeV pp data (`PDG2025:T006:ATLAS2022:dataset`). For the up-gluon production mode it gives

$$\sigma(ug \to t)\,\mathcal{B}(t \to bW)\,\mathcal{B}(W \to \ell\nu) < 3.0 \text{ pb}$$

at 95% CL (`PDG2025:T006:ATLAS2022:ug_to_t_cross_section`), with the ATLAS leptonic-W sum $\mathcal{B}(W \to \ell\nu) = 0.325$ (`PDG2025:T006:ATLAS2022:w_leptonic_sum`). The same entry gives

$$|C_{uG}^{ut}|/\Lambda^2 < 0.057 \text{ TeV}^{-2}, \qquad \mathcal{B}(t \to ug) < 6.1 \times 10^{-5}.$$

These coefficient and branching-fraction limits are recorded as `PDG2025:T006:ATLAS2022:CuGut` and `PDG2025:T006:ATLAS2022:t_ug`. The PDG block also records the CMS 7 and 8 TeV comparison result based on 5.0 fb^{-1} and 19.7 fb^{-1} (`CMS2017:T006:dataset`): $|\kappa_{tug}|/\Lambda < 4.1 \times 10^{-3}$ TeV^{-1}, corresponding to $\mathcal{B}(t \to ug) < 2.0 \times 10^{-5}$ (`CMS2017:T006:kappa_tug`; `CMS2017:T006:t_ug`). For SM context, not as a measured PDG value, Aguilar-Saavedra's charm-mode result and CKM scaling imply $\mathcal{B}(t \to ug) \simeq 3.6 \times 10^{-14}$ (`AguilarSaavedra2004:T006:t_ug_SM`).

Relevance to the RS / anarchic-flavor pipeline

This channel probes color-carrying first–third generation up-sector flavor violation rather than the neutral-meson $\Delta F = 2$ four-quark operators already audited in the quark scan. In anarchic warped flavor, the heavy color sector and the IR-localized top can generate non-universal mass-basis couplings and loop-level chromomagnetic dipoles. A $t \to ug$ limit is therefore a direct top-flavor diagnostic, complementary to D^0 mixing and to electroweak channels such as $t \to uZ$ or $t \to uH$.

Post-2008 developments

The paper-era CFW reference point, arXiv:0804.1954, predates the LHC single-top FCNC program and is used here only as RS-flavor context. Since then, CMS and ATLAS have interpreted single-top data in top-gluon FCNC operators, with the ATLAS Run-2 dataset supplying the current PDG/pdgLive entry. The resulting bounds remain far above the SM prediction, so they are new-physics limits rather than SM precision tests.

Constraint validity and model dependence

The experimental upper bound is clean because the SM signal is negligible at collider sensitivity. Its use as an RS scan cut is model dependent: ATLAS quotes a left-handed SMEFT C_{uG}^{ut} interpretation, while CMS uses an older κ_{tug} convention. The conversion from single-top production to $\mathcal{B}(t \to ug)$ assumes one FCNC operator at a time, standard top decays, and a specific normalization of the chromomagnetic interaction. A model producing both chiralities, contact terms, or correlated heavy-gluon resonance effects needs a dedicated reinterpretation.

In down-aligned or kaon-protected RS variants, top FCNCs become leading rather than secondary diagnostics.

Code coverage in this repo

NO. The required plan greps were rerun on 2026-05-16, followed by a targeted search for $t \to ug$, tug, utg, tqg, C_{uG}, and $ug \to t$ across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only implementation-directory hits were generic FCNC prose/comments at `quarkConstraints/PAPER_0710_1869.md:35` and `tests/test_paper_couplings.py:258`. The modern phenomenology surface enumerates only ϵ_K, K, B_d, B_s, and D^0 at `quarkConstraints/modern/phenomenology.py:23`, and states that it is policy-only at `quarkConstraints/modern/phenomenology.py:167`.

Implementation difficulty

HIGH. Cataloging the external limit is straightforward, but live integration would need a top chromomagnetic FCNC operator convention, RS-to-SMEFT matching, decay-width and $ug \to t$ production normalization, PDF/collider acceptance assumptions or a validated recast, and QCD-running choices not present in the existing $\Delta F = 2$ code.

Key references

`PDG2025TopFCNCTUG`; `ATLAS2022TopFCNCTUG`; `CMS2017TopFCNCTUG`; `AguilarSaavedra2004TopFCNC`; `CsakiFalkowskiWeiler2008WarpedFlavor`.

57 $t \to Hc$ FCNC Top-Higgs Decay

Process

Standard notation: $\mathcal{B}(t \to Hc)$. This entry catalogs the flavor-changing neutral-current top-Higgs decay $t \to Hc$ in the top/Higgs/electroweak family. It covers direct LHC searches for anomalous top–charm–Higgs interactions in both $t\bar{t}$ events with an FCNC top decay and associated single-top plus Higgs production.

PDG-or-equivalent value

The source-of-truth numerical entries are the `pdg_or_equivalent` block in `T007.yaml`. The PDG 2026 `pdgLive` top-quark datablock quotes the ATLAS-combination headline

$$\mathcal{B}(t \to Hc) < 3.4 \times 10^{-4}$$

at 95% CL (`PDG2026:T007:tHc_combined`), based on the ATLAS 2024 combined Higgs-decay search. The local PDG snapshot is `flavor_catalog/references/T007/pdg2026_top_hc_datablock.txt`. The ATLAS primary arXiv record gives 140 fb^{-1} at $\sqrt{s} = 13$ TeV (`ATLAS2024:T007:dataset`), a multilepton-only limit $\mathcal{B}(t \to Hc) < 3.3\times10^{-4}$ with expected 3.8×10^{-4} (`ATLAS2024:T007:tHc_multilepton`), and the combined limit $\mathcal{B}(t \to Hc) < 3.4\times10^{-4}$ with expected 2.3×10^{-4} (`ATLAS2024:T007:tHc_combined`). The CMS 2025 record uses 138 fb^{-1} at $\sqrt{s} = 13$ TeV collected in 2016–2018 (`CMS2025:T007:dataset`); its combined limit is $\mathcal{B}(t \to Hc) < 0.037\%$, expected 0.035% (`CMS2025:T007:tHc_combined`).

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models and composite-Higgs variants can generate off-diagonal scalar Yukawa couplings after rotating to the quark mass basis. The $t \to Hc$ mode probes the same top-sector localization and mixing structure that also controls the large top Yukawa, so it is complementary to light-meson $\Delta F = 2$ constraints. For this repo it is best treated as a direct top-sector diagnostic, not as a replacement for the existing kaon, B, and D-mixing gates.

Post-2008 developments

The requested baseline is arXiv:0804.1954 (`CsakiFalkowskiWeiler2008WarpedFlavor`), which predates direct LHC top-Higgs FCNC bounds. Since then, ATLAS and CMS moved the search from indirect model estimates to direct limits using many Higgs final states. ATLAS now combines $H \to WW$, $H \to ZZ$, $H \to \tau\tau$, $H \to \gamma\gamma$, and $H \to b\bar{b}$ channels in the PDG headline (`PDG2026:T007:tHc_combined`; `ATLAS2024:T007:tHc_combined`). CMS has an independent 2025 combination with $H \to \gamma\gamma$ and $H \to b\bar{b}$ inputs in addition to the same-sign lepton analysis (`CMS2025:T007:tHc_combined`). The classic top-FCNC review arXiv:hep-ph/0409342 remains useful context for why SM top FCNC Higgs amplitudes are negligible while BSM scalar flavor can be visible (`AguilarSaavedra2004TopFCNC`).

Constraint validity and model dependence

The experimental statement is a 95% CL upper limit on an assumed single anomalous top-Higgs flavor coupling. The model interpretation is operator-convention dependent: an RS implementation must choose whether the scan outputs scalar Htc, Hct, or SMEFT Wilson coefficients, and whether associated tH production is included together with $t\bar{t}$ FCNC decays. Correlations with $t \to Hu$ are analysis-dependent, so this entry keeps only the charm-channel bound as the T007 catalog value.

In down-aligned or kaon-protected RS variants, top FCNCs become leading rather than secondary diagnostics. Note: for the observed Higgs, on-shell $h \to tc$ is kinematically forbidden; Htc coupling constraints enter through $t \to Hc$ decay and associated single-top + H production.

Code coverage in this repo

Coverage is **NO**. The PKA reran the required plan greps and a targeted search for $t \to Hc$, tHc, tcH, Htc, and tqH across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only targeted non-doc hits were generic FCNC prose/comments in `quarkConstraints/PAPER_0710_1869.md:35` and `tests/test_paper_couplings.py:25` there is no $t \to Hc$ observable, likelihood, or limit check.

Implementation difficulty

Production integration is **HIGH**. Recording the external upper limit is straightforward, but using it in scans needs a new top-Higgs FCNC matching layer, a scalar-coupling convention, and a decision about combining tH production with $t\bar{t}$ FCNC-decay interpretations.

Key references

`PDG2026TopHc`; `ATLAS2024TopHcMultilepton`; `CMS2025TopHcHiggsMediated`; `AguilarSaavedra2004TopFCNC`; `CsakiFalkowskiWeiler2008WarpedFlavor`.

58 $t \to Hu$ FCNC Top-Higgs Decay

Process

Standard notation: $\mathcal{B}(t \to Hu)$. This entry catalogs the flavor-changing neutral-current top-Higgs decay $t \to Hu$, process id T008 in the top/Higgs/electroweak family. It covers direct LHC searches for anomalous Hut interactions in $t\bar{t}$ events with an FCNC top decay and in associated single-top plus Higgs production.

PDG value

The `pdg_or_equivalent` block in `T008.yaml` is the numerical source of truth. The PDG 2026 `pdgLive` top-quark datablock quotes

$$\mathcal{B}(t \to Hu) < 1.9 \times 10^{-4}$$

at 95% CL (`PDG2026:T008:tHu_headline`), based on the CMS $H \to \gamma\gamma$ result. The local PDG snapshot is `flavor_catalog/references/T008/pdg2026_top_hu_datablock.txt`. The CMS source uses 137 fb^{-1} at $\sqrt{s} = 13$ TeV and reports the same observed limit, 0.019%, with expected 0.031% (`CMS2022:T008:tHu_diphoton`). PDG also lists, below its "do not use" separator, the ATLAS multilepton limit $2.8(3.0) \times 10^{-4}$ observed(expected) and the ATLAS combination $2.6(1.8) \times 10^{-4}$ (`ATLAS2024:T008:tHu_multilepton`; `ATLAS2024:T008:tHu_combined`). The CMS 2025 same-sign-lepton search gives 0.072%(0.059%), and its combination with CMS $H \to \gamma\gamma$ and $H \to b\bar{b}$ searches gives 0.019%(0.027%) (`CMS2025:T008:tHu_samesign`; `CMS2025:T008:tHu_combined`).

Relevance to RS / anarchic flavor

Warped anarchic flavor and composite-Higgs variants can generate off-diagonal scalar Yukawa couplings after rotating to the quark mass basis. The $t \to Hu$ channel probes up-top scalar flavor, complementary to the charm channel in T007 and to light-meson $\Delta F = 2$ gates. Associated tH production is especially sensitive to an up-quark initial state, so the experimental interpretation is not identical to simply rescaling $t \to Hc$.

Post-2008 developments

The requested baseline, arXiv:0804.1954 (`CsakiFalkowskiWeiler2008WarpedFlavor`), predates direct LHC top-Higgs FCNC searches. Since then, ATLAS and CMS moved the constraint from model estimates to direct searches in diphoton, multilepton, $b\bar{b}$, and $\tau\tau$ Higgs final states. The PDG headline currently comes from the CMS diphoton search, while ATLAS and CMS now also provide multi-channel combinations that treat $t\bar{t}$ FCNC decay and associated tH production together.

Constraint validity / model dependence

The experimental statement is a direct 95% CL upper limit under a single anomalous Hut coupling assumption (`PDG2026:T008:tHu_headline`). Mapping it into this repo would require choosing a scalar-Yukawa or SMEFT convention, deciding whether Hut and Htu are independent, and preserving the analysis-specific combination of $t\bar{t}$ decay and associated-production topologies. Correlations with $t \to Hc$ are not universal, so T008 keeps only the up-channel bound.

Code coverage in this repo

Coverage is **NO**. The required plan greps and a targeted search for $t \to Hu$, tHu, tuH, Hut, Htu, and tqH were rerun across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/`. The only targeted hits outside the catalog were generic FCNC prose/comments in `quarkConstraints/PAPER_0710_1869.md:35` and `tests/test_paper_couplings.py:258`; there is no $t \to Hu$ observable, likelihood, matching layer, or limit check.

Implementation difficulty

Production integration is **HIGH**. Limit cataloging is straightforward, but a live scan needs a new top-Higgs FCNC matching layer, scalar-coupling normalization, and an analysis prescription for the tH plus $t\bar{t}$ interpretation.

Key references

`PDG2026TopHu`; `CMS2022TopHuDiphoton`; `ATLAS2024TopHuMultilepton`; `CMS2025TopHuHiggsMediated`; `CsakiFalkowskiWeiler2008WarpedFlavor`.

59 $Z \to b\bar{b}$ Pole Observables

Process

This entry covers the bottom-quark Z-pole observables R_b^0, $A_{\mathrm{FB}}^{0,b}$, and A_b. Plan v1 lists R_b as T010 and the asymmetries as T011; this entry follows the batch assignment and treats them as one $Z \to b\bar{b}$ pole-observable package.

PDG-or-equivalent value

The canonical current values are taken from the PDG Z-boson listing snapshot in `flavor_catalog/references/T01` the same snapshot is recorded in the `T010.yaml` `pdg_or_equivalent` block. It gives $R_b^0 = 0.21629 \pm 0.00066$, $A_{\mathrm{FB}}^{0,b} = 0.0992 \pm 0.0016$ after converting the listed $9.92 \pm 0.16\%$, and $A_b = 0.923 \pm 0.020$ (`pdg_2025_z_boson.txt`). The original LEP/SLC final combination is also snapshotted in `lepslc_2006_z_resonance.txt`; it records the same central values and identifies the $A_{\mathrm{FB}}^{0,b}$ comparison as the largest Z-pole pull, 2.8σ, relative to the Standard Model fit.

Relevance to the RS / anarchic-flavor pipeline

In warped models with anarchic five-dimensional Yukawas, the third-generation left-handed doublet is typically localized closer to the IR brane to account for the top mass. That same localization makes the $Zb_L\bar{b}_L$ coupling sensitive to gauge-boson and fermion mixing with Kaluza-Klein states. R_b mostly constrains the total $Z \to b\bar{b}$ partial width, while A_b and $A_{\mathrm{FB}}^{0,b}$ test the chiral balance of the bottom coupling. Custodial extensions, especially protections of $Zb_L\bar{b}_L$, are therefore not a minor detail; they determine whether this observable is a leading bound or a diagnostic cross-check.

Post-2008 developments

There has not been a new LEP/SLC-style experimental average superseding the Z-pole combination, so the PDG value remains legacy-data dominated. The post-CFW theory story did change: Casagrande et al. gave a detailed RS electroweak-precision treatment including Zbb couplings, while later SM work completed the fermionic two-loop description of Z partial widths and branching ratios (`freitas_2014_z_widths`). Future-collider studies now revisit the experimental systematics of R_b and $A^b_{\rm FB}$; the FCC-ee projection quotes relative uncertainties of order 0.01%, but it is a prospective reference rather than a current constraint (`fcc_ee_2025_zbb_projection.txt`).

Constraint validity and model dependence

This is a precision electroweak coupling constraint, not a flavor-changing amplitude by itself. It is highly relevant for RS flavor because the same fermion localizations and custodial choices that control flavor violation also set the tree-level Z-bottom coupling shifts. Its interpretation is therefore model-dependent: minimal bulk RS, custodial RS, brane kinetic terms, and different embeddings of the bottom multiplet can move the constraint substantially. $A^{0,b}_{\rm FB}$ also contains the initial-state leptonic asymmetry input, so A_b is the cleaner bottom-coupling handle.

In custodial RS, reduced Zb_Lb_L pressure can make this observable relatively less direct as a mass setter or more diagnostic of embeddings; quote RS bounds only after specifying custodial protection, fermion embeddings, and brane kinetic terms.

Code coverage in this repo

NO. The required greps found no implementation of R_b, A_b, $A^{0,b}_{\rm FB}$, or a Zbb electroweak-pole likelihood in `quarkConstraints/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `qcd/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. The only Z-related matches were generic M_Z support in `qcd/running.py:3`, `qcd/constants.py:11`, and `quarkConstraints/qcd_running.py:100`.

Implementation difficulty

HIGH. A catalog-only record is straightforward, but a live constraint would need a new electroweak-pole observable module, a correlated treatment of the LEP/SLC pseudo-observables, and model-specific matching from the 5D spectrum to δg_{Lb} and δg_{Rb}. The largest blocker is not lattice input or QCD running; it is the RS electroweak matching and custodial-symmetry bookkeeping.

Key references

`pdg_2025_z_boson`; `lepslc_2006_z_resonance`; `cfw_2008_rs_flavor`; `casagrande_2008_rs_ewpt`; `freitas_2014_z_widths`; `fcc_ee_2025_zbb_projection`.

60 $Z \to c\bar{c}$ Pole Observables

Process

Standard notation: R_c^0, $A_{\rm FB}^{0,c}$, and A_c. This entry covers the charm-quark Z-pole partial-width ratio and asymmetry observables under process id T012. Plan v1 lists R_c as T012 and the charm asymmetries as T013; the dispatch instruction asks for the combined $Z \to c\bar{c}$ pole-observable package under T012.

PDG value

The canonical current values are taken from the PDG 2025 Z-boson listing snapshot in `flavor_catalog/references` accessed on 2026-05-16 (`pdg_2025_z_boson_charm`). The measured values are

$$R_c^0 = 0.1721 \pm 0.0030, \qquad A_{\rm FB}^{0,c} = 0.0707 \pm 0.0035, \qquad A_c = 0.670 \pm 0.027.$$

For $A_{\rm FB}^{0,c}$, the PDG listing gives the Z-pole charge asymmetry as $7.07\pm0.35\%$; the dimensionless value above is the direct percent conversion (`pdg_2025_z_boson_charm`). The LEP/SLC final-combination snapshot in `flavor_catalog/references/T012/lepslc_2006_z_resonance_charm.txt` records the same three charm-observable central values and uncertainties (`lepslc_2006_z_resonance_charm`).

Relevance to RS / anarchic flavor

These observables constrain flavor-diagonal shifts in the left- and right-handed $Zc\bar{c}$ couplings. In anarchic warped models, the charm coupling shifts are usually less severe than the $Zb\bar{b}$ constraint because the charm is not tied to the same custodial-protection problem as the third-generation bottom doublet. They are still useful as a consistency check on up-sector fermion localizations, bulk multiplet assignments, and the same gauge-boson mixing effects that generate flavor-changing Z couplings. A large shift in $Zc\bar{c}$ would also feed the global Z-pole heavy-flavour fit that correlates charm and bottom pseudo-observables.

Post-2008 developments

No new LEP/SLC-style experimental average supersedes the legacy Z-pole charm combination, so the PDG value remains data-dominated by the final LEP/SLC heavy-flavour fit. Since arXiv:0804.1954 (`cfw_2008_rs_flavor`), the RS literature has made the model dependence of ordinary $Zf\bar{f}$ couplings more explicit; the Casagrande–Goertz–Haisch–Neubert–Pfoh setup treats the standard Z couplings, flavor-changing Z effects, and electroweak precision constraints in a unified warped-flavor framework (`casagrande_2008_rs_ewpt`). On the Standard-Model side, the fermionic electroweak two-loop calculation of Z partial widths and branching ratios was completed in arXiv:1401.2447 (`freitas_2014_z_widths`). Prospective FCC-ee work, including arXiv:2107.00616 (`alcaraz_2021_z_lineshape_fc` revisits how R_c and $A_{\rm FB}^c$ could be improved with modern heavy-flavour tagging, but those studies are not current experimental inputs.

Constraint validity / model dependence

This is a precision electroweak coupling constraint, not a flavor-changing decay or a stand-alone RS flavor bound. It is robust as a measured Z-pole pseudo-observable, but the translation into a bound on a 5D model is embedding-dependent: custodial symmetry, the representation of the up-sector quarks, brane kinetic terms, and fermion localization choices all affect the predicted δg_{Lc} and δg_{Rc}. R_c^0 is mostly a width constraint, while A_c and $A_{\rm FB}^{0,c}$ carry chiral information and depend on the electroweak-pole fit conventions.

Code coverage in this repo

NO. The required greps found no implementation of R_c, A_c, $A_{\rm FB}^{0,c}$, a Zcc coupling-shift calculation, or a correlated electroweak-pole likelihood in `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, or `tests/`. The only Z-related matches were generic M_Z support in `qcd/running.py:3`, `qcd/constants.py:11`, and `quarkConstraints/qcd_runn`
process-specific searches also returned unrelated R_χ hadronic helpers.

Implementation difficulty

HIGH. A live constraint would need a new electroweak-pole observable module, LEP/SLC covariance handling for heavy-flavour pseudo-observables, and model-specific matching from the warped spectrum to δg_{Lc} and δg_{Rc}. This is not covered by the existing $\Delta F = 2$ operator basis or the lepton-dipole path.

Key references

`pdg_2025_z_boson_charm`; `lepslc_2006_z_resonance_charm`; `cfw_2008_rs_flavor`; `casagrande_2008_rs_ewpt`; `freitas_2014_z_widths`; `alcaraz_2021_z_lineshape_fcc`.

61 $Z \to e\mu$ LFV Decay

Process

Standard notation: $\mathcal{B}(Z \to e^\pm\mu^\mp)$. This entry covers the charged-lepton-flavor-violating on-shell Z-boson decay into one electron and one muon, summed over charge-conjugate final states, in the top/Higgs/electroweak family.

PDG / equivalent value(s)

The source-of-truth values are in the `pdg_or_equivalent` block of `T015.yaml`. The sidecar-selected experiment-equivalent headline value is the CMS 2025 direct-search result

$$\mathcal{B}(Z \to e^\pm\mu^\mp) < 1.9 \times 10^{-7}$$

with an expected limit of 2.0×10^{-7}, at 95% CL, using 138 fb^{-1} of $\sqrt{s} = 13$ TeV data (`CMS2025ZLFV`; `CMS2025:T015:zemu_limit`; `CMS2025:T015:dataset`). The PDG 2025 Z-boson listing quotes the charge-summed limit

$$\mathcal{B}(Z \to e^{\pm}\mu^{\mp}) < 2.62 \times 10^{-7}$$

at 95% CL (`PDG2025:T015:zemu_limit`), matching the ATLAS Run-2 search with 139 fb^{-1} at $\sqrt{s} =$ 13 TeV (`PDG2025ZLFVListing`; `ATLAS2023ZLFVEmu`; `ATLAS2023:T015:zemu_limit`; `ATLAS2023:T015:dataset`). For historical comparison, the earlier ATLAS Run-1 search set $< 7.5 \times 10^{-7}$ with 20.3 fb^{-1} at $\sqrt{s} = 8$ TeV (`ATLAS2014ZLFVEmu`; `ATLAS2014:T015:zemu_limit`; `ATLAS2014:T015:dataset`).

Relevance to the RS / anarchic-flavor pipeline

Anarchic or partially composite lepton sectors can induce non-diagonal neutral currents after rotating to charged-lepton mass eigenstates. A direct $Z \to e\mu$ limit tests any effective off-diagonal $Ze\mu$ coupling from KK gauge mixing, zero-mode coupling non-universality, or lepton-sector flavor symmetry breaking. It is therefore relevant to a lepton/electroweak extension of the present pipeline, not to the existing quark $\Delta F = 2$ scan lane.

Post-2008 developments

The arXiv:0804.1954 warped-flavor baseline predates the LHC direct Z-LFV program (`CsakiFalkowskiWeiler2008` `PerezRandall2008WarpedLeptonFlavor`). Since then ATLAS established the 8 TeV $e\mu$ search at the 10^{-7} level (`ATLAS2014ZLFVEmu`), then improved it with the full Run-2 data set (`ATLAS2023ZLFVEmu`); CMS subsequently pushed the direct $e\mu$ limit below 2×10^{-7} (`CMS2025ZLFV`). On the theory side, post-LEP EFT studies of future Tera-Z facilities emphasize that indirect low-energy μ-to-e constraints can make $Z \to \mu e$ harder to observe than $Z \to \tau\ell$ even with $\mathcal{O}(10^{12})$ produced Z bosons (`CalibbiMarcanoRoy2021:T015:teraZ`).

Constraint validity and model dependence

Experimentally this is a clean direct upper limit on a forbidden Standard Model decay. Its use as a scan cut assumes SM-like inclusive Z production and total Z width. Model interpretations are less universal: in many lepton-flavor models the same operator structure is correlated with $\mu \to e\gamma$, $\mu \to 3e$, or coherent μ-to-e conversion, so this should be treated as a direct collider constraint on $Ze\mu$, not as a complete charged-lepton flavor-violation likelihood.

Code coverage in this repo

Coverage is **NO**. The required broad greps and a targeted search for `Z -> e mu`, `Z -> mu e`, `Zemu`, and related LFV-Z strings across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $Z \to e\mu$ observable implementation. The only charged-lepton LFV backend seen is $\mu \to e\gamma$ (`flavorConstraints/muToEGamma.py:75`), wired into the scan at `scanParams/scan.py:524`; the modern phenomenology policy surface is limited to ϵ_K, K, B_d, B_s, and D^0 (`quarkConstraints/modern/phenomenology.py:23`).

Implementation difficulty

Production integration is **HIGH**. Storing the external limit is straightforward, but a faithful RS constraint needs a new electroweak LFV operator convention, matching from lepton localization and gauge/Higgs-sector mixing to $Ze\mu$, and a decision on how to combine direct Z-pole limits with low-energy μ-flavor observables.

Key references

`PDG2025ZLFVListing`; `CMS2025ZLFV`; `ATLAS2023ZLFVEmu`; `ATLAS2014ZLFVEmu`; `CalibbiMarcanoRoy2021ZLFV`; `CsakiFalkowskiWeiler2008WarpedFlavor`; `PerezRandall2008WarpedLeptonFlavor`.

62 $Z \to e\tau$ LFV Decay

Process

Standard notation: $\mathcal{B}(Z \to e^{\pm}\tau^{\mp})$. This entry covers the charged-lepton-flavor-violating on-shell Z-boson decay into one electron and one tau lepton, summed over charge-conjugate final states, in the top/Higgs/electroweak family.

PDG / equivalent value(s)

The canonical current listing is the PDG 2025 Z-boson LFV row, which gives

$$\mathcal{B}(Z \to e^{\pm}\tau^{\mp}) < 5.0 \times 10^{-6}$$

at 95% CL for the charge-summed final state (`PDG2025:T016:zetau_limit`). This is the same headline limit as the combined ATLAS tau-lepton result (`ATLAS2021:T016:zetau_combined_limit`), obtained after combining the leptonically decaying tau search with previously published hadronic-tau analyses. The standalone ATLAS leptonically decaying tau channel gives $< 7.0 \times 10^{-6}$ at 95% CL (`ATLAS2021:T016:zetau_leptonic_limit`). CMS 2025 searched the same channel with 138 fb^{-1} at $\sqrt{s} = 13$ TeV and quotes an observed (expected) limit of 13.8 (11.4) $\times 10^{-6}$ at 95% CL (`CMS2025:T016:zetau_limit`; `CMS2025:T016:dataset`).

Relevance to the RS / anarchic-flavor pipeline

Warped or partially composite lepton sectors can generate non-universal neutral-current couplings before rotating to charged-lepton mass eigenstates. After that rotation, a non-diagonal $Ze\tau$ vertex can appear from gauge KK mixing, zero-mode coupling distortions, or lepton-sector flavor breaking. This process is therefore a direct collider probe of electroweak lepton-flavor violation in a lepton-sector extension, not a constraint implemented in the current quark $\Delta F = 2$ scan.

Post-2008 developments

The arXiv:0804.1954 warped-flavor baseline predates the LHC Z-LFV program. Since then ATLAS superseded the old LEP-era $Z \to e\tau$ sensitivity with Run-2 searches using hadronically and leptonically decaying tau channels; the combined ATLAS result is now the PDG-listed 5.0×10^{-6} bound (`PDG2025:T016:zetau_limit`). CMS added a full-Run-2 search covering $Z \to e\mu$, $Z \to e\tau$, and $Z \to \mu\tau$; its $e\tau$ result is weaker than the ATLAS/PDG bound but is an independent experimental cross-check. Phenomenology work on future e^+e^- colliders has also sharpened the interpretation: a Tera-Z program with $\mathcal{O}(10^{12})$ Z decays is expected to be more promising for $Z \to \tau\ell$ than for $Z \to \mu e$, because low-energy μ-flavor constraints are much more restrictive (`CalibbiMarcanoRoy2021:T016:teraZ`).

Constraint validity and model dependence

Experimentally this is a clean direct upper limit on a forbidden Standard Model decay, normalized to the inclusive Z branching fraction. As a model constraint it is not universal: the same effective $Ze\tau$ structures may also induce tau LFV decays, dipoles, or four-lepton operators depending on the UV completion. In the catalog it should be classified as a direct collider constraint on an off-diagonal $Ze\tau$ coupling, with correlations to low-energy tau observables left for a future lepton-sector likelihood.

Code coverage in this repo

Coverage is **NO**. The required broad greps and a targeted search for `Z -> e tau`, `Z -> tau e`, `Zetau`, and related LFV-Z strings across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $Z \to e\tau$ observable implementation. The only charged-lepton LFV backend seen is $\mu \to e\gamma$ (`flavorConstraints/muToEGamma` called by the scan at `scanParams/scan.py:524`; the modern phenomenology policy surface is limited to ϵ_K, K, B_d, B_s, and D^0 (`quarkConstraints/modern/phenomenology.py:23`).

Implementation difficulty

Production integration is **HIGH**. Storing the external branching-ratio limit is simple, but a live RS constraint needs a new electroweak LFV operator convention, matching from lepton localization and Z-coupling distortions to left- and right-handed $Ze\tau$ amplitudes, and a policy for combining the direct Z-pole bound with low-energy tau LFV limits.

Key references

`PDG2025ZLFVListing`; `ATLAS2021ZLFVTau`; `ATLAS2020ZLFVTauHad`; `CMS2025ZLFV`; `CalibbiMarcanoRoy2021ZLFV`; `CsakiFalkowskiWeiler2008WarpedFlavor`; `PerezRandall2008WarpedLeptonFlavor`.

63 $Z \to \mu\tau$ LFV Decay

Process

Standard notation: $\mathcal{B}(Z \to \mu^{\pm}\tau^{\mp})$. This entry covers the charged-lepton-flavor-violating on-shell Z-boson decay into one muon and one tau lepton, summed over charge-conjugate final states, in the top/Higgs/electroweak family.

PDG / equivalent value(s)

The PDG 2025 Z-boson listing quotes the charge-summed upper limit

$$\mathcal{B}(Z \to \mu^{\pm}\tau^{\mp}) < 6.5 \times 10^{-6}$$

at 95% CL (`PDG2025:T017:zmutau_limit`). This matches the strongest ATLAS combination, which combined a 139 fb^{-1}, $\sqrt{s} = 13$ TeV leptonically decaying τ search with previous hadronic-τ analyses (`ATLAS2021:T017:dataset`; `ATLAS2021:T017:zmutau_combined_limit`). The same ATLAS leptonic τ analysis alone set $\mathcal{B}(Z \to \mu\tau) < 7.2 \times 10^{-6}$ at 95% CL (`ATLAS2021:T017:zmutau_leptonic_tau_limit`). CMS 2025 is the latest direct LHC search including this channel; using 138 fb^{-1} at $\sqrt{s} = 13$ TeV, it reports an observed (expected) 95% CL limit of $12.0\,(5.3) \times 10^{-6}$ (`CMS2025:T017:dataset`; `CMS2025:T017:zmutau_limit`).

Relevance to the RS / anarchic-flavor pipeline

Warped or partially composite lepton sectors can generate non-diagonal neutral currents after the charged-lepton mass matrix is diagonalized. $Z \to \mu\tau$ is therefore a direct probe of effective off-diagonal $Z_{\mu\tau}$ couplings from lepton localization, gauge-boson mixing, or flavor-symmetry breaking. It is relevant to a lepton/electroweak extension of the pipeline, not to the present quark $\Delta F = 2$ implementation.

Post-2008 developments

The arXiv:0804.1954 warped-flavor baseline predates the LHC Z-LFV program. The main post-2008 experimental development is the ATLAS τ-channel search and combination, which established the current strongest observed $\mu\tau$ limit at the 10^{-6} branching-fraction level (`ATLAS2021:T017:zmutau_combined_li`
CMS later performed a Run-2 search covering all three charged-lepton flavor pairs in one analysis, providing an independent 2025 check of the $\mu\tau$ channel. Future-factory studies also changed the context: Tera-Z proposals with $\mathcal{O}(10^{12})$ produced Z bosons are especially promising for $Z \to \tau\ell$ modes, whereas $Z \to \mu e$ remains more tightly constrained by low-energy observables (`CalibbiMarcanoRoy2021:T017:teraZ`).

Constraint validity and model dependence

Experimentally this is a clean direct upper limit on a forbidden Standard Model decay. Applying it as a scan cut assumes the usual inclusive Z production and total-width normalization used by the collaborations. As an RS constraint it is model-dependent: the same flavor spurions may also induce $\tau \to \mu\gamma$, $\tau \to \mu\mu\mu$, or Higgs LFV, so this entry should be treated as a direct collider constraint on $Z_{\mu\tau}$, not a complete LFV likelihood.

Code coverage in this repo

Coverage is **NO**. The required broad greps and a targeted search for `Z -> mu tau`, `Z -> tau mu`, `Zmutau`, and related LFV Z strings across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $Z \to \mu\tau$ observable implementation. The only charged-lepton LFV backend seen is $\mu \to e\gamma$ (`flavorConstraints/muToEGamma` called by the scan at `scanParams/scan.py:524`; the modern quark phenomenology policy is limited to ϵ_K, K, B_d, B_s, and D^0 (`quarkConstraints/modern/phenomenology.py:23`).

Implementation difficulty

Production integration is **HIGH**. The external limit is easy to store, but a faithful RS constraint needs a new LFV Z-coupling convention, left- and right-handed lepton operator matching, model-dependent gauge/mass mixing inputs, and a policy for combining direct Z-pole searches with tau LFV and Higgs LFV constraints.

Key references

`PDG2025ZLFVListing`; `ATLAS2021ZLFVTau`; `CMS2025ZLFV`; `CalibbiMarcanoRoy2021ZLFV`; `CsakiFalkowskiWeiler2` `PerezRandall2008WarpedLeptonFlavor`.

64 $h \to \mu\tau$ LFV Higgs Decay

Process

Standard notation: $\mathcal{B}(h \to \mu^{\pm}\tau^{\mp})$. This entry catalogs the lepton-flavor-violating Higgs decay $h \to \mu\tau$ in the top/Higgs/electroweak family. It covers direct ATLAS and CMS searches for a neutral Higgs boson decaying to a muon and an oppositely charged tau lepton.

PDG-or-equivalent value

The source-of-truth entries are in the `pdg_or_equivalent` block of `T018.yaml`. The PDG 2025 Higgs review summarizes the full $\sqrt{s} = 13$ TeV Run-2 limits (`PDG2025HiggsLFVReview`):

$$\mathcal{B}(h \to \mu\tau) < 0.15\%$$

from CMS, with the same 0.15% expected limit, at 95% CL (`CMS2021:T018:mutau_limit`; `PDG2025:T018:cms_run2` The CMS analysis used 137 fb^{-1} (`CMS2021:T018:dataset`). The latest ATLAS Run-2 result gives

$$\mathcal{B}(h \to \mu\tau) < 0.18\% \quad (0.09\% \text{ expected})$$

at 95% CL using 138 fb^{-1} (`ATLAS2023:T018:mutau_limit`; `ATLAS2023:T018:dataset`). The CMS observed limit is the strongest single-experiment headline value in the local snapshots.

Relevance to the RS / anarchic-flavor pipeline

In warped or partially composite lepton sectors, fermion localization explains charged-lepton masses, but misalignment between the mass matrix and physical Higgs couplings can generate off-diagonal Yukawa interactions. The $h \to \mu\tau$ limit directly tests the μ-τ charged-lepton Yukawa entries and is complementary to dipole LFV constraints such as $\mu \to e\gamma$. It is therefore relevant only to a lepton-sector extension of the current quark scan, not to the existing $\Delta F = 2$ quark lane.

Post-2008 developments

The requested baseline is arXiv:0804.1954, before the LHC Higgs LFV program (`CsakiFalkowskiWeiler2008Warpe`
Harnik, Kopp, and Zupan later emphasized that flavor-violating Higgs decays could be a direct discovery channel and that $h \to \tau\mu$ searches could beat rare-tau bounds (`HarnikKoppZupan2012:order10`). CMS then reported the historical 8 TeV hint: a 2.4σ excess with $p = 0.010$, best fit $\mathcal{B}(H \to \mu\tau) = (0.84^{+0.39}_{-0.37})\%$, and a 95% CL limit of 1.51% using 19.7 fb^{-1} (`CMS2015:T018:hint`; `CMS2015:T018:dataset`; `CMS2015:T018:limit`). The full Run-2 CMS and ATLAS analyses did not confirm a significant signal and pushed the direct limits to the 10^{-3}-branching-fraction level (`CMS2021:T018:mutau_limit`; `ATLAS2023:T018:mutau_limit`).

Constraint validity and model dependence

Experimentally this is a direct, clean upper limit on a forbidden SM decay. The interpretation assumes the signal production mixture and total Higgs width used by each collaboration; a model that changes Higgs production or the total width needs a recast rather than a literal branching-ratio cut. Low-energy LFV bounds constrain related Yukawa products, but the direct $h \to \mu\tau$ search is the least model-dependent way to constrain this specific final state.

Code coverage in this repo

Coverage is **NO**. The required greps and a targeted search for `h->mu tau`, `H->mu tau`, `mutau`, and `Y_mu_tau` across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no observable implementation. The existing LFV backend is specifically $\mu \to e\gamma$ (`flavorConstraints/muToEGamma.py:1`; `flavorConstraints/muToEGam`
and the scanner applies that filter at `scanParams/scan.py:523`. Charged-lepton Yukawa helpers are diagonal mass-relation utilities, not an $h \to \mu\tau$ prediction.

Implementation difficulty

Production integration is **HIGH**. The external limit is easy to store, but a faithful RS constraint needs a new charged-lepton Higgs-Yukawa misalignment calculation, a convention for $\mu\tau$ versus $\tau\mu$ couplings, and a Higgs-rate/width interpretation before scan points can be accepted or rejected.

Key references

PDG2025HiggsLFVReview; CMS2021HiggsLFVMuTau; ATLAS2023HiggsLFVMuTau; CMS2015HiggsLFVMuTauHint; HarnikKoppZupan2012FlavorViolatingHiggs; CsakiFalkowskiWeiler2008WarpedFlavor; PerezRandall2008Wa

65 $h \to e\tau$ LFV Higgs Decay

Process

Standard notation: $\mathcal{B}(h \to e^{\pm}\tau^{\mp})$. This entry catalogs the lepton-flavor-violating Higgs decay $h \to e\tau$ in the top/Higgs/electroweak family. It covers direct ATLAS and CMS searches for a neutral Higgs boson decaying to an electron and an oppositely charged tau lepton.

PDG / equivalent value(s)

The source-of-truth values are in the pdg_or_equivalent block of T019.yaml. The PDG 2025 Higgs review summarizes the full $\sqrt{s} = 13$ TeV Run-2 direct limits on LFV Higgs decays (PDG2025HiggsLFVReview). For $h \to e\tau$, ATLAS gives

$$\mathcal{B}(h \to e\tau) < 0.20\% \quad (0.12\% \text{ expected})$$

at 95% CL using 138 fb^{-1} (ATLAS2023HiggsLFVETau; PDG2025:T019:atlas_run2; ATLAS2023:T019:dataset). CMS gives

$$\mathcal{B}(h \to e\tau) < 0.22\% \quad (0.16\% \text{ expected})$$

at 95% CL using 137 fb^{-1} (CMS2021HiggsLFVETau; PDG2025:T019:cms_run2; CMS2021:T019:dataset). The strongest observed single-experiment headline value in the local snapshots is therefore the ATLAS 2.0×10^{-3} branching-fraction limit (ATLAS2023:T019:etau_limit).

Relevance to the RS / anarchic-flavor pipeline

In warped or partially composite lepton sectors, flavor-dependent localization can explain charged-lepton masses while leaving the physical Higgs Yukawa matrix misaligned with the mass matrix. The $h \to e\tau$ limit directly tests the e-τ off-diagonal charged-lepton Yukawa entries. It is complementary to dipole constraints such as $\mu \to e\gamma$, but it is not part of the current quark $\Delta F = 2$ scan lane.

Post-2008 developments

The baseline arXiv:0804.1954 predates direct LHC Higgs LFV searches (CsakiFalkowskiWeiler2008WarpedFlavor Harnik, Kopp, and Zupan later emphasized that $h \to \tau e$ could be a discovery channel and that branching fractions of order 10% were still possible in generic LFV-Higgs EFT settings after low-energy constraints (HarnikKoppZupan2012FlavorViolatingHiggs). ATLAS performed an early 13 TeV search with 36.1 fb^{-1}, setting $\mathcal{B}(H \to e\tau) < 0.47\%$ (ATLAS2019HiggsLFVETau). The full

Run-2 CMS and ATLAS analyses then pushed the direct e-tau limit to the few-10^{-3} level, with no significant excess (`CMS2021HiggsLFVETau`; `ATLAS2023HiggsLFVETau`).

Constraint validity and model dependence

Experimentally this is a direct upper limit on a Standard-Model-forbidden decay. A literal branching-ratio cut assumes the Higgs production mixture and total width used by the collaborations. A model with altered Higgs production rates, new invisible width, or correlated LFV tau decays would need a recast. For the catalog, the limit is robust as an experimental observable but model-dependent as an RS acceptance criterion.

Code coverage in this repo

Coverage is **NO**. The required greps and a targeted search for `h->e tau`, `H->e tau`, `etau`, `Y_e_tau`, and `Y_tau_e` across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no observable implementation. The related LFV backend is specifically $\mu \to e\gamma$ (`flavorConstraints/muToEGamma.py:1`; `flavorConstraints/muToEGamma.py:75`), and the scan applies that filter at `scanParams/scan.py:523`–`scanParams/scan.py:524`. The charged-lepton Yukawa helper is diagonal (`yukawa/charged_lepton.py:1`; `yukawa/charged_lepton.py:36`).

Implementation difficulty

Production integration is **HIGH**. The external limit is simple to catalog, but a faithful RS constraint needs a new charged-lepton Higgs-Yukawa misalignment calculation, a convention for $Y_{e\tau}$ and $Y_{\tau e}$, and a Higgs-rate/width interpretation before scan points can be accepted or rejected.

Key references

`PDG2025HiggsLFVReview`; `ATLAS2023HiggsLFVETau`; `CMS2021HiggsLFVETau`; `ATLAS2019HiggsLFVETau`; `HarnikKoppZupan2012FlavorViolatingHiggs`; `CsakiFalkowskiWeiler2008WarpedFlavor`; `PerezRandall2008Wa`

66 $h \to e\mu$ LFV Higgs Decay

Process

Standard notation: $\mathcal{B}(h \to e^{\pm}\mu^{\mp})$. This entry catalogs the lepton-flavor-violating Higgs decay $h \to e\mu$ in the top/Higgs/electroweak family. It covers direct ATLAS and CMS searches for a 125 GeV Higgs boson decaying to an opposite-sign electron–muon pair (`PDG2025HiggsLFVReview`).

PDG-or-equivalent value(s)

The source-of-truth values are in `T020.yaml:pdg_or_equivalent`. The PDG 2025 Higgs review summarizes the full $\sqrt{s} = 13$ TeV Run-2 direct limits on LFV Higgs decays (`PDG2025HiggsLFVReview`). For $h \to e\mu$, PDG quotes

$$\mathcal{B}(h \to e\mu) < 4.4 \times 10^{-5} \quad (4.7 \times 10^{-5} \text{ expected})$$

from CMS at 95% CL using 138 fb^{-1} (`CMS2023HiggsLFVEMu`; `PDG2025:T020:cms_run2`). PDG also quotes the ATLAS limit

$$\mathcal{B}(h \to e\mu) < 6.2 \times 10^{-5} \quad (5.9 \times 10^{-5} \text{ expected})$$

at 95% CL (`PDG2025:T020:atlas_run2`). The primary ATLAS abstract reports the same search as 6.2×10^{-5} observed and 5.9×10^{-5} expected with 139 fb^{-1} (`ATLAS2020HiggsEMu`); this agrees with the PDG value kept as the catalog headline. The CMS number is the strongest direct single-experiment limit.

Relevance to the RS / anarchic-flavor pipeline

Warped or partially composite lepton sectors can generate off-diagonal physical Higgs Yukawa couplings after the charged-lepton mass matrix is diagonalized. The $h \to e\mu$ search directly probes the e-μ entries of that Higgs Yukawa matrix and is therefore complementary to the repo's existing $\mu \to e\gamma$ dipole filter. It is not a constraint on the current quark $\Delta F = 2$ scan by itself; it belongs to a future lepton-sector extension.

Post-2008 developments

The requested arXiv:0804.1954 baseline predates the LHC Higgs LFV program (`CsakiFalkowskiWeiler2008Warpe`
Harnik, Kopp, and Zupan then framed flavor-violating Higgs decays as direct discovery channels and showed that LFV Higgs branching fractions can be sizable in generic EFT settings (`HarnikKoppZupan2012FlavorViolatingHiggs`). ATLAS produced a full Run-2 $h \to e\mu$ limit with 139 fb^{-1} (`ATLAS2020HiggsEMu`). CMS later used 138 fb^{-1}, improved the direct Higgs limit to 4.4×10^{-5}, and extended the same $e\mu$ mass scan to 110–160 GeV. The largest excess was near 146 GeV, with local/global significance $3.8/2.8\sigma$; CMS did not interpret it as evidence (`CMS2023HiggsLFVEMu`; `CMS2023:T020:mass_scan`).

Constraint validity and model dependence

Experimentally this is a clean direct upper limit on a Standard-Model-forbidden Higgs decay. A literal branching-ratio cut assumes the SM-like Higgs production mixture and total width used by ATLAS and CMS. Models with altered Higgs production, exotic Higgs width, or correlated low-energy LFV signals need a recast. PDG notes that indirect $\mu \to e\gamma$-based constraints are stronger for this channel, so this entry should be used as the direct collider observable rather than the global e-μ LFV bound.

Code coverage in this repo

Coverage is **NO**. The required greps and a targeted search for `h->e mu`, `H->e mu`, `Y_e_mu`, and `Y_mu_e` across `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no observable implementation. The existing lepton LFV backend is specifically $\mu \to e\gamma$ (`flavorConstraints/muToEGamma.py:1`; `flavorConstraints/muToEGamma.py` and the scan applies that filter at `scanParams/scan.py:523`–`scanParams/scan.py:524`. The charged-lepton Yukawa helper is diagonal (`yukawa/charged_lepton.py:1`; `yukawa/charged_lepton.py:36`). See `T020.yaml:code_coverage`.

Implementation difficulty

Production integration is **HIGH**. The direct limit is easy to catalog, but a scan constraint needs a new charged-lepton Higgs-Yukawa misalignment calculation, a convention for $Y_{e\mu}$ and $Y_{\mu e}$, and a Higgs-rate or total-width interpretation before accepting or rejecting points (`T020.yaml:implementation_difficu`

Key references

`PDG2025HiggsLFVReview`; `CMS2023HiggsLFVEMu`; `ATLAS2020HiggsEMu`; `HarnikKoppZupan2012FlavorViolatingH` `CsakiFalkowskiWeiler2008WarpedFlavor`; `PerezRandall2008WarpedLeptonFlavor`.

67 Charged Lepton (LFV)

68 L001: $\mu \to e\gamma$ Dipole LFV

Process

Standard notation: $\mathcal{B}(\mu^+ \to e^+\gamma)$. This entry covers the charged-lepton-flavor-violating radiative muon decay and the dipole-style bound on the off-diagonal $e\mu$ lepton-flavor spurion used by the current RS lepton-sector scan. It is a charged-lepton process, not part of the quark $\Delta F = 2$ lane.

PDG-or-equivalent value

The current primary-experiment limit is the MEG II 2025 result recorded in `L001.yaml:primary_current_limit`,

$$\mathcal{B}(\mu^+ \to e^+\gamma) < 1.5 \times 10^{-13} \quad (90\% \text{ C.L.}),$$

from the 2021–2022 MEG II data set, with quoted search sensitivity 2.2×10^{-13}. The local snapshot is `flavor_catalog/references/L001/megii2025_arxiv2504_15711.txt`. For audit context, `L001.yaml:pdg_2025_listing` records the PDG 2025 muon-listing value

$$\mathcal{B}(\mu^+ \to e^+\gamma) < 3.1 \times 10^{-13} \quad (90\% \text{ C.L.}),$$

which is the MEG+MEG II 2021 combined limit and appears to lag the later MEG II 2025 publication.

Relevance to RS / anarchic flavor

In the lepton extension of the warped model, $\mu \to e\gamma$ probes brane-localized or loop-induced dipoles controlled by the off-diagonal combination $(\bar{Y}_N \bar{Y}_N^\dagger)_{12}$. The repo implements the Perez–Randall NDA form $\mathcal{B} \simeq 4 \times 10^{-8} \, |(\bar{Y}_N \bar{Y}_N^\dagger)_{12}|^2 (3\,\mathrm{TeV}/M_{\mathrm{KK}})^4$, converts an experimental branching-ratio limit to $C = \sqrt{\mathcal{B}_{\mathrm{lim}}/(4 \times 10^{-8})}$, and uses the updated MEG II limit in the scan default. Those numerical inputs are the `L001.yaml:paper_era_reference` and `L001.yaml:repo_default` entries: the live default gives $C \simeq 1.936 \times 10^{-3}$, compared with the paper-era $C = 0.02$.

Post-2008 developments

Relative to the contemporaneous Perez–Randall lepton paper, the experimental story changed substantially. Perez–Randall used the MEGA-era $\mathcal{B}(\mu \to e\gamma) < 1.2\times10^{-11}$ input (`L001.yaml:paper_era_reference`). The full MEG 2009–2013 data set then gave 4.2×10^{-13}, MEG II's 2021-only result gave 7.5×10^{-13}, and the MEG+MEG II combination gave 3.1×10^{-13} (`L001.yaml:prior_experimental_limits`). The MEG II 2021–2022 analysis now sets the 1.5×10^{-13} primary value.

Constraint validity / model dependence

The experimental bound is clean: a signal would be charged-lepton flavor violation with negligible Standard Model background. Its translation into this repo is model-dependent, because it relies on the chosen RS lepton spurion, NDA prefactor, and M_{KK} convention rather than a full loop calculation. The catalog should therefore classify it as a robust experimental limit but a lepton-extension-only, dipole/NDA model interpretation.

Code coverage in this repo

YES. The sidecar `code_coverage.evidence` block lists the implementation in `flavorConstraints/muToEGamma.p` the module documents the $\mu \to e\gamma$ dipole constraint, records the paper-era $C = 0.02$ coefficient, defines `PREFAC_BR` $= 4.0\times10^{-8}$, converts branching-ratio limits to C, and exposes `check_mu_to_e_gamma`. The scan default is the MEG II 2025 value: `scanParams/scan.py`:32–33 sets `BR_LIMIT_MEGII_2025` $=$ 1.5×10^{-13}, and line 264 makes it the default `ScanConfig.br_limit`. Tests at `tests/test_mu_to_e_gamma.py`:40 and `tests/test_scan.py`:50 exercise the updated limit.

Implementation difficulty

LOW for cataloging and present code coverage, because the observable is already implemented and the scan default matches the primary 2025 result. A future production refresh is **MEDIUM**: the code should audit the paper-era NDA normalization, M_{KK} convention, and whether the final catalog should expose both paper-reproduction and MEG II 2025 modes.

Key references

Process-local source keys before bibliography consolidation: `MEGII2025_MuEGamma`, `MEGII2024_MuEGammaFirst`, `MEG2016_MuEGammaFull`, `PDG2025_MuonListing`, and `PerezRandall2008_WarpedNeutrinos`.

69 L002: $\mu \to 3e$ Four-Lepton LFV

Process

Standard notation: $\mathcal{B}(\mu^+ \to e^+e^-e^+)$. This entry covers the neutrinoless charged-lepton-flavor-violating three-electron muon decay, with emphasis on four-lepton/contact operators and the off-shell dipole contribution. It is a charged-lepton process and is separate from the quark $\Delta F = 2$ scan lane.

PDG-or-equivalent value

The current PDG 2026 muon listing, datablock S004R4, gives the sidecar `L002.yaml:primary_current_limit` value

$$\mathcal{B}(\mu^+ \to e^+e^-e^+) < 1.0 \times 10^{-12} \quad (90\% \text{ C.L.}).$$

PDG attributes the limit to the 1988 SINDRUM spectrometer result of Bellgardt et al.; the same value is recorded in `L002.yaml:original_experiment`. The local PDG snapshot is `flavor_catalog/references/L` the SINDRUM metadata snapshot is `flavor_catalog/references/L002/sindrum1988_inspire_abstract.txt`. This remains an upper limit, not a measurement. The sidecar `prospects` entries keep Mu3e separate from the present bound: the phase-I technical design targets a single-event sensitivity of 2×10^{-15}, while the broader program aims at $\mathcal{O}(10^{-16})$ after future upgrades.

Relevance to RS / anarchic flavor

In a warped lepton extension, $\mu \to 3e$ probes flavor-violating neutral currents and four-lepton operators generated by KK gauge exchange, Z-like mixing, or other lepton-current misalignment. It is complementary to L001: $\mu \to e\gamma$ is dominantly dipole-like in the current repo, while $\mu \to 3e$ can be tree-level/contact dominated and therefore has different dependence on lepton localization and five-dimensional Yukawa assumptions. The Agashe–Blechman–Petriello RS LFV analysis already treated rare muon decays as part of the lepton-sector flavor test set.

Post-2008 developments

Relative to the arXiv:0804.1954 RS-flavor baseline, the experimental limit has not yet been superseded in PDG; SINDRUM still sets the catalog value. The main post-2008 change is the Mu3e program at PSI: the phase-I detector design, published in 2021, is built for 10^8 muon decays per second and a 2×10^{-15} single-event sensitivity (`L002.yaml:prospects`). The 2025 status note records a phase-I sensitivity scale of $\mathcal{O}(10^{-15})$ and an upgraded scale of $\mathcal{O}(10^{-16})$. On the theory side, the sidecar source `crivellin2017_mu_e_eft_arxiv1702_03020.txt` covers systematic EFT/RG analyses of $\mu \to e\gamma$, $\mu \to 3e$, and coherent conversion below m_W, including QED/QCD operator mixing.

Constraint validity / model dependence

The experimental interpretation is robust: the Standard-Model rate with massive neutrinos is negligible, quoted by Mu3e status material as $\mathcal{O}(10^{-54})$, so an observation would be new physics. The mapping to this repo is model-dependent because a four-lepton Wilson basis, dipole interference convention, and possible RG evolution must be chosen. This entry should be classified as lepton-extension-only and contact-operator sensitive, not as a quark-scan constraint.

Code coverage in this repo

NO. The required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $\mu \to 3e$, Mu3e, or four-lepton-contact implementation. The only adjacent charged-lepton LFV path is the dipole $\mu \to e\gamma$ checker at `flavorConstraints/muToEGamma.py`:75 and its scan call at `scanParams/scan.py`:524, which do not evaluate $\mu \to 3e$.

Implementation difficulty

MEDIUM. A minimal implementation needs a new low-energy $\mu \to 3e$ observable formula, a four-lepton Wilson/operator convention, and a matching hook from the RS lepton-sector parameters. It should not require lattice matrix elements or hadronic long-distance inputs. A production EFT treatment with QED/QCD RG mixing and correlated $\mu \to e\gamma$, $\mu \to 3e$, and μ-to-e conversion fits would raise the integration difficulty.

Key references

Process-local source keys before bibliography consolidation: `PDG2026_MuonS004R4`, `SINDRUM1988_Mu3e`, `Mu3eTDR2021`, `Mu3eStatus2025`, `CrivellinDavidsonPrunaSigner2017_MuEFT`, `AgasheBlechmanPetriello2006_R` and `CFW2008`.

70 L003: $\mu - e$ Conversion in Aluminum

Process

Standard notation: $R_{\mu e}^{\mathrm{Al}} = \Gamma(\mu^- \mathrm{Al} \to e^- \mathrm{Al})/\Gamma_{\mathrm{capture}}(\mu^- \mathrm{Al})$. This entry covers coherent neutrinoless muon-to-electron conversion on an aluminum target, the design target for Mu2e and COMET. It is a charged-LFV nuclear process and is outside the current quark $\Delta F = 2$ scan lane.

PDG-or-equivalent value

The PKA snapshots contain no published direct aluminum-target limit (`L003.yaml:pdg_or_equivalent.aluminum` The current published benchmark across targets is the SINDRUM II gold result,

$$R_{\mu e}^{\mathrm{Au}} < 7 \times 10^{-13} \quad (90\% \text{ C.L.}),$$

from `L003.yaml:pdg_or_equivalent.current_world_limit`; the local snapshot is `flavor_catalog/references/L`
This is not an aluminum measurement. Aluminum inputs in the sidecar are prospects: Mu2e quotes 2.8×10^{-17} single-event sensitivity and an expected 90% C.L. upper limit 6.7×10^{-17} (`L003.yaml:pdg_or_equivalent.aluminum_projections[0]`), Mu2e Run I quotes $R_{\mu e} < 6.2 \times 10^{-16}$ expected sensitivity and 1.2×10^{-15} discovery sensitivity (`L003.yaml:pdg_or_equivalent.aluminum_project`
COMET Phase-I quotes 3.1×10^{-15} single-event sensitivity and a 7.0×10^{-15} expected upper limit (`L003.yaml:pdg_or_equivalent.aluminum_projections[2]`), and COMET Phase-II planning quotes 2.6×10^{-17} single-event sensitivity (`L003.yaml:pdg_or_equivalent.aluminum_projections[3]`).

Relevance to RS / anarchic flavor

In a warped lepton-flavor extension, coherent conversion probes dipole, scalar, and vector lepton-quark operators. KK gauge exchange, Z-like flavor violation, Higgs-mediated scalar currents, and lepton-sector dipoles can feed the same nuclear final state. The observable is complementary to $\mu \to e\gamma$ because it can be contact-operator dominated, so it tests lepton-extension structure not covered by the current dipole checker.

Post-2008 developments

Relative to the arXiv:0804.1954 RS-flavor baseline (`L003.yaml:pdg_or_equivalent.post_2008_baseline`), the main change is experimental. Mu2e published staged aluminum sensitivity projections, and Mu2e-II records a proposed PIP-II upgrade aiming for at least one additional order of magnitude beyond Mu2e (`L003.yaml:pdg_or_equivalent.aluminum_projections[4]`). COMET published Phase-I and Phase-II aluminum strategies (`L003.yaml:pdg_or_equivalent.aluminum_projections[2]` and `L003.yaml:pdg_or_equivalent.aluminum_projections[3]`). The process therefore moved from SINDRUM-II context to a near-term dedicated aluminum search program.

Constraint validity / model dependence

The experimental signature is robust: a monoenergetic conversion electron from a stopped negative muon with no neutrinos. The mapping to RS parameters is model-dependent. It requires a lepton-quark operator basis, target-dependent overlap integrals and capture rates, and a convention for combining dipole, scalar, and vector contributions. A gold limit cannot be treated as an aluminum limit without a nuclear/operator translation.

Code coverage in this repo

NO. The required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no Mu2e, COMET, SINDRUM, $R_{\mu e}$, or $\mu - e$-conversion implementation. The only adjacent LFV code is the $\mu \to e\gamma$ dipole checker at `flavorConstraints/muToEGamma.py`:75 and its scan call at `scanParams/scan.py`:524.

Implementation difficulty

HIGH. A production implementation needs new lepton-quark Wilson coefficients, target-dependent conversion formulae for aluminum, nuclear overlap/capture inputs, and likely EFT running or matching shared with $\mu \to e\gamma$ and $\mu \to 3e$. This is beyond adding a new numerical limit to the existing dipole-only LFV path.

Key references

Process-local source keys before bibliography consolidation: `SINDRUMII2006_GoldMuE`, `Mu2e2016_FullProjection`, `Mu2eRunI2023`, `COMETPhaseITDR2020`, `COMETStrategy2018`, `Mu2eII2022`, and `CFW2008`.

71 L004: $\mu - e$ Conversion in Gold

Process

Standard notation: $R_{\mu e}^{\mathrm{Au}} = \Gamma(\mu^- \mathrm{Au} \to e^- \mathrm{Au})/\Gamma_{\mathrm{capture}}(\mu^- \mathrm{Au})$. This entry covers coherent neutrinoless muon-to-electron conversion in a gold target, the strongest published SINDRUM II conversion bound and the direct companion to the aluminum and titanium conversion entries.

PDG-or-equivalent value

The streamed PDG live muon listing, generated on 2026-05-14 and accessed on 2026-05-16, gives

$$R_{\mu e}^{\mathrm{Au}} < 7 \times 10^{-13} \qquad (90\% \text{ C.L.}).$$

These dates, the 7×10^{-13} value, and the 90% C.L. qualifier are recorded in `L004.yaml:pdg_or_equivalent.primar`
The PDG line identifies the document as BERTL 06, technique SPEC, comment SINDRUM II. The local minimal text snapshot is `flavor_catalog/references/L004/pdg2026_muon_listing_s004_au_conversion.`
The SINDRUM II article metadata is separately snapshotted from Crossref for *Eur. Phys. J. C* **47**, 337–346 (2006), DOI `10.1140/epjc/s2006-02582-x` (`L004.yaml:pdg_or_equivalent.experimental_publicatio`

Relevance to RS / anarchic flavor

In a warped lepton-flavor extension, $\mu - e$ conversion probes more than the dipole structure tested by $\mu \to e\gamma$. Tree-level or loop-induced lepton-quark vector and scalar operators, flavor-changing Z-like couplings, Higgs-mediated scalar currents, and dipoles can all contribute to the same coherent nuclear final state. A heavy target such as gold is useful as an existing benchmark, but its interpretation is target and operator dependent.

Post-2008 developments

Since the arXiv:0804.1954 RS-flavor baseline, no newer gold-target result has replaced the SINDRUM II bound in the PDG listing (`L004.yaml:pdg_or_equivalent.primary_current_limit`;

L004.yaml:pdg_or_equivalent.post_2008_baseline). The major change is the experimental program around lighter targets: the Mu2e Technical Design Report states an aluminum conversion search aiming for sensitivity approximately four orders of magnitude beyond the previous world-best conversion limits (L004.yaml:pdg_or_equivalent.post_2008_developments[0]), and the COMET Phase-I TDR quotes an aluminum sensitivity goal 3.1×10^{-15} with an expected 90% upper limit 7.0×10^{-15} (L004.yaml:pdg_or_equivalent.post_2008_developments[1]). Thus the gold limit remains the published benchmark, while future comparisons will need target-by-target translations.

Constraint validity / model dependence

The experimental statement is robust: it is a null search normalized to the ordinary muon-capture rate on gold. The mapping to RS parameters is not a single-number rescaling. It requires a lepton-quark effective operator basis, gold nuclear overlap integrals and capture normalization, and a convention for combining dipole, scalar, and vector amplitudes. Comparisons with aluminum or titanium limits are therefore model- and operator-dependent.

Code coverage in this repo

NO. The required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no SINDRUM, Mu2e, COMET, $R_{\mu e}$, gold conversion, or muon-to-electron conversion implementation. The only adjacent charged-LFV code is the dipole-only $\mu \to e\gamma$ checker at `flavorConstraints/muToEGamma.py`:75, called by the scan at `scanParams/scan.py`:524.

Implementation difficulty

HIGH. A production constraint needs a new $\mu - e$ conversion observable, lepton-quark Wilson coefficients, target-specific gold nuclear inputs, capture-rate normalization, and likely shared EFT matching/running with $\mu \to e\gamma$ and $\mu \to 3e$. This is a new mode calculation, not an input-value update to the existing dipole checker.

Key references

Process-local source keys before bibliography consolidation: `PDG2026_MuonAuConversion`, `SINDRUMII2006_GoldMuE`
`Mu2eTDR2015`, `COMETPhaseITDR2020`, and `CFW2008`.

72 L005: $\mu - e$ Conversion in Titanium

Process

Standard notation: $R_{\mu e}^{\mathrm{Ti}} = \Gamma(\mu^- \mathrm{Ti} \to e^- \mathrm{Ti})/\Gamma_{\mathrm{capture}}(\mu^- \mathrm{Ti})$. This entry covers coherent neutrinoless muon-to-electron conversion in a titanium target. The SINDRUM II titanium result should be kept separate from aluminum and gold conversion limits because nuclear overlap and capture inputs are target-dependent.

PDG / equivalent value(s)

The sidecar block `pdg_or_equivalent.primary_current_limit`, from the PDG live muon listing generated on 2026-05-14 and accessed on 2026-05-16, gives

$$R_{\mu e}^{\mathrm{Ti}} < 4.3 \times 10^{-12} \qquad (90\% \text{ C.L.}).$$

The PDG line identifies the source as DOHMEN 93, technique SPEC, comment SINDRUM II, and notes that the quoted limit assumes conversion to the nuclear ground state. The local minimal snapshot is `flavor_catalog/references/L005/pdg2026_muon_listing_s004_ti_conversion.txt`; the SINDRUM II publication metadata is snapshotted separately in `flavor_catalog/references/L005/sindrumii_19`

Relevance to the RS / anarchic-flavor pipeline

In a warped lepton-flavor extension, $\mu - e$ conversion probes lepton-quark contact operators, flavor-changing Z-like couplings, scalar currents, and dipoles. It is therefore complementary to $\mu \to e\gamma$, which tests only the dipole lane in the current repository. Titanium is useful historically because it is a real published conversion target, but its interpretation is not target-universal.

Post-2008 developments

No newer titanium-target measurement was found in the PDG stream used here after the `CFW2008` RS-flavor baseline. The main developments are prospective aluminum programs: the sidecar `post_2008_developments` entry for the Mu2e TDR records a coherent $\mu^- N \to e^- N$ program aiming roughly four orders of magnitude beyond previous published conversion limits, and the COMET Phase-I entry records an aluminum sensitivity goal 3.1×10^{-15} with an expected 90% upper limit 7.0×10^{-15}. These are not titanium measurements; they explain why the Ti number is now primarily a legacy benchmark and target-comparison input.

Constraint validity and model dependence

The experimental statement is robust as a null search normalized to the ordinary muon-capture rate in titanium. The mapping to RS parameters is model-dependent: dipole, vector, and scalar amplitudes can interfere, and the nuclear overlap integrals and capture normalization are target-specific. Direct comparison with gold or aluminum therefore requires an operator basis and nuclear-convention choice.

Code coverage in this repo

NO. The required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no SINDRUM, Mu2e, COMET, $R_{\mu e}$, titanium conversion, or general $\mu - e$-conversion implementation. The only adjacent charged-LFV code is the dipole-only $\mu \to e\gamma$ checker at `flavorConstraints/muToEGamma.py`:75, called by the scan at `scanParams/scan.py`:524.

Implementation difficulty

HIGH. A production constraint needs a new coherent conversion observable, lepton-quark Wilson coefficients, titanium nuclear overlap and capture inputs, and likely EFT matching/running shared with $\mu \to e\gamma$ and $\mu \to 3e$. This is a new target-dependent mode calculation, not a new limit for the existing dipole checker.

Key references

Process-local source keys before bibliography consolidation: `PDG2026_MuonTiConversion`, `SINDRUMII1993_Titaniu` `Mu2eTDR2015`, `COMETPhaseITDR2020`, and `CFW2008`.

73 L006: Muonium–Antimuonium Conversion

Process

Standard notation: $P_{M\bar{M}}$ and G_C/G_F, where $M = (\mu^+e^-)$ and $\bar{M} = (\mu^-e^+)$. This charged-lepton entry covers spontaneous muonium-to-antimuonium conversion, a $\Delta L_\mu = -\Delta L_e = 2$ lepton-flavor-violating oscillation process, not a dipole decay.

PDG / equivalent value(s)

The sidecar block `pdg_or_equivalent.pdg_effective_coupling_limit`, from the 2026 PDG muon listing datablock S004MC, gives the canonical effective-coupling limit

$$G_C/G_F < 0.0030 \qquad (90\%\ \text{C.L.})$$

for the stated $(V-A)\times(V-A)$ four-lepton effective Lagrangian. The same PDG datablock, mirrored in `pdg_or_equivalent.primary_current_probability_limit`, records the underlying MACS/PSI probability limit from Willmann et al.,

$$P_{M\bar{M}} < 8.3 \times 10^{-11} \qquad (90\%\ \text{C.L.})$$

in a 0.1 T magnetic field. The local canonical snapshot is `flavor_catalog/references/L006/pdg2026_muonium_an` the primary MACS article snapshot is `flavor_catalog/references/L006/willmann1999_macs_arxiv_hepex9807` These are upper limits, not measurements of a nonzero conversion rate.

Relevance to the RS / anarchic-flavor pipeline

Muonium conversion probes a four-lepton contact structure $(\bar{\mu}\Gamma e)(\bar{\mu}\Gamma e)$ that violates muon and electron family numbers by two units. In an RS lepton extension, such operators could arise from flavor-misaligned heavy neutral or doubly charged states, KK-sector lepton-current misalignment, or other non-dipole lepton interactions. It is therefore complementary to L001: the present repo checks only $\mu \to e\gamma$, while L006 would require a separate $\Delta L = 2$ four-lepton observable.

Post-2008 developments

Relative to the `CFW2008` era, the experimental bound itself has not changed: PDG still uses the 1999 MACS/PSI result. The main post-2008 movement is prospective. The sidecar `prospects` entries record the Snowmass 2021 statement that MACE is intended to improve probability sensitivity by more than two orders of magnitude and the 2024 MACE conceptual-design target beyond the 10^{-13} level. On the theory side, `post_2008_theory_context` records recent EFT work separating $\Delta L_\mu = -\Delta L_e = 2$ operators from ordinary one-unit LFV and noting that muonium conversion probes only part of that operator class.

Constraint validity and model dependence

The experimental signature is clean and lepton-sector-only: a confirmed signal would be physics beyond the Standard Model. The quoted G_C/G_F limit is not operator-universal, however; it is tied to the PDG $(V-A)\times(V-A)$ normalization. Any catalog-to-code translation must choose the Lorentz basis, magnetic-field treatment, and spin-state normalization before comparing a model Wilson coefficient to the MACS probability limit.

Code coverage in this repo

NO. A targeted grep for `muonium|antimuonium|Muonium|Antimuonium|MACE|MACS` over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` returned no hits. The adjacent charged-lepton LFV implementation is the $\mu \to e\gamma$ dipole checker at `flavorConstraints/muToEGamma.py`:75, called from `scanParams/scan.py`:524; it does not cover muonium conversion.

Implementation difficulty

MEDIUM. L006 needs a new $\Delta L_\mu = -\Delta L_e = 2$ four-lepton operator convention and a bound-state conversion-probability wrapper. It does not require lattice inputs or hadronic long-distance calculations for a first catalog-to-code implementation, but a production version should handle operator-dependent magnetic-field and spin factors explicitly.

Key references

Process-local keys before bibliography consolidation: `PDG2026_MuoniumAntimuoniumS004MC`, `Willmann1999_MACS`, `Bai2022_SnowmassMuonium`, `Bai2024_MACECDR`, `HeeckSokhashvili2024_DeltaLTwo`, `AgasheBlechmanPetriello2` and `CFW2008`.

74 L007: $\tau \to \mu\gamma$ Dipole LFV

Process

Standard notation: $\mathcal{B}(\tau^- \to \mu^-\gamma)$. This entry covers the charged-lepton-flavor-violating radiative tau decay and its use as a tau-sector dipole probe, complementary to the much tighter $\mu \to e\gamma$ constraint but involving the 2-3 lepton-flavor transition.

PDG-or-equivalent value

The current PDG/equivalent value is the sidecar `L007.yaml:primary_current_limit` upper limit

$$\mathcal{B}(\tau^- \to \mu^-\gamma) < 4.2 \times 10^{-8} \quad (90\% \text{ C.L.}),$$

from the PDGLive $S035.31$ tau decay-mode summary, locally snapshotted at `flavor_catalog/references/L007/p`
The primary experiment behind this value is the Belle full-data search using $988\,\text{fb}^{-1}$, recorded in `L007.yaml:primary_experiment`, which quotes $\mathcal{B}(\tau^\pm \to \mu^\pm\gamma) \leq 4.2 \times 10^{-8}$ at 90% C.L.; the local Belle snapshot is `flavor_catalog/references/L007/belle2021_tau_lgamma_arxiv2103_12994.txt`.

Relevance to RS / anarchic flavor

In warped lepton-flavor extensions, this mode constrains the same class of brane-localized or loop-induced charged-lepton dipoles as $\mu \to e\gamma$, but with the $\tau\mu$ flavor spurion instead of the $e\mu$ spurion. It is therefore not part of the quark $\Delta F = 2$ scan lane, but it is a useful cross-check of any future lepton-sector implementation: models that suppress $\mu \to e\gamma$ through alignment can still leave enhanced tau-sector LFV. The Perez–Randall warped-neutrino source records the lepton-MFV/GIM-like assumptions under which charged-lepton flavor violation is controlled.

Post-2008 developments

Since the arXiv:0804.1954 RS flavor baseline, the experimental limit was sharpened by the full-data B-factory searches. BaBar's post-2008 full-data search used $(963 \pm 7) \times 10^6$ tau decays and set $\mathcal{B}(\tau \to \mu\gamma) < 4.4 \times 10^{-8}$ at 90% C.L. (`L007.yaml:prior_experimental_limit`). Belle's 2021 full-data update improved the canonical $\tau \to \mu\gamma$ value to 4.2×10^{-8}. The PKA pass found no Belle II measurement superseding Belle for this channel; the sidecar records only a Belle II Physics Book projection of $< 5 \times 10^{-9}$ with $50\,\text{ab}^{-1}$, not a current bound.

Constraint validity / model dependence

Experimentally this is a clean null search for charged-lepton flavor violation, with negligible Standard Model rates. Its RS interpretation is model-dependent: the bound constrains a dipole coefficient or lepton-flavor spurion only after a choice of lepton localization, flavor symmetry, Yukawa normalization, and KK mass convention. The catalog should classify it as robust experimentally but lepton-extension-only and dipole-model-dependent for this repo.

Code coverage in this repo

NO. The required catalog grep over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $\tau \to \mu\gamma$ implementation. The adjacent code surface is only $\mu \to e\gamma$: `flavorConstraints/muToEGamma.py`:1 documents that process, lines 16–18 fix the $(1, 2)$ element, line 75 defines `check_mu_to_e_gamma`, and `scanParams/scan.py`:523–524 calls that checker in scans. Those lines are reusable design evidence, not L007 coverage.

Implementation difficulty

MEDIUM. A future implementation can reuse the existing LFV dipole pattern, but it must generalize the flavor indices and rate normalization from $\mu \to e\gamma$ to tau decays, add a tau-sector branching-ratio limit, and decide whether the same RS lepton-spurion assumptions apply. No new hadronic matrix elements, lattice inputs, or QCD running are needed.

Key references

Process-local source keys before bibliography consolidation: `PDG2025_TauMuGamma`, `Belle2021_TauLGamma`, `BaBar2010_TauLGamma`, `BelleIIPhysicsBook2019_TauLFV`, `CFW2008_RSWarpedFlavor`, and `PerezRandall2008_Wa`

75 L008: $\tau \to e\gamma$ Dipole LFV

Process

Standard notation: $\mathcal{B}(\tau^- \to e^-\gamma)$. This entry covers the charged-lepton-flavor-violating radiative tau decay and its use as a tau-sector dipole probe of the tau-electron transition. It is the electron-channel companion to L007 $\tau \to \mu\gamma$ and complements the much tighter L001 $\mu \to e\gamma$ constraint.

PDG-or-equivalent value

The current PDG/equivalent value is `L008.yaml:pdg_or_equivalent.primary_current_limit`,

$$\mathcal{B}(\tau^- \to e^-\gamma) < 3.3 \times 10^{-8} \quad (90\% \text{ C.L.}),$$

from the PDGLive $S035.32$ tau decay-mode summary, locally snapshotted at `flavor_catalog/references/L008/p`
The primary experiment behind this value is the BaBar full-data search using $(963 \pm 7) \times 10^6$ tau decays (`L008.yaml:pdg_or_equivalent.primary_experiment`), which quotes $\mathcal{B}(\tau \to e\gamma) < 3.3 \times 10^{-8}$ at 90% C.L. Belle's full-data 2021 update used $988\,\text{fb}^{-1}$ and obtained $\mathcal{B}(\tau^\pm \to e^\pm\gamma) \leq 5.6 \times 10^{-8}$ (`L008.yaml:pdg_or_equivalent.recent_belle_cross_check`); it is an important cross-check but not the strongest electron-mode limit. The Belle II Physics Book records a $50\,\text{ab}^{-1}$ prospect of 1.2×10^{-8} for this channel (`L008.yaml:pdg_or_equivalent.belle_ii_projection`), not a current bound.

Relevance to RS / anarchic flavor

In warped lepton-flavor extensions, this mode constrains brane-localized or loop-induced charged-lepton dipoles with a τe flavor spurion. It is not a constraint on the current quark $\Delta F = 2$ scan, but it is a useful catalog entry for any future charged-lepton implementation because alignment that suppresses $\mu \to e\gamma$ need not suppress the tau-to-electron transition in the same way. The Perez–Randall warped-neutrino source gives the lepton-MFV/GIM-like assumptions under which charged-lepton flavor violation is controlled.

Post-2008 developments

Relative to the arXiv:0804.1954 RS flavor baseline, the main experimental development is the BaBar 2010 full-data result, which remains the canonical PDG value for $\tau \to e\gamma$ (`L008.yaml:pdg_or_equivalent.primary` `L008.yaml:pdg_or_equivalent.rs_context`). Belle later repeated the search with its full 988 fb^{-1} data set and found no significant excess, setting the weaker electron-mode limit 5.6×10^{-8} at 90% C.L. (`L008.yaml:pdg_or_equivalent.recent_belle_cross_check`). The PKA pass found no Belle II measurement superseding the BaBar/PDG bound; the sidecar records only the 50 ab^{-1}, 1.2×10^{-8} Belle II prospect (`L008.yaml:pdg_or_equivalent.belle_ii_projection`).

Constraint validity / model dependence

Experimentally this is a clean null search for charged-lepton flavor violation, with a negligible Standard Model rate. Its RS interpretation is model-dependent: the bound constrains a dipole coefficient only after choosing the lepton localization, flavor symmetry, Yukawa normalization, and KK-mass convention. The catalog should therefore classify it as robust experimentally, but lepton-extension-only and dipole-model-dependent for this repo.

Code coverage in this repo

NO. The required catalog grep over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no $\tau \to e\gamma$ implementation. The adjacent code surface is only $\mu \to e\gamma$: `flavorConstraints/muToEGamma.py`:1 documents that process, `flavorConstraints/muToEGamma.py`:75 defines `check_mu_to_e_gamma`, and `scanParams/scan.py`:523 and `scanParams/scan.py`:524 call that checker in scans. Those lines are reusable design evidence, not L008 coverage.

Implementation difficulty

MEDIUM. A future implementation can reuse the existing LFV dipole pattern, but it must generalize the flavor indices and rate normalization from $\mu \to e\gamma$ to tau decays, add a tau-sector branching-ratio limit, and decide whether the same RS lepton-spurion assumptions apply. No new hadronic matrix elements, lattice inputs, or QCD running are needed.

Key references

Process-local source keys before bibliography consolidation: `PDG2025_TauEGamma`, `BaBar2010_TauLGamma`, `Belle2021_TauLGamma`, `BelleIIPhysicsBook2019_TauLFV`, `CFW2008_RSWarpedFlavor`, and `PerezRandall2008_Wa`

76 L009: $\tau^- \to \mu^-\mu^+\mu^-$ LFV

Process

Standard notation: $\mathcal{B}(\tau^- \to \mu^-\mu^+\mu^-)$. This entry covers the charged-lepton-flavor-violating four-lepton decay $\tau^- \to \mu^-\mu^+\mu^-$, including charge-conjugate modes when experiments quote combined limits. It is the contact-operator counterpart to the tau dipole searches in the `charged_lepton` family.

PDG-or-equivalent value

The current PDG tau listing gives

$$\mathcal{B}(\tau^- \to \mu^-\mu^+\mu^-) < 1.9 \times 10^{-8} \quad (90\% \text{ CL}),$$

with the PDG-selected entry attributed to the Belle II ADACHI 24R result using 424 fb^{-1} of e^+e^- data (`L009.yaml:pdg_or_equivalent.primary_current_limit`). The local PDG extract is `flavor_catalog/references/L009/pdg_2025_tau_listing.txt`. The Belle II primary paper reports the same 1.9×10^{-8} 90% CL limit (`L009.yaml:pdg_or_equivalent.primary_experiment`). A newer LHCb Run 2 preprint, not yet part of that PDG listing, also sets $1.9\,(2.3) \times 10^{-8}$ at 90% (95%) CL with 5.4 fb^{-1} of 13 TeV pp data (`L009.yaml:pdg_or_equivalent.post_pdg_update`). Historical benchmarks are Belle 2010 $< 2.1 \times 10^{-8}$, BaBar 2010 $< 3.3 \times 10^{-8}$, and LHCb 2015 $< 4.6\times10^{-8}$, all at 90% CL (`L009.yaml:pdg_or_equivalent.historical_experimental_limits[0]`, `L009.yaml:pdg_or_equivalent.historical_experimental_limits[1]`, and `L009.yaml:pdg_or_equivalent.his`

Relevance to RS / anarchic flavor

This mode tests lepton-sector flavor violation that need not be dipole dominated. In warped or composite-flavor extensions, flavor-nonuniversal lepton profiles, KK gauge exchange, flavor-changing Z couplings, or scalar exchange can generate effective $\tau\mu\mu\mu$ operators. The observable is therefore complementary to $\tau \to \mu\gamma$ and $\mu \to e\gamma$: dipoles can feed the three-muon final state through an off-shell photon, but the four-lepton final state is also directly sensitive to contact LFV.

Post-2008 developments

Relative to the arXiv:0804.1954 RS-flavor baseline, the experimental frontier moved to 10^{-8}-level limits (`L009.yaml:pdg_or_equivalent.rs_context`; `L009.yaml:pdg_or_equivalent.historical_experimental`
Belle and BaBar published full three-lepton tau searches in 2010, reaching 2.1×10^{-8} and 3.3×10^{-8}, respectively (`L009.yaml:pdg_or_equivalent.historical_experimental_limits[0]` and `L009.yaml:pdg_or_equ`

LHCb demonstrated a hadron-collider tau-to-three-muons program, setting 4.6×10^{-8} with Run 1 data and 1.9×10^{-8} with Run 2 (`L009.yaml:pdg_or_equivalent.historical_experimental_limits[2]`; `L009.yaml:pdg_or_equivalent.post_pdg_update`). Belle II now supplies the PDG benchmark, 1.9×10^{-8} (`L009.yaml:pdg_or_equivalent.primary_current_limit`).

Constraint validity / model dependence

Experimentally this is a clean upper limit on a forbidden charged-LFV decay. As a constraint on the current quark-scan code it is lepton-extension-only: the present quark Delta F=2 pipeline has no direct tau-lepton matching. As a future RS constraint, the translation is model dependent because a dipole-only assumption, a contact-operator assumption, and a full lepton-sector EFT can give different correlations with $\tau \to \mu\gamma$, $\mu \to e\gamma$, and $\mu \to 3e$.

Code coverage in this repo

NO. Targeted greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no implementation of $\tau \to 3\mu$, tau four-lepton LFV, or contact LFV. The adjacent LFV code is only $\mu \to e\gamma$: `flavorConstraints/muToEGamma.py:1`, `flavorConstraints/muToEGamma.py:75`, and `scanParams/scan.py:524`. The modern phenomenology surface is neutral-meson mixing only, see `quarkConstraints/modern/phenomenology.py:23`.

Implementation difficulty

MEDIUM. A first implementation needs a new four-lepton LFV observable and a convention for lepton-flavor Wilson coefficients or RS matching. It does not require lattice inputs or hadronic long-distance treatment, but a production-grade implementation should decide whether to include dipole/contact interference and correlations with the existing $\mu \to e\gamma$ lane.

Key references

Process-local raw keys: `PDG2025_Tau3Mu`, `BelleII2024_Tau3Mu`, `LHCb2026_Tau3Mu`, `LHCb2015_Tau3Mu`, `Belle2010_Tau3Leptons`, `BaBar2010_Tau3Leptons`, and `CFW2008_RSWarpedFlavor`.

77 L010: $\tau^- \to e^- e^+ e^-$ LFV

Process

Standard notation: $\mathcal{B}(\tau^- \to e^- e^+ e^-)$. This entry covers the charged-lepton-flavor-violating four-lepton decay $\tau^- \to e^- e^+ e^-$, including charge-conjugate modes when experiments quote combined limits. It is the electron-flavor counterpart of L009 $\tau \to 3\mu$ in the `charged_lepton` family.

PDG-or-equivalent value

The current PDG live tau datablock S035R58 gives

$$\mathcal{B}(\tau^- \to e^- e^+ e^-) < 2.7 \times 10^{-8} \quad (90\% \text{ CL}).$$

PDG attributes the selected limit to the Belle 2010 analysis by Hayasaka et al., based on 782 fb^{-1} of e^+e^- data at $\sqrt{s} = 10.6$ GeV (`L010.yaml:pdg_or_equivalent.primary_current_limit`). The local PDG snapshot is `flavor_catalog/references/L010/pdg2025_tau3e_pdglive_s035r58.txt`. The Belle primary paper reports the same 2.7×10^{-8} limit (`L010.yaml:pdg_or_equivalent.primary_experiment`). The BaBar 2010 analysis provides the principal independent cross-check, $\mathcal{B} < 2.9 \times 10^{-8}$ at 90% CL with 468 fb^{-1} (`L010.yaml:pdg_or_equivalent.supporting_experiment`).

Relevance to RS / anarchic flavor

This mode tests lepton-sector flavor violation that is not forced to be dipole dominated. Warped lepton extensions can generate τeee through flavor-nonuniversal lepton profiles, KK neutral-gauge exchange, flavor-changing Z couplings, or scalar exchange. It is therefore complementary to $\tau \to e\gamma$ and $\mu \to e\gamma$: an off-shell photon can contribute, but contact four-lepton operators can be the leading effect in models with tree-level lepton-current misalignment.

Post-2008 developments

Relative to the arXiv:0804.1954 RS-flavor baseline, the direct experimental bound became a B-factory 10^{-8}-level constraint in 2010 (`L010.yaml:pdg_or_equivalent.rs_context`; `L010.yaml:pdg_or_equivalen`
Belle set the current PDG-selected limit 2.7×10^{-8}, and BaBar reached the comparable 2.9×10^{-8} limit (`L010.yaml:pdg_or_equivalent.primary_current_limit`; `L010.yaml:pdg_or_equivalent.supporting_ex`
No newer LHCb electron-mode value is used here: the snapshotted LHCb source is for the all-muon companion mode only (`L010.yaml:pdg_or_equivalent.lhcb_context`). For prospects, the Belle II Physics Book quotes 50 ab^{-1} sensitivity at the $\mathcal{O}(10^{-9})$–$\mathcal{O}(10^{-10})$ level for favorable low-background tau LFV modes (`L010.yaml:pdg_or_equivalent.prospects`).

Constraint validity / model dependence

Experimentally this is a clean upper limit on a forbidden charged-LFV decay. Its interpretation in the present quark-scan repository is lepton-extension only: there is no tau-sector matching or four-lepton LFV observable in the current scan. As a future RS constraint it is model dependent because dipole-only, contact-operator, and full lepton-EFT assumptions give different correlations with $\tau \to e\gamma$, $\tau \to 3\mu$, $\mu \to e\gamma$, and $\mu \to 3e$.

Code coverage in this repo

NO. Targeted greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no implementation of $\tau \to 3e$, tau four-lepton LFV, or charged-lepton contact LFV. The adjacent implemented LFV surface is only $\mu \to$

$e\gamma$: `flavorConstraints/muToEGamma.py:75` and `scanParams/scan.py:524`. The modern phenomenology registry is limited to neutral-meson systems at `quarkConstraints/modern/phenomenology.py:23`.

Implementation difficulty

MEDIUM. A first implementation needs a new four-lepton LFV observable and a convention for lepton-flavor Wilson coefficients or RS matching. It does not require lattice inputs or hadronic long-distance calculations, but a production implementation should specify dipole/contact interference and correlations with the existing $\mu \to e\gamma$ lane.

Key references

Process-local raw keys: `PDG2025_Tau3E`, `Belle2010_Tau3Leptons`, `BaBar2010_Tau3Leptons`, `BelleIIPhysicsBook`, `LHCb2026_Tau3MuContext`, `AgasheBlechmanPetriello2006_RSLFV`, and `CFW2008_RSWarpedFlavor`.

78 L023: Neutrino Trident Production

Process

Standard notation: $\nu_\mu N \to \nu_\mu N \mu^+\mu^-$. This charged-lepton-sector entry covers muon-neutrino trident production on a nuclear target. It is flavor-sensitive but not charged-lepton-flavor violation: the relevant handle is the muon-neutrino/muon current, usually phrased as a contact interaction or a gauged $L_\mu - L_\tau$ mediator.

PDG-or-equivalent value(s)

There is no PDG or HFLAV-style average. The catalog input is the set of historical measured ratios to the Standard Model prediction recorded in `L023.yaml:pdg_or_equivalent.measured_observables`:

$$\sigma_{\mathrm{CHARM-II}}/\sigma_{\mathrm{SM}} = 1.58 \pm 0.64, \qquad \sigma_{\mathrm{CCFR}}/\sigma_{\mathrm{SM}} = 0.82 \pm 0.28, \qquad \sigma_{\mathrm{NuTeV}}/\sigma_{\mathrm{SM}} = 0.72^{+1.73}_{-0.72}.$$

These values trace to the DUNE trident compilation (`Altmannshofer2019DUNETridents`); the CCFR input is also anchored by the primary snapshot `CCFR1991TridentsWZ`, which records 37.0 ± 12.4 corrected trident events against 45.3 ± 2.3 Standard Model events. The CCFR ratio is the strongest practical historical bound; CHARM-II is the CERN-SPS observation input, and NuTeV is a weak preliminary input.

Relevance to the RS / anarchic-flavor pipeline

Trident production constrains second-generation lepton currents rather than the quark $\Delta F = 2$ system. In an RS lepton extension, KK electroweak bosons, brane kinetic mixing, or a light $L_\mu - L_\tau$-like gauge state can shift the effective $(\bar{\nu}_\mu \gamma^\alpha P_L \nu_\mu)(\bar{\mu}\gamma_\alpha \mu)$ and axial analogues. The constraint is therefore a distinct muon contact/gauge constraint, complementary to dipole LFV such as $\mu \to e\gamma$.

Post-2008 developments

Relative to the arXiv:0804.1954-era RS flavor baseline (`CsakiFalkowskiWeiler2008Flavor`), the important updates are reinterpretations and future-experiment studies. Altmannshofer–Gori– Pospelov–Yavin made the CCFR measurement a standard $L_\mu - L_\tau$ constraint, using 0.82 ± 0.28 and presenting 95% CL exclusion contours (`Altmannshofer2014TridentProbe`). Belle-II studies connected the same gauge-current parameter space to $e^+e^- \to \gamma +$ missing energy at $\sqrt{s} = 10.58$ GeV with a 50 ab^{-1} luminosity goal, which is auxiliary collider reach rather than a trident measurement (`KanetaShimomura2017BelleIITrident`). The DUNE study repeats the historical CHARM-II/CCFR/NuTeV ratios and projects a near-detector trident measurement: its abstract gives about 25% accuracy after roughly three years in each beam mode, while the body uses a 40% baseline and a 25% improved contour (`Altmannshofer2019DUNETridents`).

Constraint validity and model dependence

The experimental observable is cleanly electroweak, but the bound is not a single universal number: it depends on the target material, neutrino spectrum, selection cuts, nuclear form factors, and the assumed mediator mass relative to the momentum transfer. CCFR is robust as a measured rate ratio; translating it to a new RS mediator requires a dedicated trident cross-section calculation and cannot be borrowed from the meson-mixing or dipole backends.

Code coverage in this repo

NO. Greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no trident, CCFR, NuTeV, CHARM-II, DUNE, or $L_\mu - L_\tau$ implementation. The only "CHARM" hits are QCD charm-mass threshold constants, not this process (`L023.yaml:code_coverage`).

Implementation difficulty

HIGH. A production constraint would need a new lepton-current operator convention, mediator matching, nuclear/target treatment, neutrino-flux integration, and an experiment-specific rate or likelihood. This is a new mode calculation, not a parameter update to the existing $\Delta F = 2$ or $\mu \to e\gamma$ paths (`L023.yaml:implementation_difficulty_reason`).

Key references

Process-local source keys before bibliography consolidation: `CCFR1991TridentsWZ`; `CHARMII1990FirstObservatio` `NuTeV1998Trident`; `Altmannshofer2014TridentProbe`; `Altmannshofer2019DUNETridents`; `KanetaShimomura20` `CsakiFalkowskiWeiler2008Flavor`.

79 EDM and Neutrino

80 E001: Electron electric dipole moment

Process

Standard notation: d_e. This entry covers the flavor-diagonal CP-odd electron dipole moment, cataloged in the `edm_neutrino` family as an EDM-adjacent lepton-sector probe. It is not a quark $\Delta F = 2$ observable; its RS relevance comes from CP-violating lepton dipoles generated by heavy lepton, neutrino, Higgs, or gauge-sector dynamics.

PDG / equivalent value(s)

PDG Live 2026 (`PDG2026:ElectronEDM`) lists the current bound as

$$|d_e| < 4.1 \times 10^{-30}\ e\,\mathrm{cm} \qquad (90\%\ \mathrm{CL}),$$

from the Roussy et al. 2023 HfF^+ molecular-ion measurement; see `flavor_catalog/references/E001/pdg2026_el` The PDG datablock reports the table value as < 0.041 in units of $10^{-28}\ e\,\mathrm{cm}$, and records the corresponding measurement as $(-1.3 \pm 2.0_{\mathrm{stat}} \pm 0.6_{\mathrm{syst}}) \times 10^{-30}\ e\,\mathrm{cm}$ (`Roussy:ElectronEDM2023`). The superseded ACME/Andreev 2018 ThO benchmark is $|d_e| < 1.1 \times 10^{-29}\ e\,\mathrm{cm}$ at 90% CL, with the PDG measurement note $(4.3 \pm 3.1 \pm 2.6) \times 10^{-30}\ e\,\mathrm{cm}$ (`ACME:ElectronEDM2018`).

Relevance to the RS / anarchic-flavor pipeline

The electron EDM constrains the imaginary part of a CP-odd lepton dipole, schematically $\mathcal{L} \supset -id_e\, \bar{e}\sigma_{\mu\nu}\gamma_5 e F^{\mu\nu}/2$. In warped lepton extensions, anarchic Yukawas and new CP phases can feed this operator through loop matching. The constraint is therefore a direct diagnostic of CP alignment in the charged-lepton sector, and it is especially relevant when the neutrino sector or bulk Higgs dynamics introduce additional complex spurions. It does not directly constrain the current quark-only $\Delta F = 2$ scan.

Post-2008 developments

Relative to the 2008 CFW RS-flavor reference point (`CsakiFalkowskiWeiler:RSFlavor2008`), the experimental EDM frontier changed qualitatively. ACME 2018 (`ACME:ElectronEDM2018`) set the previous molecular ThO limit at $1.1 \times 10^{-29}\ e\,\mathrm{cm}$, already strong enough to motivate EFT treatments of CP-violating electron dipoles. Roussy et al. 2023 (`Roussy:ElectronEDM2023`) then improved the previous best upper bound by about 2.4 using trapped HfF^+ molecular ions, yielding the current PDG value above. The post-ACME EFT literature, represented here by Panico, Pomarol, and Riembau 2019 (`PanicoPomarolRiembau:ElectronEDM2019`), treats electron EDMs as constraints on CP-violating dipole operators and their two-loop anomalous dimensions, the natural language for a future Wilson-coefficient implementation.

Constraint validity and model dependence

The experimental null limit is robust as a bound on the effective electron EDM extracted from paramagnetic molecular spectroscopy. Model interpretation is less automatic: paramagnetic systems can also be sensitive to semileptonic CP-odd electron-nucleon interactions, and an RS implementation must specify which SMEFT or low-energy CP-odd operator basis is being matched. As a catalog constraint this is a lepton-extension and EDM-adjacent loop bound, not a standalone exclusion in the existing quark pipeline.

For anarchic CP phases, EDM constraints should be reviewed jointly with the flavor branching rows that share the same Wilson coefficients (relevant cross-refs: K001, K003, B011, B033, B034), not as an isolated appendix.

Code coverage in this repo

NO. The required catalog greps and a focused implementation grep for `electron.?EDM`, `eEDM`, `electric dipole`, and `d_e` over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no electron-EDM implementation. The only adjacent dipole code is the off-diagonal $\mu \to e\gamma$ bound in `flavorConstraints/muToEGamma.py:1`, `:3`, `:20--21`, and `:81`; `scanParams/scan.py:32--33` sets a MEG II branching-fraction limit for that different process.

Implementation difficulty

HIGH. A live implementation needs a new flavor-diagonal CP-odd lepton dipole observable, matching from the model's lepton/KK sector to the dipole Wilson coefficient, and likely SMEFT-to-low-energy running. The existing $\Delta F = 2$ operator basis and the $\mu \to e\gamma$ NDA helper do not provide the needed CP-odd d_e calculation.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:ElectronEDM`, `Roussy:ElectronEDM2023`, `ACME:ElectronEDM2018`, `PanicoPomarolRiembau:ElectronEDM2019`, and `CsakiFalkowskiWeiler:RSFlavor2008`

81 E002: Muon electric dipole moment

Process

Standard notation: d_μ. This entry covers the flavor-diagonal CP-odd electric dipole moment of the muon, cataloged in the `edm_neutrino` family. It is an EDM-adjacent lepton-sector process, not a charged-lepton-flavor-violating decay and not a quark $\Delta F = 2$ observable.

PDG / equivalent value(s)

The sidecar block `pdg_or_equivalent.canonical_direct_limit`, from PDG Live 2026 (`PDG2026:MuonEDM`), lists the canonical direct experimental bound as

$$|d_\mu| < 1.8 \times 10^{-19}\ e\,\mathrm{cm} \qquad (95\%\ \mathrm{CL}),$$

from the Bennett et al. BNL E821 storage-ring analysis; see `flavor_catalog/references/E002/pdg2026_muon_ed` The sidecar block `pdg_or_equivalent.pdg_combined_measurement` records the combined result as $(0.0 \pm 0.9) \times 10^{-19}$ $e\,\mathrm{cm}$. The `pdg_or_equivalent.primary_bnl_measurement` block, from `Bennett:MuonEDM2009`, reports $d_\mu = -0.1(0.9) \times 10^{-19}$ $e\,\mathrm{cm}$ and quotes $|d_\mu| < 1.9 \times 10^{-19}$ $e\,\mathrm{cm}$ at 95% CL.

Relevance to the RS / anarchic-flavor pipeline

A nonzero d_μ probes the imaginary part of the flavor-diagonal muon dipole operator, schematically $\bar{\mu}\sigma_{\alpha\beta}\gamma_5\mu F^{\alpha\beta}$. In warped lepton extensions, anarchic charged-lepton and neutrino Yukawa structures can generate such dipoles through loop matching involving KK leptons, Higgs dynamics, or gauge-sector phases. The process therefore diagnoses CP alignment in the muon sector, complementary to $\mu \to e\gamma$, but it does not directly constrain the existing quark-only scan.

Post-2008 developments

Relative to the 2008 CFW warped-flavor reference point (`CsakiFalkowskiWeiler:RSFlavor2008`), the key direct experimental development is the BNL E821 publication in 2009. That analysis used muon $g-2$ storage-ring spin-precession data and remains the basis of the PDG direct limit. The major post-2008 prospective change is the dedicated PSI muEDM program. The 2021 PSI letter of intent `auxiliary_values.psi_frozen_spin_projection` (`Adelmann:MuonEDMPSI2021`) proposes a frozen-spin search with $p = 125$ MeV/c, $|B| = 3$ T, a rest-frame electric field of 1 GV/m, and an expected sensitivity $\sigma(d_\mu) \leq 6 \times 10^{-23}$ $e\,\mathrm{cm}$. The `auxiliary_values.psi_2024_status` entry (`Renga:MuonEDMStatus2024`) describes a first phase aimed at demonstrating feasibility by 2026, with an ultimate goal of improving the sensitivity by a factor of 100 by the early 2030s.

Constraint validity and model dependence

The PDG number is a robust direct storage-ring bound on a static muon EDM. Its interpretation in RS is model-dependent: a live constraint requires the lepton-sector CP phases, the dipole matching scale, and the treatment of SMEFT/low-energy running to be fixed. The PSI entries are projections and should be treated as roadmap information, not current exclusions.

Code coverage in this repo

NO. Focused greps for `muEDM`, `muon.?EDM`, `muon electric dipole`, `d_mu`, and `dmu` over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no muon-EDM implementation. The nearby dipole code is the off-diagonal $\mu \to e\gamma$

helper in `flavorConstraints/muToEGamma.py:3`, `:21`, and `:81`, with the MEG II branching-fraction default at `scanParams/scan.py:33`; that is a different observable.

Implementation difficulty

HIGH. A production implementation needs a new flavor-diagonal CP-odd lepton dipole observable, loop matching from the lepton/KK sector, and likely SMEFT-to-low-energy running. The existing $\Delta F = 2$ basis and the $\mu \to e\gamma$ NDA helper do not calculate d_μ.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:MuonEDM`, `Bennett:MuonEDM2009`, `Adelmann:MuonEDMPSI2021`, `Renga:MuonEDMStatus2024`, and `CsakiFalkowskiWeiler:RSFlavor2008`.

82 E004: Neutron electric dipole moment

Process

Standard notation: d_n. This entry covers the permanent electric dipole moment of the neutron, cataloged in the `edm_neutrino` family as a hadronic EDM and CP-violation benchmark. The observable is flavor diagonal, but it probes CP-odd quark EDMs, quark chromo-EDMs, the QCD θ term, and the Weinberg three-gluon operator.

PDG / equivalent value(s)

PDG Live 2026 (`PDG2026:NeutronEDM`) lists the current direct neutron EDM limit as

$$|d_n| < 1.8 \times 10^{-26}\ e\,\mathrm{cm} \qquad (90\%\ \mathrm{CL}),$$

from Abel et al. 2020 using ultracold neutrons and Ramsey separated oscillatory fields at PSI (`PDG2026:NeutronEDM`). The PDG table records this as < 0.18 in units of $10^{-25}\ e\,\mathrm{cm}$; see `flavor_catalog/referen`
The associated experimental result is

$$d_n = (0.0 \pm 1.1_{\mathrm{stat}} \pm 0.2_{\mathrm{sys}}) \times 10^{-26}\ e\,\mathrm{cm},$$

recorded both by PDG and in the arXiv snapshot `flavor_catalog/references/E004/abel2020_arxiv2001_11966`
(`Abel:NeutronEDM2020`). PDG also lists the superseded Pendlebury et al. 2015 direct limit $|d_n| < 3.0 \times 10^{-26}\ e\,\mathrm{cm}$ at 90% CL (`PDG2026:NeutronEDM`).

Relevance to the RS / anarchic-flavor pipeline

In anarchic warped or composite-Higgs completions, new complex Yukawa spurions and KK fermion/gauge loops can generate flavor-diagonal CP-odd dipoles even when tree-level $\Delta F = 2$ observables are controlled. The neutron EDM is therefore a standard hadronic CP benchmark for quark EDMs and chromo-EDMs, and for threshold-induced Weinberg-operator contributions. It does not map onto the existing neutral-meson mixing basis; it would test a different CP-odd operator lane.

Post-2008 developments

The 2008 CFW reference point (`CsakiFalkowskiWeiler:RSFlavor2008`) focused on warped-flavor bounds such as KK-gluon constraints. Since then, the direct neutron EDM experimental limit improved from the PDG-listed Pendlebury-era $3.0 \times 10^{-26}\ e\,\mathrm{cm}$ bound to the Abel et al. 2020 $1.8\times10^{-26}\ e\,\mathrm{cm}$ bound (`PDG2026:NeutronEDM`; `Abel:NeutronEDM2020`). On the model side, Koenig, Neubert, and Straub 2014 (`KoenigNeubertStraub:DipoleComposite2014`) made quark electromagnetic and chromomagnetic dipoles explicit in composite-Higgs and warped extra-dimensional language, including the role of "wrong-chirality" Yukawas and the neutron EDM. On the hadronic side, lattice work such as Bhattacharya et al. 2022 (`Bhattacharya:NeutronEDMLattice2022`) now directly targets the neutron EDM induced by the QCD topological term, the Weinberg three-gluon operator, and the isovector quark chromo-EDM.

Constraint validity and model dependence

The experimental limit is a robust null bound on d_n, but translating it into RS parameters is hadronic- and basis-dependent. A live interpretation must specify the UV matching to quark EDM, quark chromo-EDM, θ, and Weinberg coefficients; their QCD running and threshold matching; and the low-energy neutron matrix elements. This is an EDM-adjacent loop constraint, not a direct $\Delta F = 2$ or lepton-universality observable.

For anarchic CP phases, EDM constraints should be reviewed jointly with the flavor branching rows that share the same Wilson coefficients (relevant cross-refs: K001, K003, B011, B033, B034), not as an isolated appendix.

Code coverage in this repo

NO. The required catalog greps and a focused search for `neutron[ -]?EDM`, `nEDM`, `d_n`, `chromo-EDM`, `CEDM`, `Weinberg`, and `three-gluon` over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no neutron EDM implementation. The only nearby dipole code is the off-diagonal $\mu \to e\gamma$ helper in `flavorConstraints/muToEGamma.py:3`, `:21`, and `:81`, which is not a hadronic EDM calculation.

Implementation difficulty

HIGH. A production constraint needs new CP-odd dipole and gluonic operator matching, QCD running with thresholds, and hadronic neutron matrix elements. The existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ infrastructure does not cover quark EDMs, chromo-EDMs, or the Weinberg operator.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:NeutronEDM`, `Abel:NeutronEDM2020`, `CsakiFalkowskiWeiler:RSFlavor2008`, `KoenigNeubertStraub:DipoleComposite2014`, and `Bhattacharya:Neut`

83 E006: Mercury-199 electric dipole moment

Process

Standard notation: d_{Hg}, for neutral ^{199}Hg. This entry covers the permanent EDM of the diamagnetic ^{199}Hg atom, cataloged in the `edm_neutrino` family as a hadronic/nuclear CP probe. It is a flavor-diagonal observable, but its interpretation constrains CP-odd quark, gluon, semileptonic, and nuclear Schiff-moment sources.

PDG / equivalent value(s)

The canonical direct experimental limit is the Graner et al. 2016 ^{199}Hg result (`Graner:HgEDM2016`):

$$|d_{\mathrm{Hg}}| < 7.4 \times 10^{-30}\ e\,\mathrm{cm} \qquad (95\%\ \mathrm{C.L.}).$$

The local snapshot `flavor_catalog/references/E006/graner2016_arxiv1601_04339.txt` also records the arXiv-v4 central measurement and uncertainties; because the PKA flagged the erratum cross-check for the signed central value, the direct limit is the catalog value. PDG Live 2026 does not expose a standalone Hg-atom datablock here, but the neutron and proton EDM datablocks cite the same Hg measurement for theory-derived nucleon limits. In the neutron datablock, PDG lists Graner 2016 as a ^{199}Hg-plus-theory row, < 0.16 in units of $10^{-25}\ e\,\mathrm{cm}$ at 95% CL, and Sahoo 2017 as < 0.22 in the same units and confidence level (`PDG2026:NeutronEDMHgCrossref`); see `flavor_catalog/references/E006/pdg2026_neutron_edm_hg_crossref.txt`.

Relevance to the RS / anarchic-flavor pipeline

Warped or composite anarchic flavor models generically contain new CP phases. Loop matching can generate quark EDMs, quark chromo-EDMs, the Weinberg three-gluon operator, and semileptonic CP-odd operators. In a diamagnetic closed-shell atom such as ^{199}Hg, the dominant hadronic sensitivity is usually phrased through the nuclear Schiff moment and CP-odd pion-nucleon interactions, rather than through a simple elementary fermion EDM. The observable is therefore complementary to the neutron EDM: it probes a different linear combination of CP-odd low-energy operators and nuclear matrix elements.

Post-2008 developments

Relative to the 2008 CFW warped-flavor baseline (`CsakiFalkowskiWeiler:RSFlavor2008`), the main experimental update is the 2016 Washington ^{199}Hg measurement (`Graner:HgEDM2016`), which improved the previous Hg bound by a factor of 4 and remains the standard diamagnetic-atom benchmark. The post-2008 theory update represented here by Sahoo 2017 (`Sahoo:HgEDMTheory2017`) combines the Hg atom limit with relativistic atomic calculations and quotes derived limits including $d_n < 2.2 \times 10^{-26}\ |e|\,\mathrm{cm}$, $d_p < 2.1 \times 10^{-25}\ |e|\,\mathrm{cm}$, $|\bar{\theta}| < 1.1 \times 10^{-10}$, and $|\tilde{d}_u - \tilde{d}_d| < 5.5 \times 10^{-27}\ |e|\,\mathrm{cm}$ (`Sahoo:HgEDMTheory2017`).

Constraint validity and model dependence

The experimental atom-EDM bound is robust as a null search for a P- and T-violating energy shift. Translating it into RS parameters is model-dependent: the result depends on the chosen CP-odd operator basis, QCD running and thresholds, atomic many-body coefficients, nuclear Schiff moments, hadronic matrix elements, and possible cancellations among sources. It should be treated as an EDM-adjacent loop/nuclear-translation constraint, not as a direct $\Delta F = 2$ bound in the current quark scan.

For anarchic CP phases, EDM constraints should be reviewed jointly with the flavor branching rows that share the same Wilson coefficients (relevant cross-refs: K001, K003, B011, B033, B034), not as an isolated appendix.

Code coverage in this repo

NO. The required catalog greps and a focused search for `Hg`, `mercury`, `199Hg`, `d_Hg`, `Schiff`, `diamagnetic`, `chromo-EDM`, `Weinberg`, and generic `EDM/dipole` terms over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no ^{199}Hg EDM implementation. The only nearby dipole code is the off-diagonal $\mu \to e\gamma$ helper in `flavorConstraints/muToEGamma.py:3`, `:21`, and `:81`.

Implementation difficulty

HIGH. A live constraint would require new CP-odd quark/gluon and semileptonic operator matching, RG running to hadronic scales, and an atomic/nuclear translation layer for ^{199}Hg. The existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ basis and the $\mu \to e\gamma$ NDA helper do not cover this observable.

Key references

Process-local source keys before bibliography consolidation: `Graner:HgEDM2016`, `PDG2026:NeutronEDMHgCrossref` `Sahoo:HgEDMTheory2017`, and `CsakiFalkowskiWeiler:RSFlavor2008`.

84 E007: Radium and xenon electric dipole moments

Process

Standard notation: d_{Ra} for ^{225}Ra and d_{Xe} for ^{129}Xe. This entry covers atomic EDM searches in radium and xenon, cataloged in the `edm_neutrino` family as complementary diamagnetic-atom and nuclear CP probes beyond the neutron and ^{199}Hg benchmark constraints.

PDG / equivalent value(s)

No standalone PDG Live Ra/Xe atom-EDM datablock was located for this PKA deposit, so the direct experiment papers are used as PDG-equivalent sources. For ^{225}Ra, the canonical Argonne direct limit is in the sidecar snapshot `bishof2016_arxiv1606_04931.txt`,

$$|d(^{225}\mathrm{Ra})| < 1.4 \times 10^{-23}\ e\,\mathrm{cm} \qquad (95\%\ \mathrm{C.L.}),$$

from `flavor_catalog/references/E007/bishof2016_arxiv1606_04931.txt`. The earlier proof-of-principle Ra measurement in `parker2015_arxiv1504_07477.txt` gave $|d(^{225}\mathrm{Ra})| < 5.0\times10^{-22}\ e\,\mathrm{cm}$ at 95% confidence and is kept as a superseded post-2008 milestone.

For ^{129}Xe, the strongest direct limit in the deposited sources is `sachdeva2019_arxiv1909_12800.txt`,

$$d_A(^{129}\mathrm{Xe}) = (1.4 \pm 6.6_{\mathrm{stat}} \pm 2.0_{\mathrm{syst}}) \times 10^{-28}\ e\,\mathrm{cm}, \qquad |d_A(^{129}\mathrm{Xe})| < 1.4 \times 10^{-27}\ e\,\mathrm{cm} \quad (95\%\ \mathrm{C.L.}).$$

An independent Heidelberg-centered ^{129}Xe measurement in `allmendinger2019_arxiv1904_12295.txt` gives $(-4.7 \pm 6.4) \times 10^{-28}\ e\,\mathrm{cm}$ and $|d_{\mathrm{Xe}}| < 1.5 \times 10^{-27}\ e\,\mathrm{cm}$ at 95% C.L.

Relevance to the RS / anarchic-flavor pipeline

Warped anarchic flavor models generically contain new CP phases. Loop matching can generate quark EDMs, chromo-EDMs, the Weinberg three-gluon operator, and semileptonic CP-odd operators. Diamagnetic atoms do not map directly onto one elementary dipole; their dominant hadronic response is usually expressed through nuclear Schiff moments and CP-odd hadronic interactions. ^{225}Ra is especially useful because octupole deformation enhances Schiff-moment sensitivity, while ^{129}Xe offers a mature noble-gas co-magnetometer platform with different nuclear and atomic systematics.

Post-2008 developments

Relative to the 2008 warped-flavor baseline of `CsakiFalkowskiWeiler:RSFlavor2008`, Ra and Xe EDMs moved from mostly future context into direct experimental inputs. Argonne reported the first ^{225}Ra EDM measurement in 2015 `parker2015_arxiv1504_07477.txt` and improved it in 2016 by a factor of 36 `bishof2016_arxiv1606_04931.txt`. The 2016 Ra paper also records planned upgrades aimed at substantially better statistical reach with controlled systematics `bishof2016_arxiv1606_04931.txt`. The deposited Argonne program page `Argonne:Ra225EDMPage` states the physical motivation: ^{225}Ra has experimentally useful nuclear properties and enhanced sensitivity to T-odd, P-odd effects relative to ^{199}Hg.

For Xe, the 2019 comagnetometer measurements `sachdeva2019_arxiv1909_12800.txt` and `allmendinger2019_ar` established $10^{-27}\ e\,\mathrm{cm}$-level direct limits. The DFG Heidelberg upgrade proposal `DFG:Xe129Upgrade2026` aims to improve the ^{129}Xe EDM reach. KU Leuven's RaF/AcF molecular project `KULeuven:RaFMoleculeProject2` is not a direct Ra-atom limit, but it is relevant future context because embedding octupole-enhanced Ra or Ac nuclei in dipolar molecules can enhance Schiff-moment sensitivity.

Constraint validity and model dependence

The direct atom-EDM limits are robust null searches for P- and T-violating energy shifts. Their interpretation in an RS pipeline is highly model-dependent: one must choose a CP-odd low-energy operator basis, run and match through QCD thresholds, and adopt hadronic, nuclear, and atomic coefficients. These entries should therefore be treated as EDM-adjacent loop/nuclear-translation probes, not as direct $\Delta F = 2$ quark-scan constraints.

Code coverage in this repo

NO. The sidecar `code_coverage` block records the required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no Ra, Xe, Schiff-moment, diamagnetic-atom, chromo-EDM, or Weinberg-operator implementation. The only nearby dipole code is the off-diagonal $\mu \to e\gamma$ helper at `flavorConstraints/muToEGamma.py:` `flavorConstraints/muToEGamma.py:21`, and `flavorConstraints/muToEGamma.py:81`; that code is not an atomic or hadronic EDM calculation.

Implementation difficulty

HIGH. A live E007 constraint needs new CP-odd quark/gluon and semileptonic matching, RG evolution to hadronic scales, and separate nuclear/atomic response layers for Ra and Xe. The existing SLL/SLR/VLL/VRR/LR $\Delta F = 2$ basis and the $\mu \to e\gamma$ dipole helper do not cover this observable.

Key references

Process-local source keys before bibliography consolidation: `Bishof:Ra225EDM2016`, `Parker:Ra225EDM2015`, `Sachdeva:Xe129EDM2019`, `Allmendinger:Xe129EDM2019`, `Argonne:Ra225EDMPage`, `DFG:Xe129Upgrade2026`, `KULeuven:RaFMoleculeProject2023`, and `CsakiFalkowskiWeiler:RSFlavor2008`.

85 E008: Quark chromo-EDM bounds

Process

Standard notation: $\tilde{d}_q$, the flavor-diagonal quark chromo-electric dipole moment in the CP-odd operator $\bar{q}T^a\sigma^{\mu\nu}\gamma_5 q\, G^a_{\mu\nu}$. This entry covers EFT-level benchmark bounds on light-quark qCEDMs inferred from neutron and ^{199}Hg EDM searches. These are interpretation-level translations, not directly measured particle properties.

PDG / equivalent value(s)

The experimental anchors are the PDG Live 2026 neutron limit

$$|d_n| < 1.8 \times 10^{-26}\ e\,\mathrm{cm} \quad (90\%\ \mathrm{CL})$$

from Abel 2020 (`PDG2026:NeutronEDM`), and the Graner et al. 2016 mercury result

$$d_{\mathrm{Hg}} = (2.20 \pm 2.75_{\mathrm{stat}} \pm 1.48_{\mathrm{syst}}) \times 10^{-30}\, e\,\mathrm{cm}, \qquad |d_{\mathrm{Hg}}| < 7.4 \times 10^{-30}\, e\,\mathrm{cm} \quad (95\%\ \mathrm{CL}).$$

The mercury anchor is `Graner:HgEDM2016`. Using the Pospelov–Ritz QCD-sum-rule relation with the quark EDMs set to zero gives the central neutron translation

$$|\tilde{d}_d + 0.5\, \tilde{d}_u| < 1.6 \times 10^{-26}\ \mathrm{cm},$$

with the quoted (1 ± 0.5) hadronic normalization uncertainty (`PospelovRitz:NeutronCEDM2000`). Using the standard mercury estimate

$$d_{\mathrm{Hg}} = 7 \times 10^{-3} e(\tilde{d}_u - \tilde{d}_d) + 10^{-2} d_e + \cdots$$

and setting non-qCEDM sources to zero gives the central equivalent bound

$$|\tilde{d}_u - \tilde{d}_d| < 1.1 \times 10^{-27}\ \mathrm{cm}.$$

This mercury translation uses `OlivePospelovRitzSantoso:HgCEDM2005`. Both qCEDM numbers are one-source-at-a-time benchmarks, not correlated global EDM-fit limits. The local snapshots are `flavor_catalog/references/E008/pdg2026_neutron_edm_datablock.txt` and `flavor_catalog/references/E0` the translation formulae are in the Pospelov–Ritz and Olive–Pospelov–Ritz–Santoso snapshots.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped models contain new CP phases and chirality flips in KK-fermion, Higgs, and gauge loops. Even when tree-level $\Delta F = 2$ mixing is under control, these loops can generate flavor-diagonal quark EDMs and qCEDMs. The qCEDM lane is therefore a direct probe of RS KK-loop CP violation and wrong-chirality Yukawa structures.

Post-2008 developments

Relative to the CFW 2008 RS-flavor baseline, the experimental inputs tightened: the direct neutron anchor is now the 2020 PDG limit above, and the 2016 ^{199}Hg result improved the prior mercury limit by a factor of 4 (`PDG2026:NeutronEDM`; `Graner:HgEDM2016`; `CsakiFalkowskiWeiler:RSFlavor2008`). Koenig, Neubert, and Straub 2014 recast electromagnetic and chromomagnetic quark dipoles in partial-compositeness and warped-language Wilson coefficients, finding strong neutron-EDM pressure when wrong-chirality Yukawas are present (`KoenigNeubertStraub:DipoleComposite2014`). On the theory side, Bhattacharya et al. 2024 target the isovector qCEDM neutron matrix element with lattice QCD and explicit control of power-divergent operator mixing (`Bhattacharya:IsoVectorQCEDM2024`).

Constraint validity and model dependence

The direct neutron and mercury limits are robust null measurements. The qCEDM bounds are not standalone PDG averages: they assume a low-energy CP-odd basis, neglect cancellations with quark EDMs, semileptonic sources, four-quark operators, $\bar{\theta}$, and the Weinberg operator, and inherit QCD, nuclear, and atomic matrix-element uncertainties. Mercury is especially powerful for the isovector combination but is also more nuclear-model dependent.

For anarchic CP phases, EDM constraints should be reviewed jointly with the flavor branching rows that share the same Wilson coefficients (relevant cross-refs: K001, K003, B011, B033, B034), not as an isolated appendix.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no qCEDM, neutron EDM, mercury EDM, or Weinberg-operator constraint. The only nearby dipole implementation is the unrelated $\mu \to e\gamma$ helper in `flavorConstraints/muToEGamma.py:3`, `:21`, and `:81`.

Implementation difficulty

HIGH. A live E008 constraint would need new CP-odd quark EDM/qCEDM operators, loop matching from the RS spectrum, QCD running and threshold mixing, plus hadronic and nuclear matrix-element choices. The existing $\Delta F = 2$ SLL/SLR/VLL/VRR/LR basis does not cover this observable.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:NeutronEDM`, `Graner:HgEDM2016`, `PospelovRitz:NeutronCEDM2000`, `OlivePospelovRitzSantoso:HgCEDM2005`, `CsakiFalkowskiWeiler:RSFlavor` `KoenigNeubertStraub:DipoleComposite2014`, and `Bhattacharya:IsoVectorQCEDM2024`.

86 E009: Weinberg three-gluon operator

Process

Standard notation: w or $C_{\tilde{G}}$, multiplying the CP-odd gluonic operator schematically $f^{abc}\tilde{G}^a_{\mu\nu}G^{b\,\nu\rho}G^{c\,\mu}_{\rho}$. This entry covers EDM constraints on the flavor-diagonal Weinberg three-gluon operator, not the lepton-number Weinberg operator for neutrino masses.

PDG / equivalent value(s)

There is no directly measured PDG value for w. The experimental anchor is the PDG Live 2026 neutron EDM limit (`PDG2026:NeutronEDM`)

$$|d_n| < 1.8 \times 10^{-26}\, e\,\text{cm} \quad (90\%\ \text{CL}),$$

based on Abel et al. 2020 (`Abel:NeutronEDM2020`), with

$$d_n = (0.0 \pm 1.1_{\text{stat}} \pm 0.2_{\text{sys}}) \times 10^{-26}\ e\,\text{cm}.$$

The local snapshots are `flavor_catalog/references/E009/pdg2026_neutron_edm_datablock.txt` and `flavor_catalog/references/E009/abel2020_neutron_edm_arxiv2001_11966.txt`. The sidecar records the convention-dependent Wilson-coefficient translations as auxiliary theory inputs rather than as PDG observables.

Relevance to the RS / anarchic-flavor pipeline

Anarchic warped or partial-compositeness models carry new CP phases in the colored sector. Integrating out KK fermions, heavy quarks, Higgs-sector states, or colored resonances can generate quark chromo-EDMs and finite heavy-threshold contributions to the Weinberg operator. The operator is therefore a compact EDM-adjacent diagnostic of colored CP violation that complements E008.

Post-2008 developments

Relative to the CFW 2008 RS-flavor baseline (`CsakiFalkowskiWeiler:RSFlavor2008`), the neutron anchor is now the Abel 2020 result adopted by PDG Live 2026 (`Abel:NeutronEDM2020`; `PDG2026:NeutronEDM`). Composite and warped dipole analyses after 2008 emphasized that heavy-quark CEDM running generates a finite threshold correction to the three-gluon Weinberg operator (`KoenigNeubertStraub:CompositeDipoles2014`). In the Pospelov–Ritz normalization, $|d_n(w)| \simeq e\,22\ \mathrm{MeV}\,w(1\ \mathrm{GeV})$, which gives the central one-source benchmark $|w(1\ \mathrm{GeV})| < 4.1\times10^{-11}\ \mathrm{GeV}^{-2}$ from the current neutron limit when other CP-odd sources are set to zero (`PospelovRitz:EDMReview2005`; `PDG2026:NeutronEDM`). Haisch and Hala 2019 quote, in their O_6 convention, $(d_n/e)_{O_6} = 74(1 \pm 0.5)$ MeV, corresponding to the central benchmark $|C_6| < 1.2\times10^{-11}\ \mathrm{GeV}^{-2}$ (`HaischHala:WeinbergSumRules2019` `PDG2026:NeutronEDM`). Lattice work using gradient flow now studies the dimension-six gluonic CP-violating matrix element and its operator mixing (`Bhattacharya:WeinbergLattice2022`).

Constraint validity and model dependence

The neutron EDM limit is robust. The Weinberg-operator bound is not: it depends on operator normalization, RG running, heavy-quark thresholds, hadronic matrix elements, and assumptions about cancellations with $\bar{\theta}$, quark EDMs, qCEDMs, semileptonic terms, and four-quark operators. In global EDM language, the same neutron and diamagnetic data probe more CP-odd sources than there are independent low-energy observables.

For anarchic CP phases, EDM constraints should be reviewed jointly with the flavor branching rows that share the same Wilson coefficients (relevant cross-refs: K001, K003, B011, B033, B034), not as an isolated appendix.

Code coverage in this repo

NO. Required greps over `quarkConstraints/`, `qcd/`, `flavorConstraints/`, `neutrinos/`, `yukawa/`, `warpConfig/`, `solvers/`, `scanParams/`, and `tests/` found no Weinberg, neutron-EDM, or CP-odd gluonic operator constraint. As recorded in the sidecar code-coverage block, the only nearby dipole implementation is the unrelated $\mu \to e\gamma$ code in `flavorConstraints/muToEGamma.py:3`, `:21`, and `:81`.

Implementation difficulty

HIGH. A live E009 constraint needs new CP-odd colored operators, RS loop or threshold matching, QCD running and mixing into other EDM sources, and a chosen hadronic matrix-element convention. The existing $\Delta F = 2$ basis and muon LFV dipole path do not cover this observable.

Key references

Process-local source keys before bibliography consolidation: `PDG2026:NeutronEDM`, `Abel:NeutronEDM2020`, `Weinberg:ThreeGluon1989`, `PospelovRitz:EDMReview2005`, `ChuppRamseyMusolf:EDMGlobal2014`, `KoenigNeubertStraub:CompositeDipoles2014`, `HaischHala:WeinbergSumRules2019`, `Bhattacharya:Weinberg`
and `CsakiFalkowskiWeiler:RSFlavor2008`.